Fabric Evolution in 3D Discrete Element Modelling of Soil Liquefaction

by

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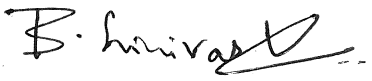
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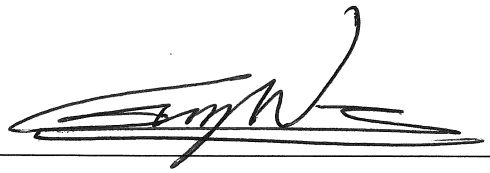
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This is to certify that I have examined the above MPhil thesis
and have found that it is complete and satisfactory in all respects,
and that any and all revisions required by
the thesis examination committee have been made.



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ABSTRACT

Liquefaction refers to the dramatic loss of soil stiffness and strength during earthquake shaking. Knowing the particle-scale information of granular particles and void space, collectively termed as "fabric", is crucial for obtaining a fundamental understanding of liquefaction in granular soils. In this study, fabric evolution of granular soils in cyclic liquefaction is investigated using 3D Discrete Element Method (DEM). A set of new fabric indices are developed for quantifying particle-void distribution (E_d,A_d), which are found to have an excellent correlation with the post-liquefaction behavior, as well as irreversible changes in fabrics developed in liquefaction processes, and finally proposed a fabric-based criterion for jamming transition in flow deformation. The descriptor E_d, is found to be a good indicator for the compressibility of the sample in the post-liquefaction stage. DEM simulations have been conducted to identify critical factors that influence the granular soil's resistance to multiple liquefaction. It is observed that re-liquefaction resistance is greatly affected by the fabric anisotropy of the sample by surpassing the effect of relative density. Within an unloading cycle in the post-liquefaction stage, completely different soil fabric was formed if the liquefied soil is consolidated at different unloading points, resulting in vastly different resistance to re-liquefaction.

In addition, medium dense granular assembly is subjected to different multi-directional loading paths (Circular/Oval, Figure-8, Uni-directional) to explore the changes in soil micromechanical structure. The trajectory of evolution in contact-based fabric tensor components is found to

follow the shape of the loading path in the initial cycles of loading, whereas the trajectory of evolution in new void-based fabric tensor components follow the variation in shear strain corresponding to different directions. A possible new-criterion to decide soil liquefaction triggering based on energy dissipation is discussed, which has more physical meaning compared to the available criterion in case of multi-directional loading paths. Overall, the discrete element modelling provides many insights that link microscopic fabric evolution to the macroscopic cyclic soil behavior, which can be crucial information to develop micro-mechanically based constitutive models.

CHAPTER 1
Introduction

1.1 Background of the research

The dramatic loss of strength in saturated sands during an earthquake shaking is commonly referred to as soil liquefaction. During seismic (cyclic) loading, the normal load that is initially carried by sand structure is gradually transferred to the pore water, resulting in a decrease of effective confining pressure and increase of pore water pressure. As pore water cannot resist shear forces, saturated sands tend to flow uncontrollably, causing soil liquefaction. The terminology of soil liquefaction first came into existence after the occurrence of the Niigata and Alaska earthquakes in 1964. The damages reported following these earthquakes includes slope failures, sinking of bridge piers, tilted houses (Figure 1.1), embankments, foundations and exposure of buried structures. Loss of shear strength in liquefied sands led to such substantial damages. After 1964, numerous other examples of massive destruction due to soil liquefaction are reported. Disastrous earthquakes such as the 1971 San Fernando earthquake, the 1989 Loma Prieta earthquake, the 1995 Kobe earthquake, the 2010 Haiti earthquake, and the recent 2018 Palu earthquake have all come with consequences of soil liquefaction. Reports show that in the recent Palu earthquake, the flow slides in coastal areas triggered the tsunamis of a maximum height of about 6.8 meters (Sassa and Takagawa 2019).

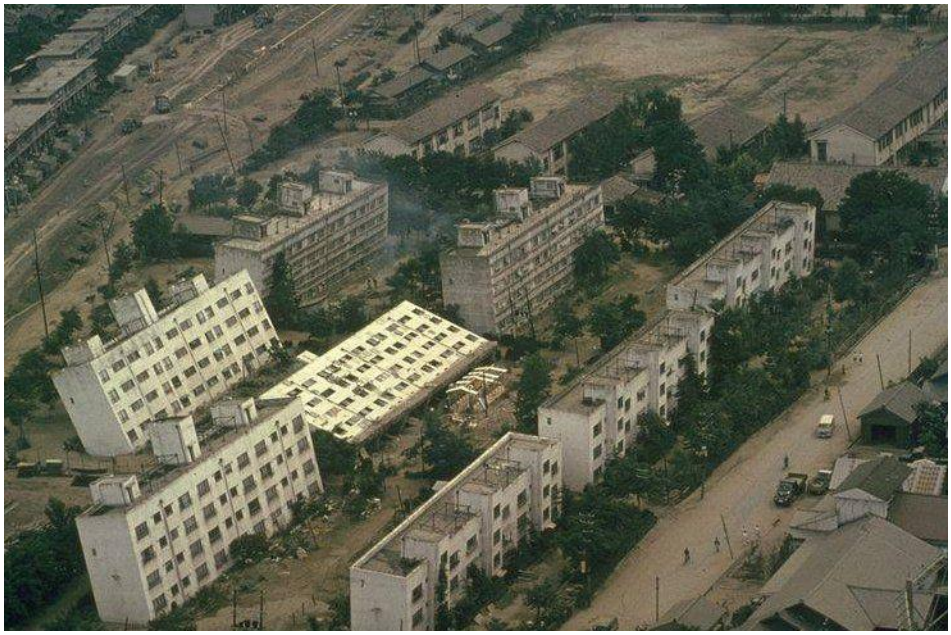


Figure 1.1 Apartment buildings tilted because of the 1964 Niigata earthquake (photo: National Information Service for Earthquake Engineering, EERC, University of California, Berkeley)

In general, sands tend to contract upon cyclic loading. Earthquake shaking is a strong cyclic loading but for a short period. There is no enough time for the water to dissipate, thus making the loading conditions to be mostly undrained. In undrained cyclic loading conditions as the volume remains constant, the contraction in sand structure leads to the transfer of load-carrying ability from the sand structure to the pore water. As a result, pore water pressure increases, and effective confining pressure decreases. Ultimately soil liquefaction is triggered when the effective stress state is reduced to zero. After liquefaction, depending on the density of sand layers, deformation can be either uncontrollably large with continuous flow or can be an accumulative deformation with the limited flow in each cycle of loading. Loose sands undergo flow type of failure, thereby causing large, rapid, and uncontrollable deformations in the ground often referred to as lateral spreading (Figure 1.2). This phenomenon is called flow liquefaction. Whereas medium dense to dense sands upon liquefaction also tends to flow but may regain strength after a certain level of deformation in the ground. These deformations are generally accumulative with every cycle of loading; thus, this phenomenon is commonly referred to as cyclic mobility.

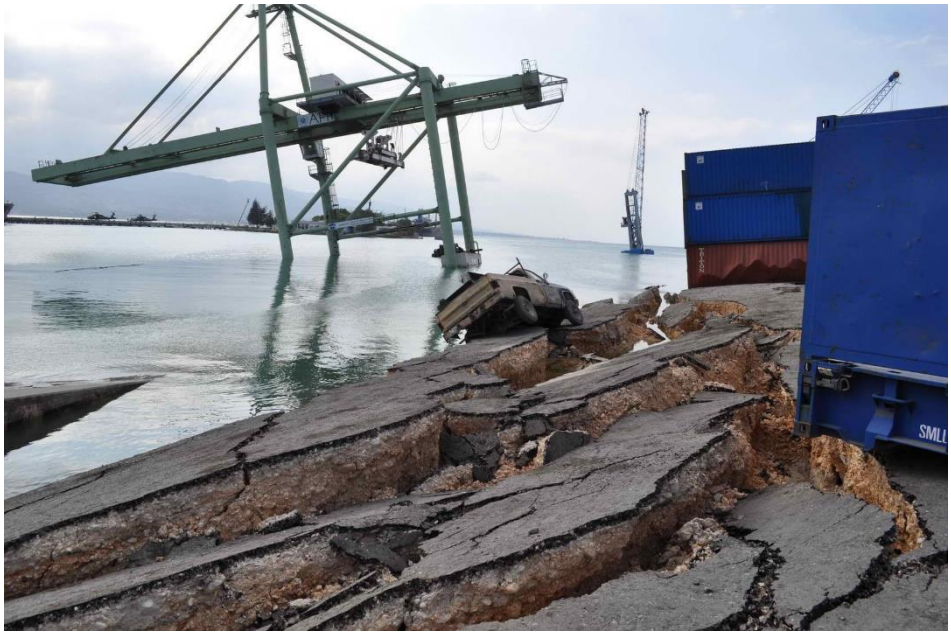


Figure 1.2 Liquefaction induced lateral spreading in the 2010 Haiti Earthquake (photo: Earthquake Engineering Research Institute (EERI))

Early research mostly concentrated on identifying sites that are prone to liquefaction. Researchers developed methodologies to estimate liquefaction potential based on empirical analysis of case histories. Both in-situ testing and laboratory testing on field retrieved samples were focused on determining liquefaction resistance. In common practice, liquefaction resistance is usually correlated to the SPT-N value (No. of blows required for standard penetration using standard hammer) obtained from the in-situ testing. In experimental studies, it is correlated to the relative density. In the early 1980s, with the rise in performance-based design framework in earthquake engineering, research community recognized the need for understanding the performance implications of the soil liquefaction. The development of tools to predict the deformation behavior in soil liquefaction has become the new focus of research. Bartlett and Youd (1995) proposed a deformation prediction equation based on the statistical analyses of case histories where deformations were measured using aerial photography with an accuracy of 1m. By Incorporating additional earthquake data, new empirical prediction equations were proposed by Youd et al. (2002) to predict deformations during liquefaction. Usually, large deformations such as lateral spreading because of flow liquefaction can be well

predicted by these equations but fails to predict the moderate range of deformations (<1m) caused by cyclic mobility.

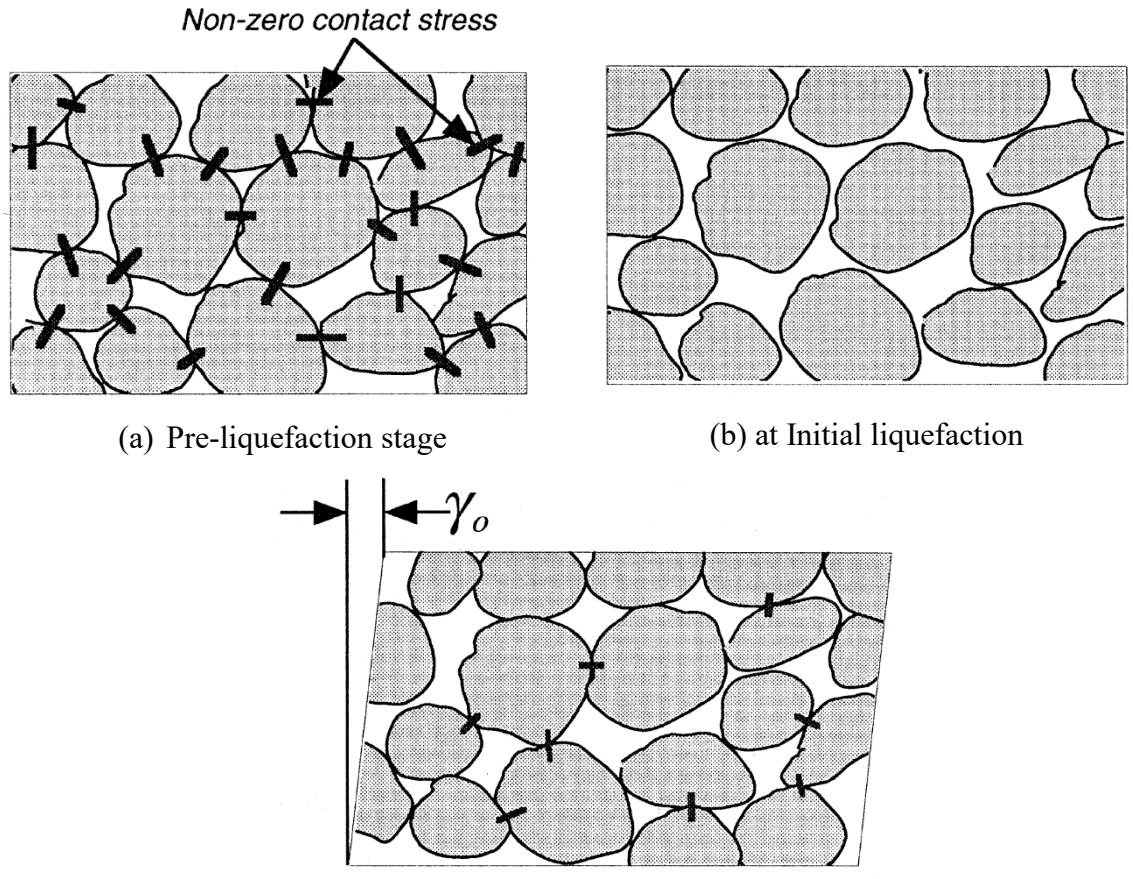
Constitutive models have been a promising tool to predict soil behavior, thereby predicting both large and moderate deformations during soil liquefaction. It also predicts the variation in effective stress and pore water pressure during the liquefaction process. Constitutive models, when incorporated in finite element framework, can be a powerful tool to predict the response of even the most complex sites which is practically not possible otherwise. In the last three decades, many constitutive models have been developed to predict soil liquefaction behavior (e.g., Boulanger and Ziotopoulou 2013; Elgamal et al. 2003; Li and Dafalias 2004; Park and Byrne 2004; Pastor et al. 1990; Taiebat and Dafalias 2008; Wang and Xie 2014; Zhang and Wang 2012; Zienkiewicz et al. 1999). Most of these models are validated with available experimental results and case histories. The accuracy and reliability of models are not up to par. This is mainly because of two reasons, one, the available constitutive models to predict liquefaction are mostly phenomenological in nature involving various parameters which have no physical basis of measuring, and two, assuming the sand to be continuum and predicting its behavior using a constitutive model disrespects the very discrete nature of sands, i.e., soil fabric. Recent studies are focused on incorporating soil fabric properties into constitutive models (e.g., Bauer et al. 2004; Dafalias and Manzari 2004; Gao et al. 2010; Gao and Zhao 2015; Li and Dafalias 2011; Oda and Nakayama 1989; Wan and Guo 2001). Understanding the fundamental mechanism of soil liquefaction can pave a way to develop more robust constitutive models and also it has been an ultimate motivation for this study.

Soil fabric refers to the arrangement of particles, particle group, and void space distribution (Mitchell and Soga 2005). During liquefaction, the soil fabric undergoes appreciable changes. Shamoto et al. (1998) illustrated the probable internal structure at different stages of liquefaction and is shown in Figure 1.3. At different stages of liquefaction, the internal soil structure attains different configurations, i.e., load-bearing soil fabric in the pre-liquefaction stage (refer to Figure 1.3a), suspended soil fabric with no contacts at initial liquefaction (refer to Figure 1.3b), and new load-bearing soil fabric with less contacts in the post-liquefaction deformation (refer to Figure 1.3c). Though the geotechnical community has long known that liquefaction results in significant changes in the load-bearing fabric of granular soils, a clear characterization of these changes have been lacking. Laboratory works related to fabric quantification include 2D biaxial tests using photo-elastic disks (Oda et al. 1985) and rods (O'Sullivan 2011). Most recently, the X-ray micro-CT technique has been used to obtain the

micro-scale images of fabric in the granular assemblage of real sands (Fonseca et al. 2012). This technology has not been used to study cyclic liquefaction of granular soils. Numerical simulations based on the Discrete Element Method (DEM) provide a more convenient and less costly way to observe the fabric directly (e.g., Guo and Zhao 2013; O'Sullivan 2011; Rothenburg and Bathurst 1989; Thornton 2000). The DEM is capable of simulating the cyclic soil behavior and cyclic liquefaction (e.g. Ng and Dobry 1994; Sitharam et al. 2009; Wang and Wei 2016; Wei et al. 2018; Wei and Wang 2016). Traditional studies on fabric quantification are mostly based on inter-particle contacts (e.g. Guo and Zhao 2013; Kuhn 1999; Satake 1992; Thornton 2000; Xia and Xiang-Song 2009), which are easily implemented in DEM simulations using contact normal vectors or branch vectors. Coordination number is another indicator of the number of contacts per particle. The contact-based fabric has been extensively used in quantifying the load-bearing capacity of the soil under monotonic loading (O'Sullivan 2011). However, fabric quantification of liquefied soils is particularly challenging because a liquefied soil loses nearly all of its contacts, so the contact-based fabric may not a reliable indicator in such a case. Alternatively, the void-based fabric can be an option for studying cyclic liquefaction. A few void-based fabric have been proposed in very limited studies, including the "void cells" (e.g. Kuhn 1999; Satake 1992) and "solid/void cell system" (Li and Li 2009). However, the construction of the void cells or void cell systems are still based on inter-particle contacts, which are only useful for quantifying the anisotropy of the void space in a "solid-like" state, but not a "fluid-like" state (e.g. Wang and Wei 2016; Wei et al. 2018; Wei and Wang 2016). There is a great need to develop new void-based fabric to quantify particle-void structure change in the post-liquefaction stage.

Laboratory tests also indicated that the post-liquefaction consolidation is directly related to the post-liquefaction shear deformation (e.g., Ishihara et al. 2016; Ishihara and Yoshimine 1992; Nagase and Ishihara 1988), which implies that the change in internal soil structure effects the post-liquefaction deformation/settlement. Re-liquefaction of sites struck by multiple earthquakes also underscores the importance of studying fabrics in the liquefied and reconsolidated soils, as exemplified by the 2010-2011 Canterbury earthquakes in New Zealand, the 2011 Tohoku earthquake in Japan and a couple of experimental tests (Wahyudi et al. 2015).

The shear waves propagated during an earthquake shaking are inherently irregular. In consequence, the cyclic load experienced by a soil element is multi-directional with varying magnitude and direction. Because of the limitation in apparatuses to perform the multi-directional loading shear tests on soil specimen, most commonly used apparatuses to study cyclic liquefaction are cyclic simple shear test apparatus, cyclic triaxial test apparatus or a cyclic torsional simple shear test apparatus, in which shear stresses were generated only in one direction. The limited studies (e.g. Boulanger and Seed 1995; Ishihara and Yamazaki 1980; Kammerer 2002; Seed et al. 1975) had found that both excess pore water pressure and shear deformation would increase faster when sands were sheared under multi-directional loading. In this study, the characteristics of fabric evolution during multi-directional loading is understood with an aim to link the macroscopic response to the microscopic evolution. This would provide new insights into understanding liquefaction phenomenon in more general loading paths.



(a) Pre-liquefaction stage

(b) at Initial liquefaction

(c) post-liquefaction load bearing structure

Figure 1.3 Soil internal structure at different stages of cyclic liquefaction (after Shamoto et al. 1998)

1.2 Objectives of the research

In light of issues discussed in the above section, the objectives of the current research are as follows:

1) To propose new 3D void-based fabric descriptors, i.e., applicable to both liquefied and non-liquefied states.

2) Modelling cyclic liquefaction phenomenon using 3D discrete element method; Characterizing fabric evolution in pre- and post-liquefaction stages and connecting microscale information to the macroscale behavior; Understanding soil fabric effect on post-liquefaction deformation and re-liquefaction behaviors.

3) To simulate stress-controlled multi-directional loading tests on granular assembly; Characterizing soil fabric in multi-directional loading paths and connecting microscale information to the macroscale response.

1.3 Organization of the work

The focus of this research is to improve the fundamental understanding of cyclic liquefaction. This thesis consists of six chapters. The initial chapter has focused on relevant background knowledge of soil liquefaction and the objectives of this thesis.

Chapter 2 presents a thorough literature review of cyclic liquefaction. More specifically, it includes the current understanding of soil liquefaction, numerical modelling techniques, Soil-fabric characterization methods, literature on post-liquefaction settlements, reliquefication studies, and review of research on multi-directional loading paths.

Chapter 3 introduces 3D Discrete Element Method and fabric quantification methods. The formulation for the new particle-void descriptors proposed in this study is also discussed in this chapter.

Chapter 4 discusses the evolution of different soil fabric quantities during cyclic liquefaction and connecting the microscale observations to the macroscale behavior. The effect of soil fabric in post-liquefaction volumetric compression and reliquefication resistance is also highlighted.

Chapter-5 discusses the method to generate stress-controlled multi-directional loading in the discrete element method. It also discusses the evolution of different microscale quantities, both contact-based and void-based fabric descriptors, and connect the microscale information to the macroscale response of the granular assembly.

In the final chapter, the summary of current findings and the ideas for future research are presented.

CHAPTER 2
Literature Review

2.1 Cyclic liquefaction in granular soils

Saturated granular soils upon cyclic loading undergo cyclic liquefaction. In laboratory, element level cyclic loading tests on sands are predominantly performed using a triaxial apparatus, simple shear apparatus or a hollow cylinder torsional shear apparatus. Regardless of the apparatus used, saturated sands in undrained cyclic loading conditions undergo cyclic liquefaction. Cyclic liquefaction phenomenon can be divided into two stages, i.e., the response of sand before initial liquefaction (Pre-liquefaction) and response after initial liquefaction (Post-liquefaction). Initial liquefaction is when the load-bearing capacity of soil is totally collapsed for the very first time during the cyclic loading.

2.1.1 Initial/Pre-liquefaction

The basic mechanism of the onset of cyclic liquefaction is first understood by Seed and Lee (Seed et al. 1966). In their study, saturated sands were consolidated under a confining pressure and later subjected to repeated constant-amplitude cyclic deviatoric loads in a triaxial apparatus under undrained conditions. It is a known fact that though the triaxial setup applies load in principal directions, a plane inclined in 45° to the soil specimen experiences shear stress that is equal to the half of deviatoric stress. This plane is generally assumed to be the horizontal ground during earthquakes. A typical result of such an experiment is shown in Figure 2.1. It was observed that with repeated cyclic loads, pore water pressure increases gradually and

eventually approaches the initial confining pressure, thereby, producing a double amplitude axial strain of about 4%. Such a state is referred to as 'initial liquefaction'. The increase in pore-water pressure in undrained cyclic tests is analogous to the volumetric compression in drained cyclic test (Idriss and Boulanger 2008).

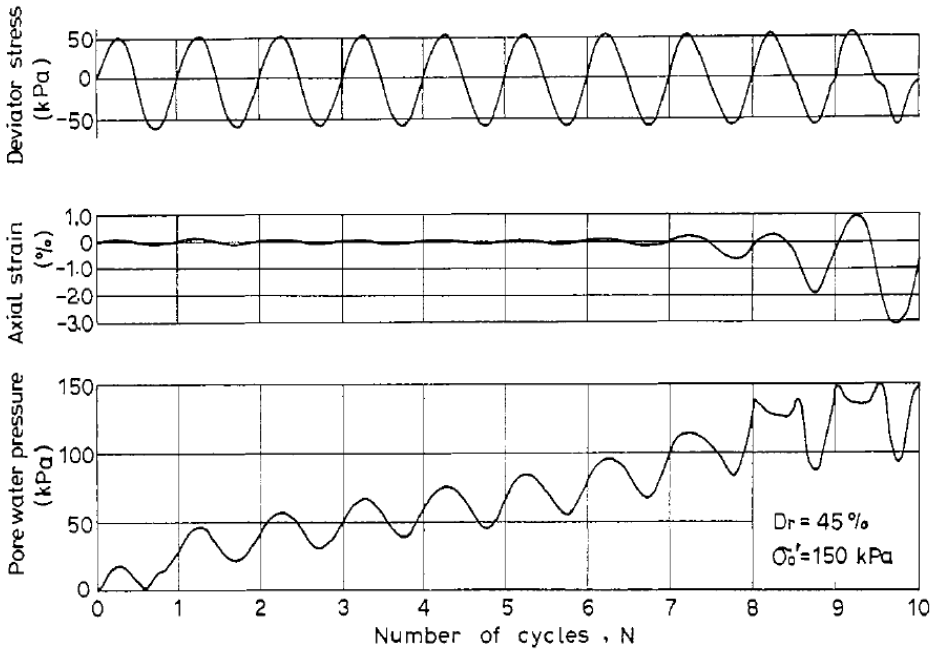


Figure 2.1 Axial strain generation and pore water pressure buildup in the cyclic triaxial test (after Ishihara 1996)

In a drained cyclic test, the soil assembly contracts upon cyclic loading. But for an undrained test, as the volume of the assembly is constant, the sand particles rearrange in a manner that the normal stresses are transferred through pore water rather than sand particles. As shown in Figure 2.2, if the initial state of the sample is at 'A', for a drained cyclic load the sand reaches to point 'B' with Δe volume change but for an undrained cyclic load the sand reaches to point 'C' which is a rebound from point 'B' under reduced effective stress. The decrease in effective stress from point 'A' to point 'C' causes to increase the pore water pressure in order to support the total load. Initial liquefaction is the state when pore water pressure reaches to the initial confining pressure. If excess pore water pressure ratio (r_u) is defined as the ratio of excess pore pressure (Δu) to the initial mean confining pressure (p'), i.e., $r_u = \Delta u/p'$, then initial liquefaction is defined as $r_u = 1$. Note that in some exceptional cases, the excess pore water pressure may not reach the value 1, for example, multi-directional loading or unidirectional

loading with static shear stress higher than the cyclic stress or cyclic loading in silty sands (Boulanger et al. 1991; Ishihara 1996). In such cases the definition for initial liquefaction, i.e., $r_u = 1$, is not applicable. Ishihara and Yamazaki (1980) suggested a new criterion based on axial strain amplitude, i.e., point at which the sample experiences 3% single amplitude axial strain for the first time is identified as initial liquefaction. Figure 2.3 shows the cyclic undrained triaxial loading behavior of an anisotropically consolidated specimen of a medium dense clean sand, loaded maximum up to 3% axial strain (Idriss and Boulanger 2008). Upon cyclic loading in a sinusoidal fashion the effective mean stress ratio (p'/p'_c) was observed to reduce after every cycle along with increase in pore water pressure gradually. After about 27 cycles, the initial liquefaction state is identified where the excess pore water pressure ratio reaches 1. Under only one cycle from initial liquefaction, 3% single amplitude axial strain is also achieved. Thus, either of the definitions can hold true to define initial liquefaction in case of unidirectional loading. It is important to note that during initial cycles the axial strains (ϵ_a) remained relatively very small for about most of the time but started to evolve as the sample is approaches to initial liquefaction.

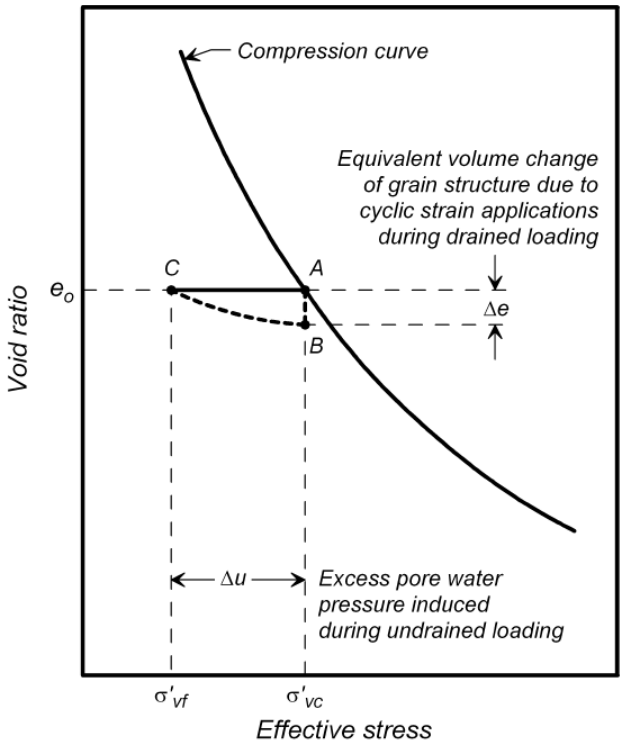


Figure 2.2 Mechanism of excess pore water pressure generation in undrained cyclic loading (after Idriss and Boulanger 2008)

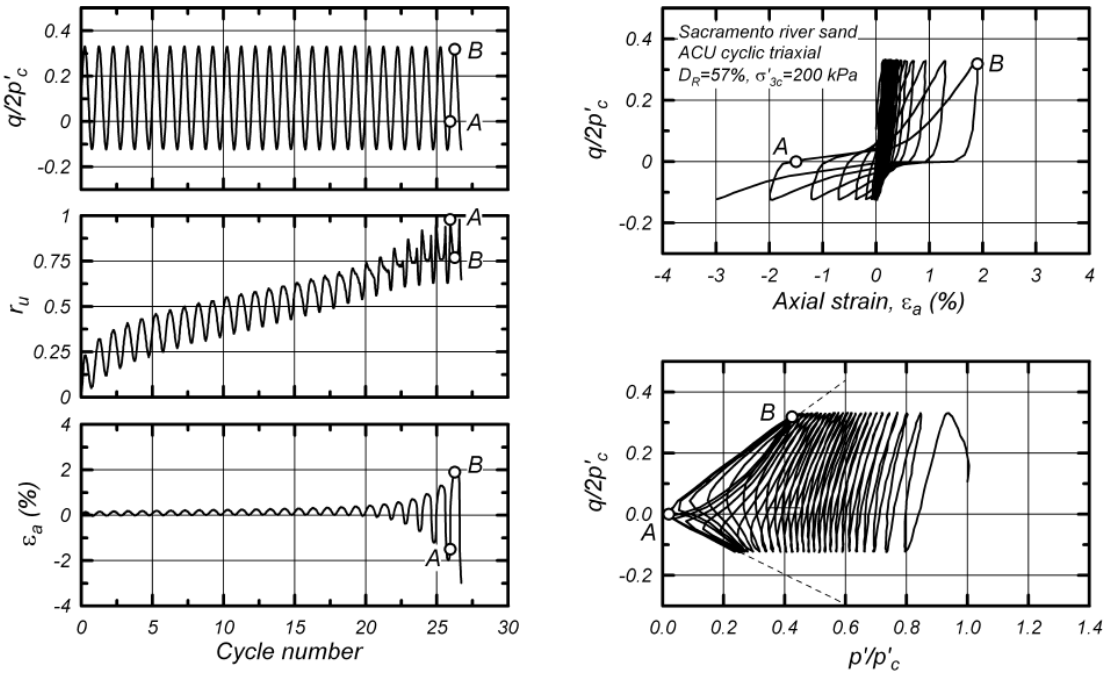
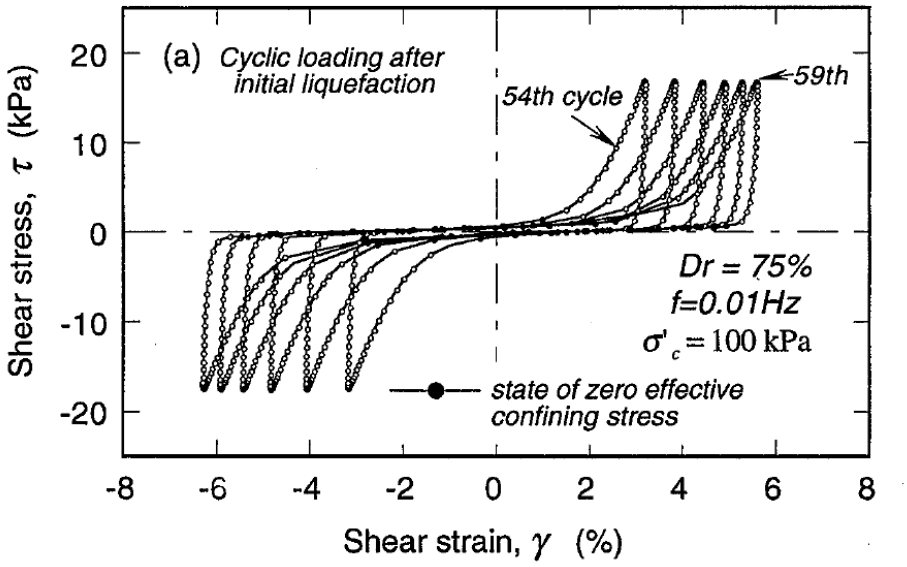


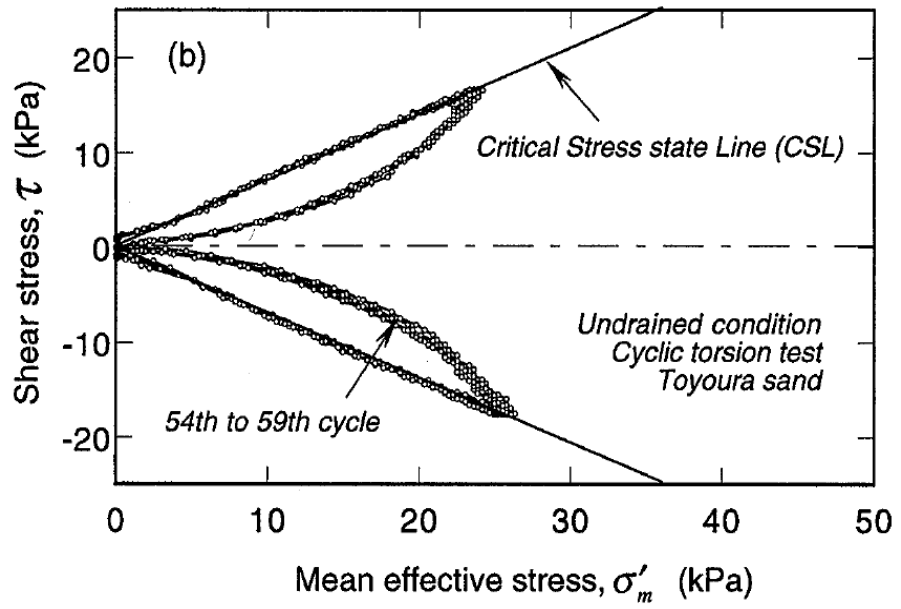
Figure 2.3 Response of Sacramento river sand with Dr=57% to undrained cyclic triaxial loading (after Idriss and Boulanger 2008)

2.1.2 Post-liquefaction behavior

Depending on the relative density of the sand specimens, the nature of post-liquefaction behavior can be either a flow behavior or a cyclic softening behavior. For loose sands, the initial liquefaction is certainly a state of softening where large deformations occur with the sudden loss of strength, during which the soil behaves like a fluid (Steven L. 1996). For medium dense to dense sand, after initial liquefaction, a state of softening is attained, however, the deformation does not grow indefinitely large rather limited strains are accumulated cycle after cycle. Figure 2.4 shows the post-liquefaction behavior of a medium dense Toyoura sand in an undrained cyclic torsional test. In the post-liquefaction stage, the effective stress point keeps oscillating through the same stress path every cycle and is accompanied by significant development in the double amplitude of shear strain, as shown in Figure 2.4. After every cycle of load, the sand specimen is little more softer than it was in its previous cycle, such kind of behavior is referred to cyclic softening or cyclic mobility (Casagrande 1976; Casto 1975).



(a) Stress-strain response



(b) Effective stress path

Figure 2.4 Typical soil behavior in the post-liquefaction stage (after Shamoto et al. 1997)

The shear strain in post liquefaction stage has mainly two components, one that is dependent on the effective stress (hardening strain) and the other that is independent of the effective stress (Flow strain) (Shamoto et al. 1997). The strain developed while the stress path travels through the zero effective stress state (Flow stage) is the flow strain. Whereas, the strain corresponding to the non-zero effective stress state (hardening stage) is the hardening strain (refer to Figure 2.4). The hardening strain in every cycle is almost of the same magnitude, but the flow strain keeps increasing with the increase in cycle number, hence developing high post-liquefaction shear strains. For an undrained cyclic loading test on medium dense to dense sand specimens, it is prevalent to observe the "butterfly" loops in stress path and "inverted S-shaped" loops in stress-strain curves.

2.2 Numerical simulation of cyclic loading

2.2.1 Constitutive modelling of cyclic liquefaction

Constitutive modelling is the systematic mathematical formulations that can be used to predict the soil response under a given loading. Such models can be incorporated into finite element or finite difference framework to analyze the performance of large-scale engineering structures. The development of accurate and reliable constitutive models that can predict cyclic liquefaction behavior is a popular area of research interest. Many constitutive models to capture cyclic liquefaction behavior were developed in last three decades based on different mathematical theories such as, elastoplasticity theories (e.g. Park and Byrne 2004; Pastor et al. 1990; Ye and Wang 2015; Zienkiewicz et al. 1999), multi-yield surface models (e.g. Elgamal et al. 2003; Prévost 1981; Yang et al. 2003), and bounding surface hypoplasticityity models (e,g, Boulanger and Ziotopoulou 2013; Dafalias 1986; Dafalias and Manzari 2004; Li 2002; Manzari and Dafalias 1997; Taiebat and Dafalias 2008; Wang and Xie 2014; Wang et al. 1990; Zhang and Wang 2012). All these models are validated either with the experimental tests, centrifuge tests, or the case histories.

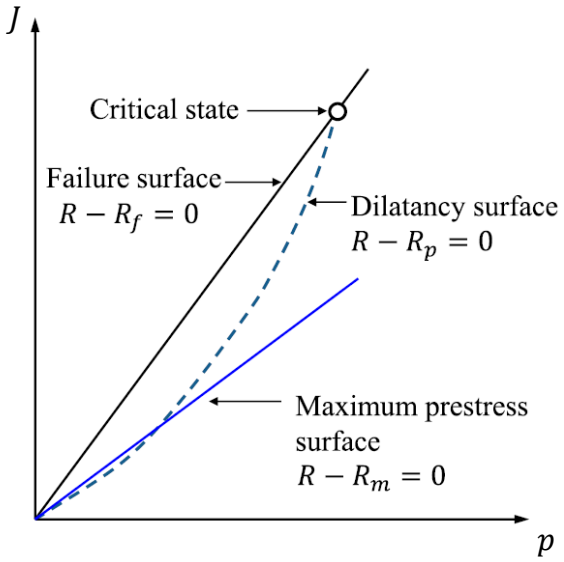
To get an idea of the basic structure of a constitutive model, the concepts of the bounding surface hypoplasticity model and its formulation is discussed here. Bounding Surface hypoplasticity model for sands was initially developed by Wang et al. (1990) but was later modified by many researchers to incorporate different characteristics of sand's response. The original model was able to simulate the behavior of loose sands reasonably well but fails to

predict the dense sand behavior. Wang and Xie (2014) proposed a modified formulation to the original bounding surface hypoplasticity model which incorporated features that could predict dense sand behavior as well.

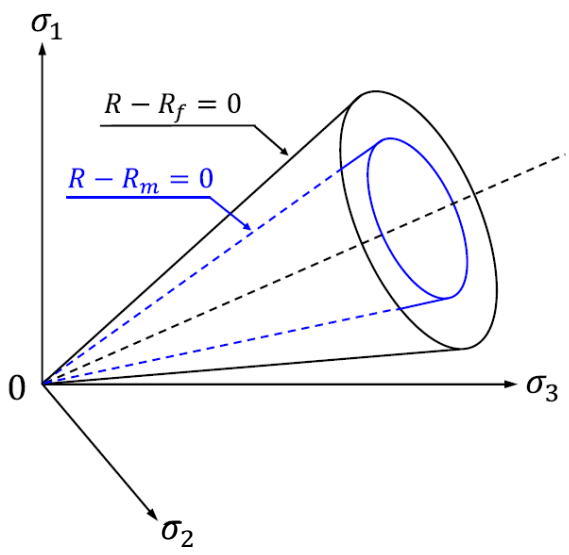
The mathematical formulation of Wang and Xie (2014) model is briefly discussed here. If $\boldsymbol{\sigma}$ is the effective stress tensor and $\boldsymbol{I}$ is the second order identity tensor then $p = \frac{1}{3} tr(\boldsymbol{\sigma})$ is the mean effective stress, $\boldsymbol{s} = \boldsymbol{\sigma} - p\boldsymbol{I}$ is the deviatoric stress tensor and $\boldsymbol{r} = \boldsymbol{s}/p,$ is the deviatoric stress ratio tensor. The stress rate ($\dot{\boldsymbol{\sigma}}$) is given as:

$$\dot{\boldsymbol{\sigma}} = p\dot{\boldsymbol{r}} + \dot{p}\frac{\boldsymbol{\sigma}}{p} \tag{2.1}$$

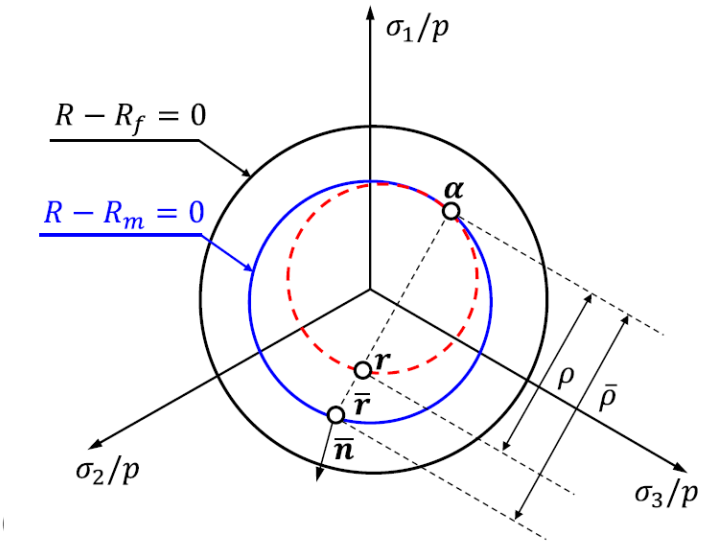
From the deviatoric stress tensor ($\boldsymbol{s}$), the isotropic invariant (J) can be defined as $J = \sqrt{\frac{1}{2}\boldsymbol{s}:\boldsymbol{s}}$, where the tensor operation $\boldsymbol{a}:\boldsymbol{b} = a_{ij}b_{ji}$. R is the stress ratio invariant given from the isotropic invariant as, $R = J/p$. The stress ratio invariant defines different bounding surfaces that control the soil behavior in $J - p$ space of stress response. Figure 2.5(a) shows the Failure surface ($R - R_f = 0$), Maximum prestress surface ($R - R_m = 0$) and the Dilatancy surface ($R - R_p = 0$) in the $J - p$ space. Failure surface is the stress ratio at critical state (Roscoe et al., 1958; Schofield and Wroth, 1968), maximum prestress surface defines the maximum stress ratio that was ever experienced by the material, and the Dilatancy surface defines the location where soil response changes from contractive to dilative. The dilatancy surface is defined from the failure surface as $R_p = R_f e^{m\psi}$, where ψ is the state parameter that depends on critical state void ratio. Bounding surfaces when plotted on 3D principal stress space is shown in Figure 2.5(b) and when plotted on stress ratio space is shown in Figure 2.5(c). Wang and Xie (2014) represented the bounding surfaces as conical and circular shapes for easy understanding. From Figure 2.5(c), if $\boldsymbol{r}$ is the current stress state and $\boldsymbol{\alpha}$ is the reversal point in the last reversal unloading then using a unique mapping rule, $\bar{\boldsymbol{r}}$ can be defined on the maximum prestress surface. The distances ($\rho, \bar{\rho}$) shown in Figure 2.5(c) are important for further plastic modulus calculation.



(a) Key surfaces in $J - p$ space



(b) Key surfaces in stress space



(c) Parameters controlling bounding surface model

Figure 2.5 Definition of bounding surface constitutive model (after Wang and Xie 2014)

The strain rate ($\dot{\boldsymbol{\epsilon}}$) is decomposed into elastic part ($\dot{\boldsymbol{\epsilon}}^e$) and plastic part ($\dot{\boldsymbol{\epsilon}}^p$) and are defined as:

$$\dot{\boldsymbol{\epsilon}}^e = \dot{\boldsymbol{e}}^e + \frac{1}{3}(tr\dot{\boldsymbol{\epsilon}}^e)\boldsymbol{I} = \frac{1}{2G}p\boldsymbol{r} + \left(\frac{1}{2G}\boldsymbol{r} + \frac{1}{3K}\boldsymbol{I}\right)\dot{p} \tag{2.2}$$

$$\dot{\boldsymbol{\epsilon}}^p = \dot{\boldsymbol{e}}^p + \frac{1}{3}(tr\dot{\boldsymbol{\epsilon}}^p)\boldsymbol{I} = \left(\frac{1}{H_r}\overline{\boldsymbol{n}} + \frac{1}{3K_r}\boldsymbol{I}\right)p\dot{\boldsymbol{r}}:\overline{\boldsymbol{n}} \tag{2.3}$$

where K, G are the elastic bulk and shear modulus respectively. K_r and H_r are the plastic bulk and shear modulus respectively. $\overline{\boldsymbol{n}}$ is the unit vector normal to the maximum prestress surface at the image point $\overline{\boldsymbol{r}}$. $\dot{\boldsymbol{e}}^e$, $\dot{\boldsymbol{e}}^p$ are the elastic and plastic deviatoric strain rates respectively.

The constitutive relation is defined between the stress rate and the strain rate as follows:

$$\dot{\boldsymbol{\sigma}} = \boldsymbol{\Lambda}:\dot{\boldsymbol{\epsilon}} \tag{2.4}$$

where,

$$\boldsymbol{\Lambda} = \boldsymbol{E} - \frac{\boldsymbol{p}^r \otimes \boldsymbol{Q}^p}{A_r B_p - A_p B_r} \tag{2.5}$$

where the tensor operation $\boldsymbol{a} \otimes \boldsymbol{b} = a_{ij}b_{kl}$ and,

$$\boldsymbol{E} = E_{ijkl} = K\delta_{ij}\delta_{kl} + G(\delta_{ij}\delta_{kl} + \delta_{ij}\delta_{kl} - \frac{2}{3}\delta_{ij}\delta_{kl}) \tag{2.6}$$

$$\boldsymbol{p}^r = \frac{2G}{H_r}\overline{\boldsymbol{n}} + \frac{K}{K_r}\boldsymbol{I} \tag{2.7}$$

$$\boldsymbol{Q}^p = B_p\overline{\boldsymbol{n}} - B_r\boldsymbol{I} \tag{2.8}$$

$$A_r = \frac{1}{2G} + \frac{1}{H_r};\ \ A_p = \frac{1}{K_r} \tag{2.9}$$

$$B_r = \frac{1}{2G}\boldsymbol{r}:\overline{\boldsymbol{n}};\ \ B_p = \frac{1}{K} \tag{2.10}$$

The elastic moduli K, G are given as follows:

$$K = p_a\frac{1+e_{in}}{\kappa}\left(\frac{p}{p_a}\right)^{\frac{1}{2}}\ and\ G = p_a G_0\frac{2.973 - e_{in}}{1+e_{in}}\left(\frac{p}{p_a}\right)^{\frac{1}{2}} \tag{2.11}$$

where p_a is the atmospheric pressure, e_{in} is the initial void ratio, κ is the slope of unloading-reloading curve and G_o is the small strain shear modulus.

The plastic shear moduli H_r is given as follows:

$$H_r = Gh_r C_H(\xi_q)\left[\frac{R_f}{R_m}\left(\frac{\bar{\rho}}{\rho}\right)^m - 1\right]\left(\frac{p}{p_m}\right)^{1/2} \tag{2.12}$$

where

$$C_H(\xi_q) = \frac{1}{1+\alpha\xi_q};\ \xi_q = \int_0^{\boldsymbol{e}^p}\sqrt{\frac{2}{3}d\boldsymbol{e}^p : d\boldsymbol{e}^p} \tag{2.13}$$

$$m = \frac{2R_m}{\bar{\rho}} \tag{2.14}$$

where h_r is a dimension material constant, $d\boldsymbol{e}^p$ is the deviatoric plastic strain increment and α is a dimensionless quantity that controls the cyclic mobility.

The plastic bulk modulus K_r is given as:

$$K_r = p_a\frac{1+e_{in}}{w\kappa}\left(\frac{p}{p_a}\right)^{1/2} = \frac{K}{w} \tag{2.15}$$

where

$$w = \begin{cases} w_1 = \dfrac{1}{k_r}\left(\dfrac{R_m}{R_f}\right)^b\left(\dfrac{R_p - R}{R_f - R_m}\right), if\ R = R_m\ or\ R > R_p\ and\ \dot{R} > 0 \\ w_2 = C_K(\xi_v)\left(\dfrac{R_m + sign(\dot{R})R}{R_f}\right)\left(\dfrac{R_p + sign(\dot{R})R}{R_p + R_m}\right), otherwise \end{cases} \tag{2.16}$$

and

$$C_K(\xi_v) = d_1 + d_2\tanh(100\xi_v) \tag{2.17}$$

$$\xi_v = \int_0^{\epsilon_v^p}\langle -d\epsilon_v^p\rangle \tag{2.18}$$

where ξ_v is the plastic volumetric strain, d_1 and d_2 are model parameters.

Overall, 13 parameters are identified in the above constitutive model and can be categorized based on their control on different factors, as shown in Table 2.1.

Table 2.1 Parameters controlling the constitutive model

Factors	Controlling Parameters
Critical State	$e_\Gamma, \lambda, \xi, M$
Phase Transformation	m
Elastic Moduli	G_0, κ
Plastic Shear Modulus	h_r, α
Plastic bulk modulus	k_r, b, d_1, d_2

Most of the controlling parameters of the constitutive model have no physical meaning and are phenomenologically based. Figure 2.6 shows the controlling effect of parameters (b, k_r, d_1, d_2) on different parts of the cyclic loading response. Thus, after careful examination and trial and error procedure, the reasonable parameters should be selected to accurately fit the behavior according to experimental results.

Traditional constitutive models for cyclic liquefaction, like the one discussed above, are unusually formulated in stress space without considering the internal soil fabric. It is well known in the geotechnical community that internal soil structure and its evolution plays an essential role in the overall soil response. Researchers argued that proper fabric tensors which characterize the internal structure of sands are necessary to be incorporated into the constitutive models, making them more sophisticated to capture realistic sand behaviors (Bauer et al. 2004; Dafalias and Manzari 2004; Gao et al. 2010; Oda and Nakayama 1989; Wan and Guo 2001). Dafalias and Manzari (2004) introduced a fabric-dilatancy related tensor quantity into the bounding surface hypoplasticity model to account for the effect of sand fabric changes during loading. Li and Dafalias (2004) proposed an anisotropic critical state theory where a fabric tensor and its evolution to critical state value was introduced in order to account for the role of anisotropic fabric at the critical state. Gao and Zhao (2015) discusses a comprehensive bounding surface model in the framework of Anisotropic critical state theory using a void fabric tensor by Li and Li (2009). The importance of internal soil structure and incorporating it in the form fabric tensors into constitutive models is found necessary. Currently, if not impossible, it is quite challenging to physically observe the evolution of internal soil structure during liquefaction. However, it is possible to conduct the state-of-the-art numerical

simulations to quantify microscopic soil fabric in liquefaction using discrete element modelling techniques.

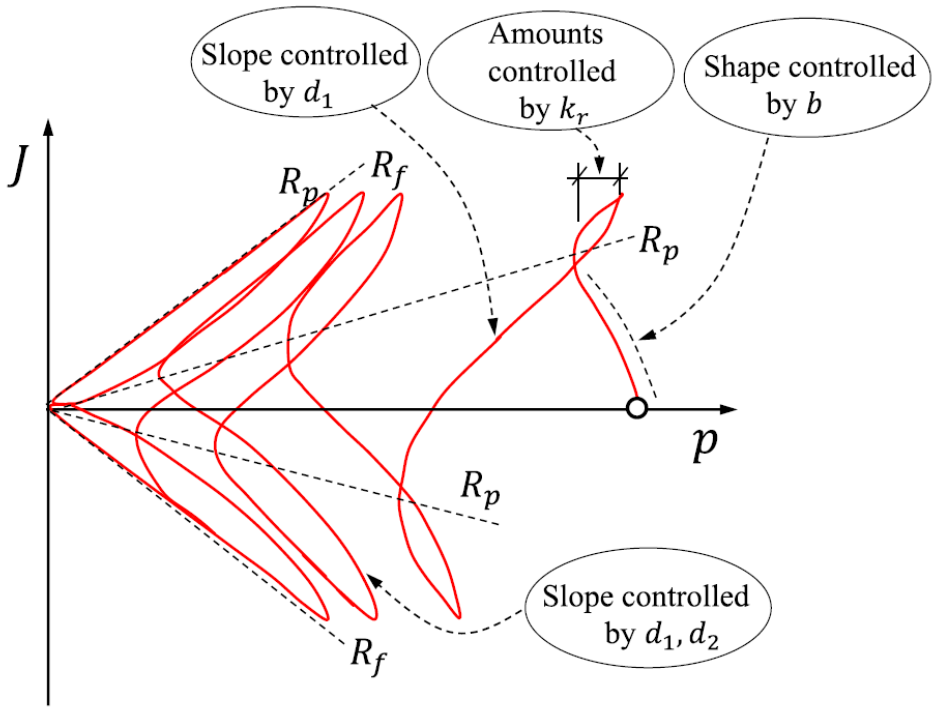


Figure 2.6 Controlling parameters of bounding surface model (after Wang and Xie 2014)

2.2.2 Discrete Element Method for cyclic liquefaction

Discrete Element Method (DEM) was first proposed by Cundall and Strack (1979) to simulate the behavior of granular assemblies. In this technique, the discrete particles of the granular assembly are individually modeled and can interact with neighboring particles to generate contact forces and displacements. The macroscopic behavior of the whole granular assembly is obtained from the integration of the forces and displacements at the particle level. In DEM, constitutive models are defined at the contact level to calculate contact forces (normal and tangential). Unlike constitutive models of continuum modelling, these models are relatively elementary and mostly physics-based. Even considering such simple contact models and even simpler particle shapes (like spheres and discs), the qualitative macroscopic response under different loading conditions can be well captured using DEM (Sitharam 2003). The major advantage of DEM from continuum modelling and experimental studies is that using DEM the

internal fabric of the granular assembly can be observed and quantified, which is the critical information to predict the macroscopic soil behavior.

DEM simulations for cyclic liquefaction are first discussed by Ng and Dobry (1994). In their study, undrained cyclic simple shear tests were performed on an assembly of disks (2D) and spheres (3D) using periodic boundaries. The qualitative behavior was well captured in these simulations, and the influence of particle rotations, i.e., a micro parameter, on the macroscale response was highlighted. Sitharam (2003) conducted a series of cyclic biaxial tests in undrained conditions with constant stress and strain amplitudes on polydisperse and monodisperse assemblies. They observed that qualitative macroscopic behavior and cyclic liquefaction phenomenon could be well captured using DEM. Their study validated DEM simulations with the experimental understanding of cyclic liquefaction, such as increased susceptibility to liquefaction with the increase in cyclic stress or strain amplitude, increased resistance to liquefaction with the increase in confining stress and increased resistance in a polydisperse system compared to a monodisperse packing. Also, the liquefaction phenomenon was correlated to the cumulative loss of coordination number during stress reversals in every cycle. Ashmawy et al. (2003) studied the effect of particle morphology on cyclic liquefaction using DEM simulations. They found that for very loose specimens, the particle shape played no significant role, but for intermediate to dense specimens at same void ratio the particle morphology has a significant effect on the cyclic resistance to liquefaction. Also, in their study, a new method to input arbitrary shapes, i.e., a clump of spheres, into DEM is highlighted. Sitharam et al. (2009) studied the monotonic loading behavior of already liquefied samples in DEM. In their study, two series of tests were carried out, one in which undrained monotonic tests were performed on specimens prepared after dissipation of excess pore water pressure after initial liquefaction and the other where no dissipation was allowed. In series one tests, it was observed that no significant influence of axial strain amplitude that was required for initial liquefaction on the post-liquefaction undrained monotonic behavior whereas, for the series two tests, it played a major role. Wang and Wei (2016) conducted undrained cyclic simple-shear tests on a 2D granular assembly. The qualitative behavior of cyclic liquefaction is well captured, and a scalar void-based fabric indicator is introduced to understand the post-liquefaction behavior of the granular assembly. In their following study, Wei et al. (2018) proposed a new set of void-based fabric indicators for 2D granular assemblies, which helped to understand the strain accumulation behavior in the post-liquefaction stage. Wei and Wang (2017) studied the influence of initial fabric on the cyclic liquefaction resistance. It was

observed that at same void ratio isotropic samples having high resistance to liquefaction whereas highly anisotropic samples having the least resistance to liquefaction.

In summary, all the authors claim that the qualitative behavior of cyclic liquefaction can be simulated using DEM. Experimental findings such as dependence of liquefaction resistance on relative density, confining pressure, stress ratio, and so forth are similarly observed in DEM. Also, the strain accumulation behavior in post-liquefaction is well captured. DEM technique has quite a few advantages compared to experiments. The non-uniform deformations at large strain is a serious issue in experiments which can be overcome using periodic boundaries in DEM simulation. Also, DEM preserves the fabric of the initial sample for parametric studies, whereas it is challenging to obtain in the experiments. DEM provides a viable means to quantify the micro-scale information, i.e., particle contacts, voids, and their evolution, which is quite challenging in experiments.

2.3 Soil fabric and quantification techniques

The term “soil fabric” is commonly used to represent the arrangement of soil particles, particle groups, and void space distribution (Mitchell and Soga 2005). The close synonym “soil structure” represents the soil fabric along with the particle’s composition and inter-particle contact forces. In soil mechanics, the term ‘fabric’ and ‘structure’ are used interchangeably. Soil fabric has a profound influence on the macroscopic properties of sands such as small-strain stiffness (e.g. Kuwano and Jardine 2002; Wang and Mok 2008; Zdravković and Jardine 2001), permeability (e.g. Chan and Kenney 1973), peak strength (e.g. Arthur and Menzies 1972; Ventouras and Coop 2009) and dilatancy (e.g. Cresswell and Powrie 2004). For the same void ratio and effective confining pressure, samples having different soil fabric can be obtained by using different sample preparation methods or by subjecting to different pre-shearing histories (e.g. Ishibashi and Capar 2003; Ishihara and Okada 1978; Mulilis et al. 1977; Sze and Yang 2013; Vaid et al. 1989; Yimsiri and Soga 2010). Kuwano and Jardine (2002) highlighted that though the sample is under isotropic confining stresses, the small strain stiffness is still anisotropic, i.e., different stiffness value is observed in different directions of loading. This behavior is attributed to the anisotropic soil fabric formed from the sample preparation. Vaid and Sivathayalan (2000) found that the soil fabric plays an important role in determining the liquefaction susceptibility, for example, samples with same void ratio and confining pressure but prepared using moist tamping method are observed to be easily liquefiable compared to

samples prepared using water-pluviation method. Overall, the quantification of soil fabric is found to be necessary.

The quantification of fabric necessarily means evaluating the particle positions, geometries, contacts, and contact forces. It is challenging to access all the microscale information through experiments. Several laboratory tests were attempted to look at the soil fabric and understand its relationship with the macroscale response. Tests on photo-elastic disks are particularly common (e.g., Bi et al. 2011; Majmudar et al. 2007; Majmudar and Behringer 2005; Oda et al. 1985; Zhang et al. 2010). Photo-elastic disks are special birefringent materials which rotate the polarization of light according to the amount of local stress that is experienced. Thus, allowing it to make both qualitative and quantitative measurement of forces at the microlevel and depicting the soil fabric. However, tests using photo-elastic disks are limited to 2D conditions with disk-shaped particles. Most recently, high resolution computed tomography (CT) scan technique that is common in medical field applications, has been used to scan the 3D microscale images revealing the soil fabric information (e.g. Atwood et al. 2004; Desrues 2004; Desrues et al. 2006; Fonseca et al. 2012). Using image processing techniques on these microimages, relevant contact information and particle shapes can be obtained. However, the limitation of this technique is that contact force transmission at inter-particle contacts cannot be obtained.

Numerical simulation using DEM is much cost-effective and time-efficient way to look at the fabric of a granular assembly. All the essential fabric information, i.e., inter-particle contacts, contact forces, particle orientations, and so forth, can be easily tracked using DEM. In this thesis, all the numerical tests are performed using DEM, and the evolution of internal soil fabric in undrained cyclic loading is studied. Soil internal fabric is quantified generally in terms of descriptors using the contact-based information or the void-based information. The overall picture of soil fabric is obtained by understanding the evolution of both these fabrics.

2.3.1 Contact-based fabric indicators

Contact-based fabric indicators are the most widely used indicators which are based on inter-particle contacts. Some of the most commonly used indicators are coordination number, contact-force distributions, and contact-based fabric tensors.

Coordination number, Z, is defined as the ratio of the total number of contacts to the total number of particles:

$$Z = \frac{2N_c}{N_p} \tag{2.19}$$

where N_c is the total number of contacts and N_p is the total number of particles. Factor 2 is because each contact is shared by two particles. In general, not all the particles in contact, transmit stresses. Hence a modified definition of coordination number, i.e., "mechanical coordination number (Z_m)" was proposed by Thornton (2000) and is defined as:

$$Z_m = \frac{2(N_c - N_p^1)}{N_p - (N_p^1 + N_p^0)} \tag{2.20}$$

where N_p^0 is the number of particles with no contact and N_p^1 are the particles with just one contact. such particles are termed as "rattlers" or "floaters". Kuhn (1999) identified groups of particles that do not participate in load transmissions, such as the pedant particles, the isolated particles, the island particles and, peninsular particles, as shown in Figure 2.7. Thus, a new coordination number termed effective coordination number (Z_p) is defined by excluding all these particle group with no stress transmission.

Coordination number highlights only about the number of active contacts that transmit load but do not give any information about the contact force magnitudes. Many researchers analyzed the contact force information by considering the probability distribution of contact force magnitudes. The force that is transmitted at each contact is different. The plot showing the distribution of probable contacts transmitting a given force against the normalized contact force (generally normalized using average contact force) gives the probability distribution of contact-forces (O'Sullivan 2011). The average contact force itself carries information about the load-bearing ability of a granular assembly.

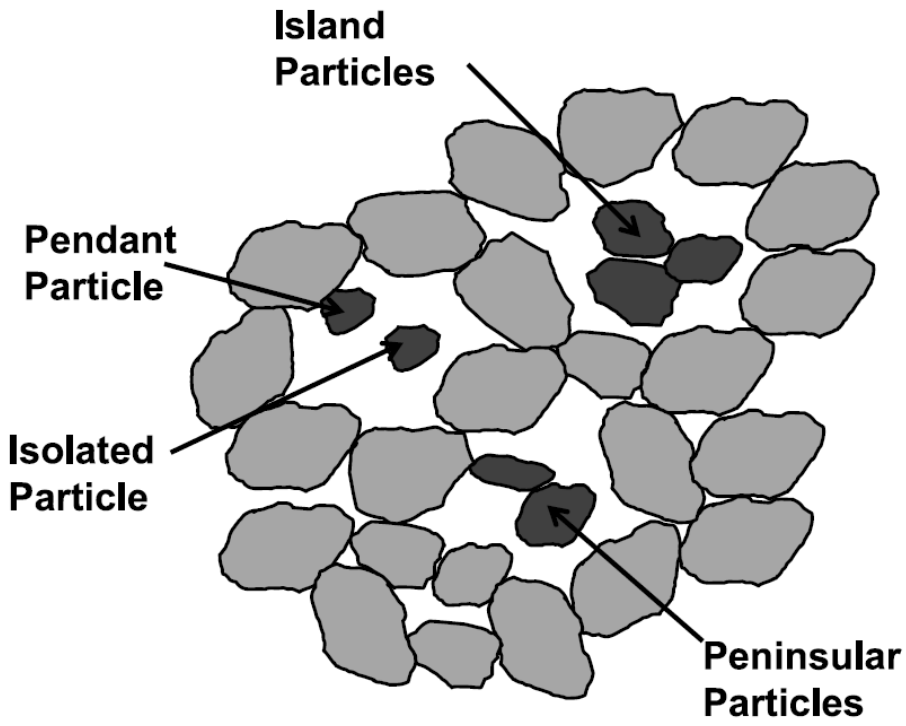


Figure 2.7 Particle configurations (after O'Sullivan 2011)

Neither the coordination number nor the probability distribution of contact force gives any information about anisotropy in granular assembly. To compute the anisotropic features of the granular assembly, fabric tensors are generally used. Fabric tensors are defined as the directional indicators for numerous vectors that are distributed in space. The contact normal vectors, branch vectors (joining the centroids of two contacting particles), and long-axis orientations are the geometrical characteristics of a granular assembly (Fonseca et al. 2013), as shown in Figure 2.8. Computing fabric tensors relating to each of these vectors would provide significant insights into the understanding of internal soil structure. The definition of second-order fabric tensor ($\boldsymbol{\phi}$) proposed by Satake (1982) and Oda (1982) is given as:

$$\boldsymbol{\phi} = \frac{1}{N}\sum_{k=1}^{N_c} \boldsymbol{n}^k \otimes \boldsymbol{n}^k \tag{2.21}$$

where N is the total number of vectors and $\boldsymbol{n}^k$ is the kth unit vector. Depending on the unit vector under consideration, different fabric tensors can be computed to depict different characteristics of the granular assembly. In DEM, most commonly used fabric tensor is the contact normal fabric tensor and is obtained from contact normal directions. Other fabric tensors such as contact normal force fabric tensors and contact tangential force fabric tensors (e.g., Guo and Zhao 2013; Ouadfel and Rothenburg 2001; Rothenburg and Bathurst 1989) are also commonly used by researchers. It is important to note that contact force fabric tensors are obtained from the contact force vectors which are usually not unit vectors. Thus, their definition of fabric tensor is slightly different. In general, fabric tensors is particularly crucial because the anisotropy of the constituent vectors can be easily computed and their evolution during a loading process can also be studied. Such information is essential in the formulation of constitutive models.

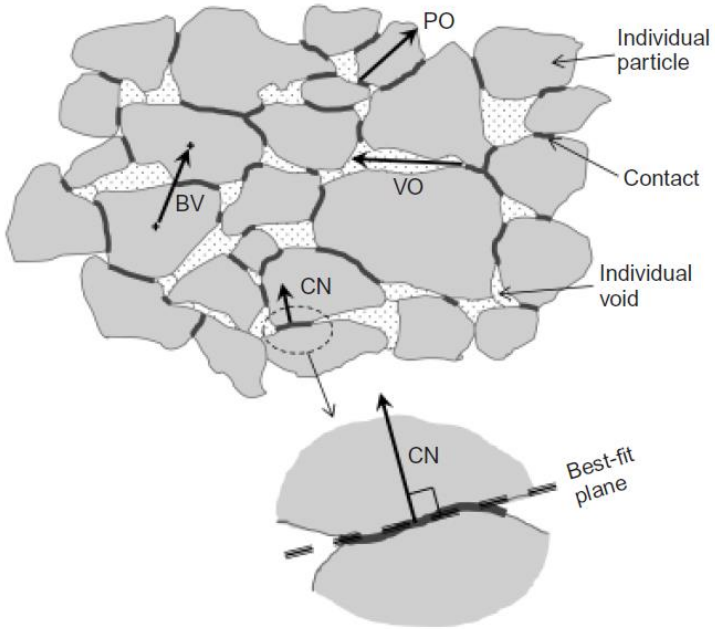


Figure 2.8 Main features in an intact sample and vectors used for fabric quantification; Branch Vectors (BV), Contact Normal (CN), Particle Orientation (PO), Void Orientation (VO) (after Fonseca et al. 2013)

2.3.2 Void-based fabric indicators

Void spaces are an important part of soil fabric. Yet, quantification of the topology of the void spaces is very less common. In the early study by Oda et al. (1985), the void space for a granular assembly is quantified using the scanning line method. In this method, the mean path length of

the void cut by a continuous scanning line in the direction of unit vector $\boldsymbol{p}$ is assumed as $\bar{l}(\boldsymbol{p})$, then the void fabric tensor in 3D is given as:

$$\boldsymbol{V} = \frac{15}{2}\left(\frac{1}{L}\sum_{k=1}^{m} \bar{l}^k(\boldsymbol{p}^k \otimes \boldsymbol{p}^k) - \frac{1}{5}\boldsymbol{E_3}\right) \tag{2.22}$$

where m is the total number of scanning lines, $\boldsymbol{E_3}$ is the identity tensor and L is the total mean length given as $L = \int_{\Omega} \bar{l}\,(\boldsymbol{p})\, d\Omega$. The average mean path length is then defined as, $l(\boldsymbol{p}) = \bar{l}(\boldsymbol{p})/L$. Ghedia and O'Sullivan (2012) discussed an approach to compute the quantities of scanning line method on a 2D digital image and later analyzed the void characteristics in shear bands formed during biaxial compression simulations in DEM. Note that the $l(\boldsymbol{p})$ defined in the scanning line method characterizes the orientation and the length of the voids in an average sense. The understanding of individual voids paces is not possible using this method.

Satake (1992) introduced a graph-theoretical approach where points, branches, and loops replace the discrete granular assembly. Points are the particle centers, branches are the branch vectors, and loops are the void space surrounded by branch vectors. Kuhn (1999) used this analogy to define the loops as the 'void cells', as shown in Figure 2.9. The fabric tensor of a void cell is then defined as:

$$\boldsymbol{F} = \frac{1}{2}\sum_{k=1}^{m} \boldsymbol{b}^k \otimes \boldsymbol{b}^k \tag{2.23}$$

where $\boldsymbol{b}^k$ is the kth branch vector corresponding to a void-cell of 'm' branch vectors (refer to Figure 2.9). The elongation ratio of that particular void cell is later defined as $\alpha = \sqrt{F_{22}/F_{11}}$. The average elongation factor, i.e., $\hat{\alpha}$ defined as the mean of α values in all the void cells is used as an anisotropy indicator.

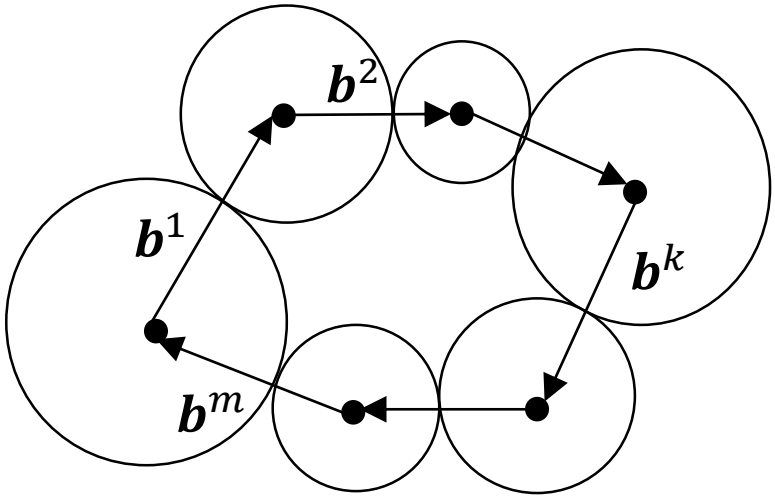


Figure 2.9 Void cell constructed from the branch vectors

Li and Li (2009) proposed a unique tessellation technique to construct a solid cell system and a void cell system of a granular assembly. Using a modified Delaunay-Voronoi tessellation technique on the contact points, these cell systems are formed. Later, the anisotropy of the void cell system is defined by a fabric tensor as:

$$\boldsymbol{G} = E_0 \oint_{\Omega} \bar{v}(\boldsymbol{n})\, \boldsymbol{n} \otimes \boldsymbol{n} \tag{2.24}$$

where E_0 is the normalization factor, equals to 2π for a 2D space and equals to 4π for a 3D space. $\bar{v}(\boldsymbol{n})$ is the average length of all the void vectors '$\boldsymbol{v}$' along the direction $\boldsymbol{n}$. $\boldsymbol{v}$ are the vectors connecting the void cell center and the contact points at the boundary.

However, in this thesis, we deal with the liquefied state of granular assembly where the packing has very few to no contacts in the flow stage. The void cells or void cell system are constructed based on the contact point information. In a fully liquefied state, contacts are entirely lost. Thus, it becomes difficult to construct such cells.

Wang et al. (2016) proposed an indicator, Mean Neighboring Particle Distance (MNPD), to capture the void features governing liquefaction behavior of sand. The definition is given as follows:

$$MNPD = \frac{1}{N}\sum_{k=1}^{N} \frac{D_1^k + D_2^k + D_3^k}{3} \tag{2.25}$$

where N is the total number of particles in the assembly. D_1^k, D_2^k and D_3^k are defined as the distance between the particle k and its three closest neighbor particles (refer to Figure 2.10). Closest three distances are considered because it is assumed that at least three contacts are required for a 2D particle to become part of the load-bearing structure. Thus, the basic idea of MNPD is that how far the granular assembly is away from its stable configuration. Suppose if all the particles in the granular assembly are in contact with at least three neighboring particles, then the MNPD value is zero. The limitation of this descriptor is that it's a scalar value; it doesn't give any sense about the void distribution around a particle.

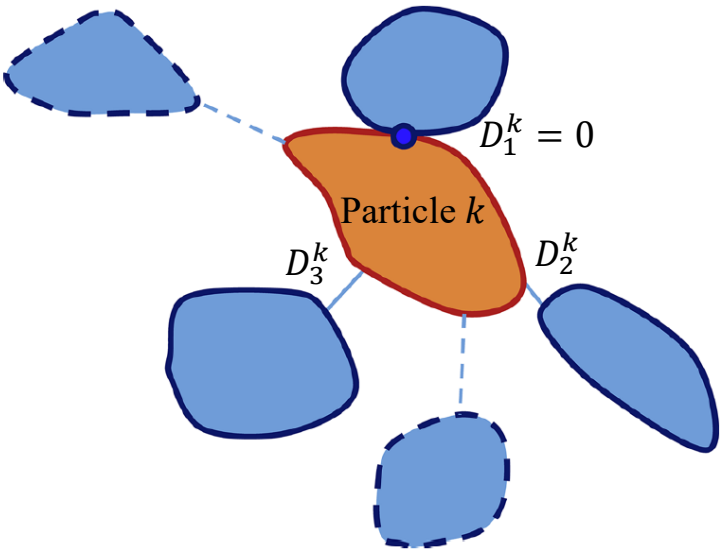


Figure 2.10 Mean Neighboring Particle Distance (after Wang et al. 2016)

Wang and Wei (2016) had a different view of the voids. It was proposed that each particle in a granular assembly is bound to a void space around it and the movement of the particle is restricted only to this space during any deformation in the granular assembly. This void space around the particle was determined from the Voronoi cell construction. A new descriptor, termed Centroid Distance D_c, was proposed to characterize the whole particle-void space. As shown in Figure 2.11, the void space around the particle 'i', is given by convex polygon $C_1C_2C_3C_4C_5$. The centroid distance is defined as:

$$D_c = \frac{1}{N_p}\sum_{i=1}^{N_p} D_c^i \quad and \quad D_c^i = \frac{\left|\boldsymbol{P}^i - \boldsymbol{O}^i\right|}{R_{50}} \tag{2.26}$$

where $\boldsymbol{P}^i$ is the mass centroid of the ith particle, $\boldsymbol{O}^i$ is the mass centroid of Voronoi cell corresponding to the ith particle and N_p is the total of particles / Voronoi cells. No contact information is required to compute the centroid distance, D_c, thus it can be used to characterize the void space even in a liquefied state.

Wei et al. (2018) introduced a special type of particle-void fabric descriptors for a 2D granular assembly by constructing "particle-void cells" around the particles using weighted Voronoi tessellation. The particle-void cells are approximated using ellipses in order to compute the aspect ratio and the direction of the ellipse. Two global descriptors (E_d, A_d) are defined in their study, where E_d is the homogenized aspect ratio of local particle void cells and A_d is the degree of anisotropy in the local particle-void cells directions.

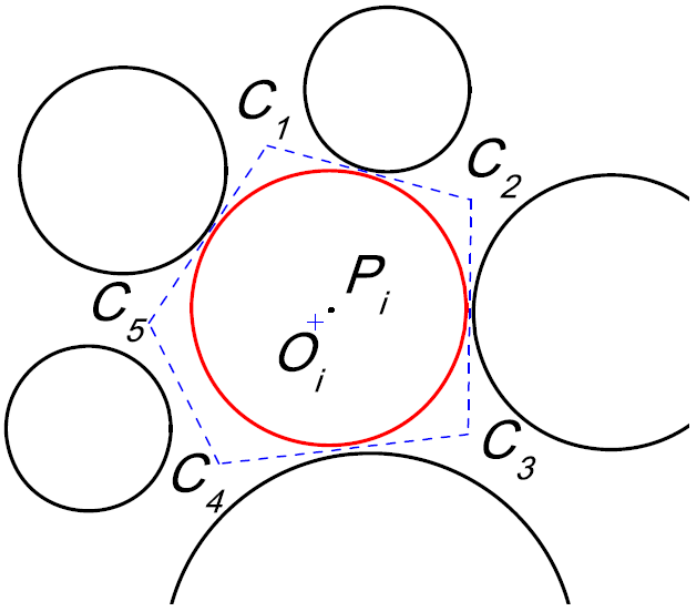


Figure 2.11 Voronoi cell construction from the neighboring particles
(after Wang and Wei 2016)

2.4 Post-liquefaction settlement and re-liquefaction behavior

2.4.1 Post-liquefaction settlement

After Initial liquefaction, the dissipation of excess pore water pressure causes settlements in ground. Lee and Albaisa (1974) were the first to comment about post-liquefaction settlements. They found that the sites which have undergone liquefaction are followed by higher ground settlement, i.e., about 3-4%, whereas the sites have not gone through liquefaction are followed

with low settlements of about 0.5%. Yoshimi et al. (1975) have performed compression and swelling tests on sands at different confining pressures and found that even at very low confining pressures, sands that have not undergone liquefaction showed less compressibility compared to liquefied soils. Tatsuoka et al. (1984) observed that the maximum shear strain experienced by the soil with a certain relative density during cyclic loading has a clear correlation with the reconsolidation volumetric strain. Also, with the increase in relative density of the soil, for an identical cyclic loading, the volumetric settlement in reconsolidation has decreased. Many other studies that followed also highlighted the correlation of volumetric strain with maximum shear strain and the relative density (e.g., Nagase and Ishihara 1988; Tokimatsu and Seed 1987). Ishihara and Yoshimine (1992) after a series of tests on different soils using different loading conditions, arrived at a simple chart for the estimation of volumetric settlements, as shown in Figure 2.12. The increasing trend of volumetric strain with maximum shear strain is evident from Figure 2.12, but after about 8% maximum shear strain, the volumetric settlement gets saturated and is unaffected with further increase in maximum shear strain. Shamoto et al. (1996) proposed that in a log-log graph the 'relative compression,' i.e., $\Delta e/(e_i - e_{min})$ is approximately linear over a range of maximum shear strains and valid for different densities of different sands. Tsukamoto et al. (2004) proposed a procedure for the evaluation of settlements after reconsolidation in silty sands.

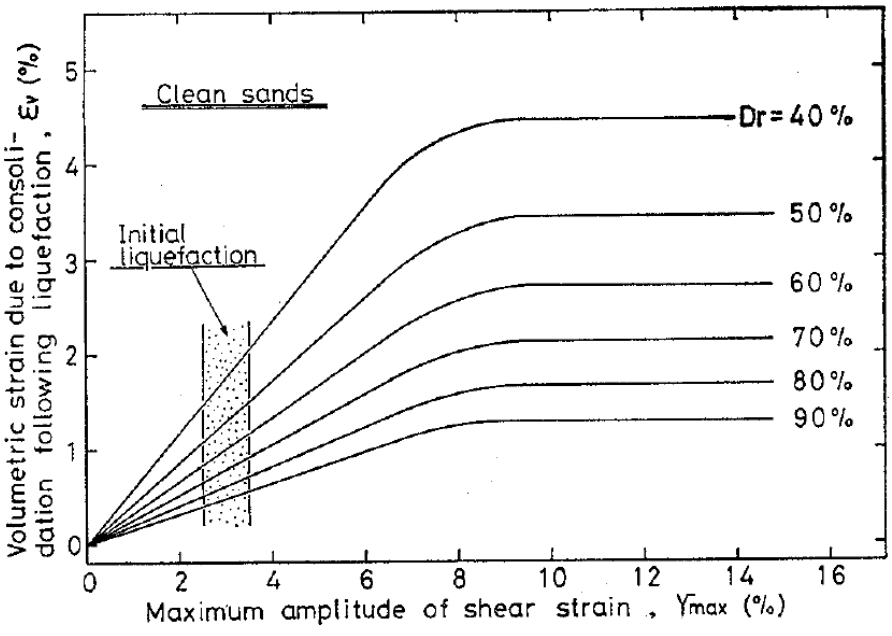


Figure 2.12 Empirical correlation between volumetric strain and maximum shear strain (after Ishihara and Yoshimine 1992)

2.4.2 Re-liquefaction behavior

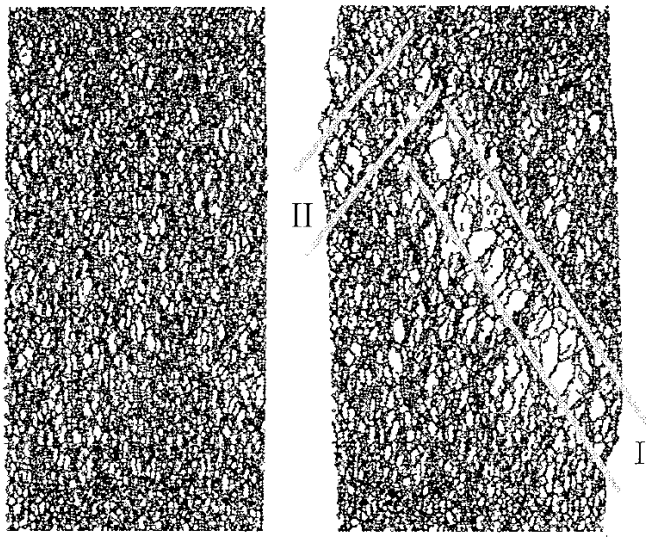
Reconsolidation after cyclic liquefaction increases the relative density of the sand. It is a common belief that denser sand has a relatively higher strength to liquefaction. The observation from case-histories and laboratory experiments proved otherwise. Several factors that affect resistance to liquefaction can also be attributed to re-liquefaction strengths. Factors such as pre-strain/stress histories and fabric anisotropy of the sand are found to have more considerable influence on liquefaction resistance than the relative density.

Finn et al. (1970) for the very first time, pointed out that the effect of pre-strain history on the soil's liquefaction resistance. It was observed that because of the small pre-strains in the soil, the local instabilities get eliminated, and the soil becomes much resistive and shows improved resistance to liquefaction. Whereas for soils with high pre-strain histories, the soil tends to attain a non-uniform structure resulting in decreased resistance to liquefaction. Similar observations were made by Suzuki and Toki (1984), but the discussion was put in terms of pre-shearing. The limiting value of about 0.65-0.75 of the undrained shear strengths is observed. Samples that are initially pre-sheared to the limiting value or above tends to show lower resistance to liquefaction, but if the pre-shear stress is lower than the limiting value, the resistance seems to increase. Ishihara and Okada (1978) observed the preferential directions in the triaxial sample with large pre-shear stresses, i.e., in the direction of initial large pre-shear stress, the sample is observed to have more stiff behavior, and in the other direction, it is much softer. Several other studies also highlighted the differences in soil liquefaction resistance for low and high pre-shear strain or stress histories (e.g., Porcino et al. 2009; Shahnazari et al. 2015; Wahyudi et al. 2015)

The major limitation of correlating pre-strain or pre-stress with the liquefaction resistance is that they cannot be measured beforehand for a given specimen, thus making it difficult to estimate the strength changes in the soil. The more radical way is to measure the internal soil fabric and to correlate it to the liquefaction resistance. The fabric quantification would reflect any changes in soil intrinsic state caused by pre-stress or pre-strains histories.

Oda et al. (2001) proposed that the significant changes due to pre-shearing are because of the formation of shear bands in the pre-sheared specimens. As shown in Figure 2.13, multiple shear bands are found with column-like structured contact forces and large local voids at selected sections in a pre-sheared specimen (Iwashita and Oda 2000). Thus, the specimens are stiffer when loaded in the direction of the column fabric but highly unstable if it is loaded in the

perpendicular direction. The effects of pre-strain on liquefaction resistance and the formation of preferential orientations in the soil fabric (fabric anisotropy) are also highlighted by Yamada et al. (2010) and Ye et al. (2015) using triaxial and hollow cylinder torsional shear tests respectively. From image analysis by Wahyudi et al. (2015), it was found that the samples that experienced low pre-strain levels had a more uniform distribution of local shear strains and has no significant fabric change thus increasing the liquefaction resistance because of increase in relative density. However, on the other hand, the samples which experienced high pre-shear strains showed significant nonuniformity of local shear strain distribution and is more prone to strain localization. The local relative density at the sections of strain localization is found to be very low, thus making it more susceptible for liquefaction. Ye et al. (2018) have performed multiple small-scale shake table tests to understand the mesoscopic mechanism of liquefaction and multiple-liquefaction behaviors. 2D mesoscopic images of the specimen were taken before the 1st, 2nd, 3rd, and 4th events of shaking of a specimen and are shown in Figure 2.14. It was observed that the long-axis directions of the sand particles are more horizontal in the initial sample (refer Figure 2.14a) whereas after multiple liquefaction and reconsolidation the sand particles tend to be more vertical, thus forming an unstable granular assembly. The reason for lower liquefaction resistance was attributed to the formation of unstable granular structure.



(a) At failure (b) After failure

Figure 2.13 Columnar structure in 2D assembly (after Iwashita and Oda 2000)

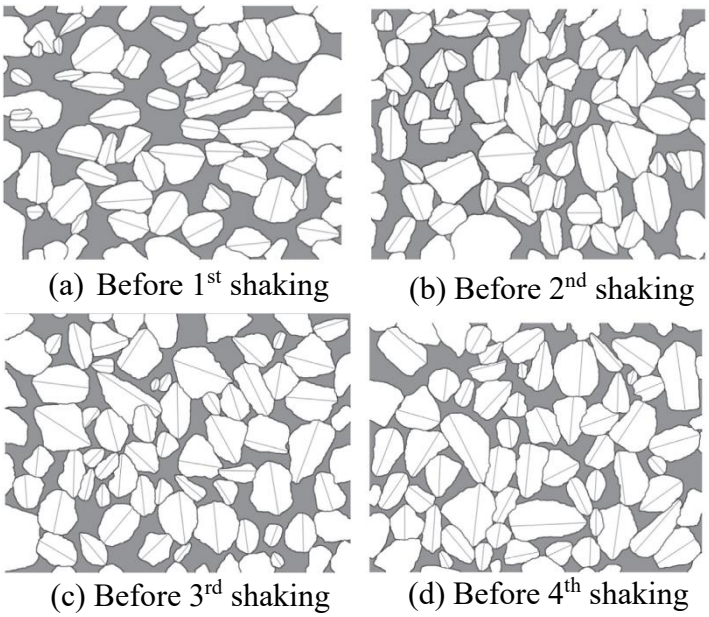


Figure 2.14 Sand granular edge images at different stages (after Ye et al. 2018)

2.5 Multi-directional loading

The seismic excitations are multi-directional loads with varying amplitudes, thus, making it a very irregular and complex loading situation. Geographical features, like sloping grounds, which induce an initial static shear on the soil element increases the level of complexity further. Hence conventional uni-directional laboratory tests are not viable to capture the real soil behavior under such complex loads. Only a very few studies were conducted in understanding the effect of multi-directionality in loads. In practice, the effects of multi-directionality of loads are considered by introducing a reduction factor into the resistance calculation for cyclic loads. Pyke et al. (1975) conducted the very first multi-directional shearing test on dry sands by mounting two shake tables transversely. It was found that the settlements of sand layers in uni-directional load is way less than the settlements in multi-directional load. Seed et al. (1975) observed that the saturated sand, when subjected to 20 cycles of uniform loading in a gyratory manner, will have 15% smaller liquefaction resistance when compared to uni-directional loading with the same number of loading cycles. Ishihara and Yamazaki (1980) in their study designed a bi-directional simple shear test device that could majorly produce two types of loading, one incorporated a stress path with no rotation (Alternate shear) occurred during loading and unloading while the other incorporated a full rotation (Rotational shear) of stresses during each cycle. In this study, the limitations in the definition of liquefaction, i.e.,100%

excess pore water pressure, was highlighted. A new failure criterion, i.e., 3% single amplitude shear strain, was considered. It was found that for all the cases, the additional shear in a second direction lowered the resistance to liquefaction proportionally to the amount of load in that direction. They also found that tests with rotational shear loading had the lowest resistance to liquefaction and followed by tests with alternate shear loading and then by the uni-directional tests. Years later, using the same bi-directional simple shear apparatus, Ishihara and Nagase (1988) have conducted several series of tests in order to understand the dual effects of irregularity and the multi-directionality of the loads on samples with different densities. It is found that the effect of multi-directionality on different relative density samples is not significantly varying, whereas the irregularity in load has a considerable effect on different relative density samples. Yamada and Ishihara (1983) have conducted true triaxial tests on a cubical saturated sand specimen in undrained conditions by considering the octahedral plane of the three-dimensional stress space to be the horizontal plane in an in-situ sand deposit. The principal stresses were cyclically applied with varying amplitudes in order to generate the circular, elliptical, and crisscrossing type of loadings on the octahedral plane. It was found that in all the cases, the specimen behaved like an elastic body in the initial stages of cyclic loading, whereas when approached to the initial liquefaction, the specimen behaved more like a perfectly plastic body. Boulanger et al. (1993) designed and constructed a bidirectional simple shear apparatus with an ability to apply both monotonic and cyclic simple shear loading to a cylindrical specimen in either direction. Using this apparatus, Boulanger and Seed (1995) tried to understand the effect of cyclic loads on slopes when applied in the direction of dip and strike. Usually, the soil underneath a sloping ground can be approximated to a specimen with initial static shear stress in dip direction. It was found that cyclic loading, when applied in strike direction, has lower resistance when compared to resistance for the load in dip direction which led to a conclusion that the degree of stress rotation is typically more important than the amount of maximum shear stress. Kammerer (2002) with an aim to generate high quality laboratory test data, conducted a series of bi-directional simple shear tests on clean sands with medium to higher relative densities using a variety of loading paths (linear, oval/circular and figure-8) on specimens replicated for both level (with no static shear stress) and sloping (with an initial static shear stress) grounds. The test results provided valuable experimental data to calibrate constitutive models and can be adopted as benchmarks for numerical simulations. Matsuda et al. (2011) carried out strain-controlled multi-directional cyclic simple shear tests under constant volume on clean sands in order to investigate the effect of multi-directionality of loads on effective vertical stress reduction and post-liquefaction settlements. Different strain paths

like oval/circle were generated by controlling amplitude and phase difference of sinusoidal strain loads in individual directions. It was found that the strain path does not influence effective vertical stress reduction for a higher level of strain amplitudes, whereas, for low amplitude strains, the effect is significant. It was observed that for the same cumulative shear strain, the strain paths has no influence on the post-liquefaction settlement, but for the same cumulative shear strain, if the shear strain amplitude is higher, the settlements are higher.

Overall most of the studies conducted to understand the effects of multi-directionality are experimentally based, and the common conclusion was that the addition of another component in shear stress direction, decreased the resistance to liquefaction. Understanding the fabric changes in multi-directional loading may provide new insights into the fundamental understanding of the soil behavior, which was lacking in previous studies but will be attempted in our study.

2.6 Summary

In the last five decades, much has been learned about the soil liquefaction phenomenon. Saturated sands, when subjected to repeated cyclic loadings, undergo soil liquefaction. The tendency to densify under cyclic loading conditions leads to increase in pore-water pressure and decrease in effective confinement under undrained conditions. A significant amount of generation in excess pore water pressure can cause large deformations in ground and hence soil liquefaction. Depending on the density of soil, the deformations in the ground can be either uncontrollably large or limited displacements. Medium dense and dense sands exhibit a behavior often referred to as “cyclic mobility” in which large strains can be generated but are eventually limited by hardening or “lock-up” of the material. In cyclic mobility, the shear strains keep accumulating after every cycle where the mechanical status of sands transform from a flow-like state to a hardening like state upon loading and vice versa upon unloading. Available constitutive models are not sophisticated enough to model the post-liquefaction deformation accurately. This is because most of these models are phenomenological in nature and are governed by variables which have no physical basis of measuring. It is believed that traditional constitutive models can be refined further by incorporating the soil fabric quantities and their evolution during soil liquefaction phenomenon.

If not impossible, quantification of soil fabric through experiments is quite challenging. Thus, numerical simulations using discrete element method can be used where the soil fabric information can be easily obtained. Soil fabric is traditionally quantified in two ways, i.e., contact-based fabric quantification and void-based fabric quantification. The available void-based fabric quantification also relies on the contact information in order to differentiate voids. But, in a liquefied state, no contacts are present in the granular packing thus quantification of soil fabric would not be possible using the current methods. Hence a new method for quantification of soil fabric is required to understand the evolution of internal structure during soil liquefaction. Soil fabric significantly influences the volumetric compression after liquefaction and the re-liquefaction resistance of the sands. From the limited studies of multi-directional shear testing, it is found that the addition of another component in shear stress direction, decreased the resistance to liquefaction. Understanding the fabric changes in multi-directional loading may provide new insights into the fundamental understanding of the soil liquefaction in more general loading paths.

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CHAPTER 3

Discrete Element Method and Soil Fabric Quantification

For a better understanding of the behavior of complex systems, numerical modelling techniques are widely used. Continuum modelling is the most common numerical method which is applicable in a variety of fields of research. The particle system like sands is assumed to be a complex pseudo-solid while modelling with continuum technique. The behavior of such pseudo systems is defined using complex constitutive models which may vary for different systems. To incorporate different behavioral aspects of particle systems under different loading conditions, the constitutive models must be very sophisticated. However, with the increased power in computation in the last few decades, it is now possible to appreciate the discrete nature of soils and the interactions at the particle level can be modeled. Such a discrete approach is now popular in the field of geomechanics and is referred to as Discrete Element Method (DEM).

DEM was first proposed by Cundall and Strack (1979) to simulate the behavior of granular assemblies. In this technique, the discrete particles of the granular assembly are individually modeled and can interact with neighboring particles to generate contact forces and displacements. The macroscopic behavior of the whole granular assembly is obtained from the integration of the forces and displacements at the particle level. In DEM, constitutive models are defined at contact level to calculate forces, unlike constitutive models of continuum modelling these models are relatively very simple and mostly physics based. Even considering

such simple contact models and even simpler particle shapes (like spheres and discs) the qualitative macroscopic response under different loading conditions can be well captured using DEM. The major advantage of DEM from continuum modelling and experimental studies is that using DEM the internal fabric of the granular assembly can be accessed, which is crucial information to predict the macroscopic behavior.

In this thesis, the open-source DEM program, Yade (Šmilauer et al. 2011) is used for conducting the numerical simulations. Yade is being used by a large community of researches and is proven very reliable for running DEM simulations. This chapter discusses the general formulation of DEM and the methods used for fabric quantifications using DEM. New fabric descriptors are proposed in this study, and the detailed formulation procedure is also mentioned.

3.1 DEM formulation

Each particle in a granular assembly experience the forces and torques form the contacts in which it participates. The motion of individual particles, i.e., the translational and rotational velocities, are governed by the equilibrium equations of motion. The traditional leapfrog integration scheme is used to compute the velocities and accelerations of the particles. The name leapfrog comes from the fact that the even derivatives of position/orientation (i.e., translational/angular acceleration) are known at on-step point whereas the odd derivatives (i.e., translational/angular velocities) are known at the mid-step points. In the current thesis, the particles are considered to be spherical, and the formulation is also mentioned for the same.

3.1.1 Algorithm

From equations of motion for body 'i', the translational acceleration at a time 't' is given as:

$$\ddot{\boldsymbol{u}}^{i}{}_{t} = \frac{\boldsymbol{F}_{t}^{i}}{m^{i}} \tag{3.1}$$

where $\boldsymbol{F}$ is the total force acting on the body, which includes the contact forces, body forces, and numerical damping. m is the mass of the spherical body.

Using the central-difference time integration approach with a time increment Δt,

$$\ddot{\boldsymbol{u}}^i{}_t = \frac{\boldsymbol{u}^i_{t-\Delta t} - 2\boldsymbol{u}^i_t + \boldsymbol{u}^i_{t+\Delta t}}{\Delta t^2} \tag{3.2}$$

Rearranging the terms of Eq-3.2 we get,

$$\boldsymbol{u}^i_{t+\Delta t} = \boldsymbol{u}^i_t + \Delta t\left(\frac{\boldsymbol{u}^i_t - \boldsymbol{u}^i_{t-\Delta t}}{\Delta t} + \ddot{\boldsymbol{u}}^i{}_t \Delta t\right) \tag{3.3}$$

where $\frac{\boldsymbol{u}^i_t - \boldsymbol{u}^i_{t-\Delta t}}{\Delta t} \approx \dot{\boldsymbol{u}}^i{}_{t-\Delta t/2}$, which is the mean velocity of the particle in the previous timestep. Thus position $\boldsymbol{u}^i_{t+\Delta t}$ is given as:

$$\boldsymbol{u}^i_{t+\Delta t} = \boldsymbol{u}^i_t + \Delta t\left(\dot{\boldsymbol{u}}^i{}_{t-\Delta t/2} + \ddot{\boldsymbol{u}}^i{}_t \Delta t\right) \tag{3.4}$$

Thus, the formulation can be written in two equations, i.e., first, calculating the mean velocity of the next time step and use (3.3) to calculate the position at $t + \Delta t$.

$$\dot{\boldsymbol{u}}^i{}_{t+\Delta t/2} = \dot{\boldsymbol{u}}^i{}_{t-\Delta t/2} + \ddot{\boldsymbol{u}}^i{}_t \Delta t \tag{3.5}$$

$$\boldsymbol{u}^i_{t+\Delta t} = \boldsymbol{u}^i_t + \Delta t * \dot{\boldsymbol{u}}^i{}_{t+\Delta t/2} \tag{3.6}$$

The rotational displacement is also similarly calculated but from the equation of motion,

$$\dot{\boldsymbol{\omega}}^i{}_t = \frac{\boldsymbol{M}^i_t}{I^i_o} \tag{3.7}$$

where $\dot{\boldsymbol{\omega}}^i{}_t$ is the angular velocity of body 'i' at time t and $\boldsymbol{M}^i_t$ is the total moment on the body 'i' at time t. I^i_o is the moment of inertia of sphere, which is $I^i_o = \frac{2}{5} m^i (r^i)^2$.

The sequence of calculations in a whole timestep of a DEM simulation is shown in Figure 3.1.

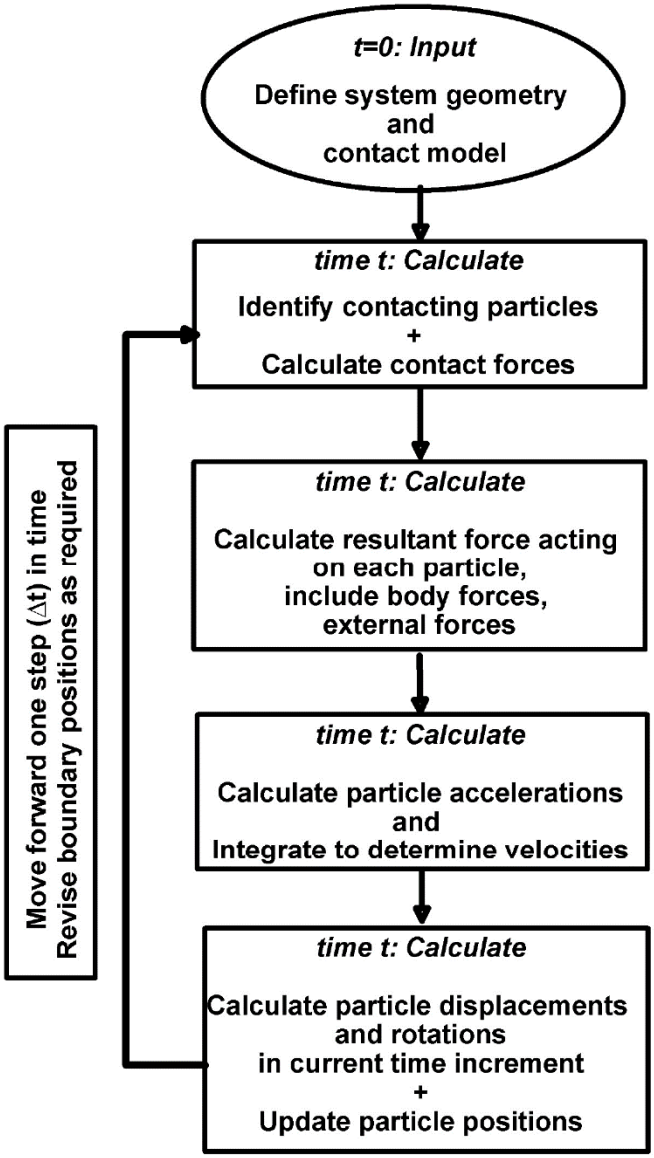


Figure 3.1 DEM algorithm (after O'Sullivan 2011)

3.1.2 Contact models

The contact forces in normal and shear directions are calculated using the Hertz-Mindlin nonlinear contact model (Tsuji et al. 1992). The total vector sum of normal and shear forces account to the total force ($\boldsymbol{F}_t$) and is used to calculate the acceleration, velocity, and displacements (refer Eq. 3.1 to 3.6). The contact moments are computed using the rolling resistance model proposed by Belheine et al. (2009).

3.1.2.1 Normal direction

The normal force at a contact ($\boldsymbol{f}_n^c$) is given by,

$$\boldsymbol{f}_n^c = k_n \delta^{1.5} \boldsymbol{n}^c \tag{3.8}$$

where δ is the overlap between two particles and $\boldsymbol{n}^c$ is the outward contact normal. k_n is the contact normal stiffness defined as,

$$k_n = \frac{2E\sqrt{r}}{3(1-v^2)} \tag{3.9}$$

where E is the young's modulus, and v is the poisons ratio of the particles. r is the equivalent radius, determined from the radii contacting particles as (say $r_a$and r_b),

$$r = \frac{r_a r_b}{r_a + r_b} \tag{3.10}$$

3.1.2.2 Tangential direction

The contact force in the tangential direction ($\boldsymbol{f}_t^c$) involve the coulomb criterion to describe the frictional stick-slip behavior and is given by,

$$\boldsymbol{f}_t^c = k_t \boldsymbol{U}_t^c, \quad if\ |\boldsymbol{f}_t^c| \le \mu |\boldsymbol{f}_n^c|\ (stick) \tag{3.11}$$

$$\boldsymbol{f}_t^c = \frac{\mu |\boldsymbol{f}_n^c| \boldsymbol{U}_t^c}{|\boldsymbol{U}_t^c|}, \quad otherwise\ (slip) \tag{3.12}$$

where $\boldsymbol{U}_t^c$ is the overall tangential displacement at the contact, and μ is the coefficient of friction. k_t is the tangential contact stiffness defined as,

$$k_t = \frac{2E\sqrt{r}\sqrt{\delta}}{(1+v)(2-v)} \tag{3.13}$$

3.1.2.3 Rolling and twisting resistance

The stick and roll behavior at contact is modeled by defining the contact moment ($\boldsymbol{M}_t^c$) as:

$$\boldsymbol{M}_t^c = k_r \boldsymbol{\theta}_r^c \quad if\ |\boldsymbol{M}_t^c| \le \eta r |\boldsymbol{f}_n^c|\ (stick) \tag{3.14}$$

$$\boldsymbol{M}_t^c = \frac{\eta r |\boldsymbol{f}_n^c| \boldsymbol{\theta}_r^c}{|\boldsymbol{\theta}_r^c|}, \quad otherwise\ (roll) \tag{3.15}$$

where $\boldsymbol{\theta}_r^c$ is the angular vector of the total rolling part and η is a dimensionless coefficient for the plastic moment. k_r is the contact rolling stiffness which is obtained from the tangential contact stiffness and is given as,

$$k_r = \beta k_t r^2 \tag{3.16}$$

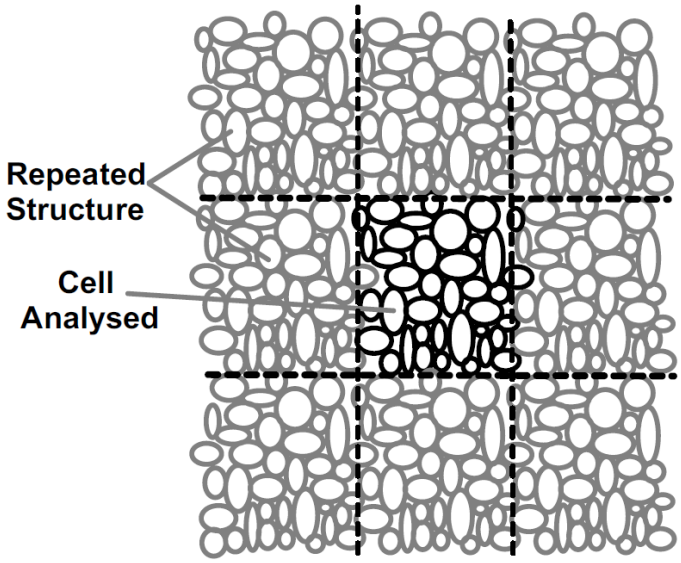
where β is a dimensionless coefficient for rolling stiffness.

3.1.3 Boundary conditions

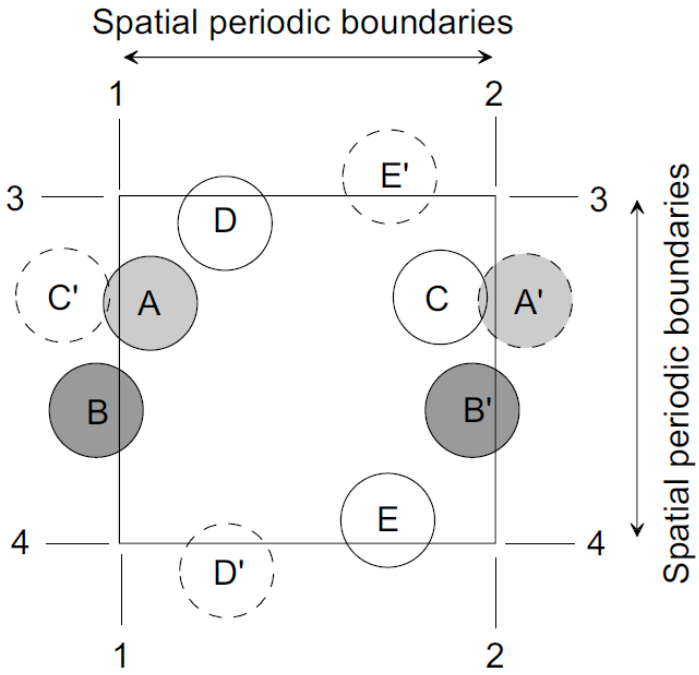
The specification of boundary conditions in DEM simulation is equally essential as it is in continuum modelling technique. In general, the boundaries conditions can either be displacement-based or traction-based. In DEM, the displacement boundary conditions are achieved by specifying the positions of selected particles (boundary particles) for the next timestep. Similarly, for the traction boundary conditions, a force is added to selected particles (boundary particles) as their contact force. The resultant displacement and accelerations of every other particle in the system are calculated because of the applied boundary conditions (O'Sullivan 2011). Note that the selected particles to which the positions or forces are specified may change every timestep as the configuration of the whole assembly changes every step. The most popularly used boundary conditions are Rigid boundaries, Periodic boundaries, and Membrane boundaries. In this thesis, periodic boundary conditions are used to simulate undrained cyclic simple shear tests.

Periodic boundaries or periodic solution space are applied on a periodic cell which is generally in the form of a parallelepiped (Cundall 1988). These boundaries are imaginary and can help to mimic the idea of infinite space, as shown in Figure 3.2a. The basic idea is that the opposite boundaries of this parallelepiped are linked, a particle moving out of one boundary is assumed to come back from its opposite boundary. Thus, at any point in time, the number of particles inside a periodic cell is always the same. As shown in Figure 3.2b, particle A that is leaving out from cell through boundary 1-1 comes back inside as pseudo- particle A' though boundary 2-2 (Cui 2006). The pseudo-particle is the exact image of the real particle but are separated by a cell-dimension. Any force acting on pseudo-particle will be equally applied on the real-particle and vice versa. This allows the particles along a boundary to form new contacts with the particles in the opposite boundary and move across. For example, though Particle A and C

are far away from each other with periodic boundary assumption, their movements are influenced by each other.



(a) Periodic space of 2D granular packing



(b) Mechanism of particles at periodic boundaries

Figure 3.2 Periodic boundaries (after O'Sullivan 2011 and Cui 2006)

The periodic boundaries are strain controlled. To deform a periodic cell, a strain gradient tensor needs to be defined as follows:

$$\dot{\boldsymbol{\epsilon}} = \begin{bmatrix} \dot{\epsilon}_{11} & \dot{\epsilon}_{12} & \dot{\epsilon}_{13} \\ \dot{\epsilon}_{12} & \dot{\epsilon}_{22} & \dot{\epsilon}_{23} \\ \dot{\epsilon}_{13} & \dot{\epsilon}_{23} & \dot{\epsilon}_{33} \end{bmatrix} \tag{3.17}$$

where $\dot{\epsilon}_{11}$, $\dot{\epsilon}_{22}$, and $\dot{\epsilon}_{33}$ are rates of axial strains in the periodic cell in X, Y, and Z direction. $\dot{\epsilon}_{12}$, $\dot{\epsilon}_{13}$, and $\dot{\epsilon}_{23}$ are the strain rates of XY-plane, XZ-plane, and the YZ-plane. For an isotropic consolidation of a periodic cell the strain gradient tensor ($\dot{\boldsymbol{\epsilon}}_{iso}$) must be of the form:

$$\dot{\boldsymbol{\epsilon}}_{iso} = \begin{bmatrix} v & 0 & 0 \\ 0 & v & 0 \\ 0 & 0 & v \end{bmatrix} \tag{3.18}$$

For a simple shear loading in XY-plane, the strain gradient tensor ($\dot{\epsilon}_{SS}$) is of the form:

$$\dot{\boldsymbol{\epsilon}}_{ss} = \begin{bmatrix} 0 & v & 0 \\ v & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{3.19}$$

where v is the strain gradient and for a quasi-static loading v can be as low as 0.01/sec.

Within a periodic cell, the displacements of the particles are governed by the global strain rate tensor. Based on the location of the particle, velocity is imparted because of the periodic cell deformation. The resultant velocity ($\dot{\boldsymbol{u}}_{res}{}^{p}$) of the of a particle '$p$' is given by,

$$\dot{\boldsymbol{u}}_{res}{}^{p} = \dot{\boldsymbol{u}}^{p} + \dot{\boldsymbol{\epsilon}}\boldsymbol{u}^{p} \tag{3.20}$$

where $\dot{\boldsymbol{\epsilon}}$ is the strain gradient tensor, $\boldsymbol{u}^{p}$ is the position vector of particle 'p' and $\dot{\boldsymbol{u}}^{p}$ is the velocity of particle 'p'. $\dot{\boldsymbol{u}}^{p}$ is obtained from the contact forces and $\dot{\boldsymbol{\epsilon}}\boldsymbol{u}^{p}$ is from the cell deformation. $\dot{\boldsymbol{u}}_{res}{}^{p}$ will be used to calculate the displacements for the next timestep (refer Eq-3.6).

In general, the stress tensor of a periodic cell is calculated using Love's formula (Love 1927) as,

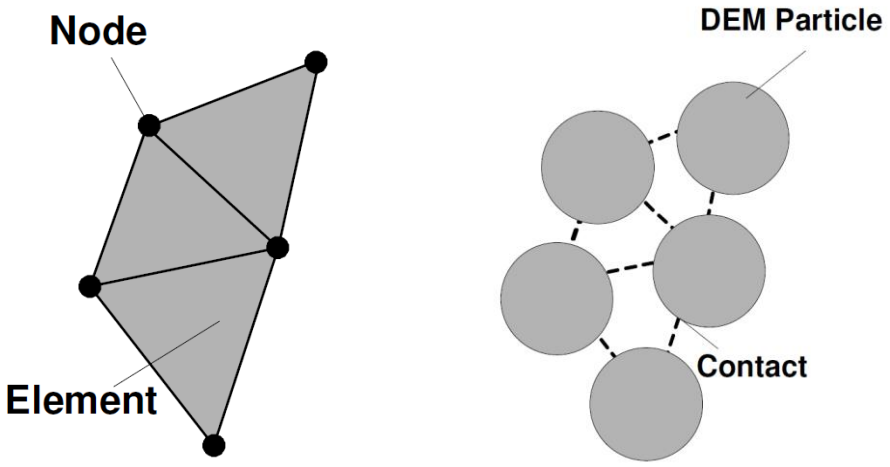
$$\boldsymbol{\sigma} = \frac{1}{V}\sum_{N_c} \boldsymbol{b} \otimes \boldsymbol{f} \tag{3.21}$$

where V is the volume of the periodic cell, N_c is the number of inter-particle contacts, $\boldsymbol{b}$ is the branch vector, and $\boldsymbol{f}$ is the contact force vector. The strain tensor is obtained from the displacement of boundaries ($\boldsymbol{u}$) as:

$$\boldsymbol{\epsilon} = \frac{1}{2}\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^{\mathrm{T}}\right) \tag{3.22}$$

3.1.4 Time step

The central difference time integration scheme is conditionally stable, to ensure the numerical stability a critical value for the timestep should be defined. In DEM, the critical timestep is calculated based on a simple analogy between DEM and finite element method (FEM). As shown in Figure 3.3, the particles are assumed to be analogous to the nodes of FEM and the inter-particle contacts to finite elements. This analogy allows the critical timestep to be determined as $\Delta t_{crit} = 2/\omega_{max}$, where ω_{max} is the maximum frequency and the highest eigenvalue of the $\boldsymbol{M}^{-1}\boldsymbol{K}$ matrix of the system. For more details of $\boldsymbol{M}^{-1}\boldsymbol{K}$ determination for a DEM system, refer to O'Sullivan (2011). Note that critical timestep determined by this method is possible only when the system has inter-particle contacts, but in a liquefied state, all the contacts are lost. A relation for critical timestep which could relate contact stiffness and particle's properties is required such that it can be applicable for any system.



(a) FEM elements (b) DEM contacts

Figure 3.3 Analogy between a DEM model and a FEM mesh (after O'Sullivan 2011)

An alternative approach to calculate the critical timestep is by wave propagation speed. If DEM particles are assumed to be continuous and having continuum like quantities such as Young's modulus, E, and density, ρ, then the wave propagation speed is defined as,

$$v = \sqrt{\frac{E}{\rho}} \tag{3.23}$$

To calculate the critical time step, it assumed that the wave must not travel farther than the minimum radius of the particle i.e.

$$\Delta t_{crit} = r_{min} \sqrt{\frac{\rho}{E}} \tag{3.24}$$

It is important to note that the critical time step is dependent on the density and Young's modulus of the particle. Higher the density or lower the Young's modulus, higher is the critical timestep.

3.1.5 Model setup

As mentioned earlier, an open-source DEM code, Yade, is used to conduct numerical simulations to simulate undrained cyclic shear tests of granular soils. A three-dimensional representative volume element (RVE) is generated, as shown in Figure 3.4. A total number of 10000 spheres are used to generate the granular assembly. The chosen size of the RVE is large enough and is sufficient to maintain the microstructure information. The response of an RVE for a given loading is representative of the material in general. It is also important to make sure the size of the RVE is small enough to prevent internal localization (Drugan and Willis 1996).

The general sample preparation method used in the current study is acquired from Makse et al. (1999). In this method, all the particles are randomly generated in space with no initial contacts in a 3D cubic RVE. Periodic boundaries are used as boundary conditions. Using an isotropic strain gradient tensor as in Eq. 3.18 the RVE is squeezed, where particles come in contact with each other, and the confining pressure starts to develop. To generate specimens with different void ratios, during the consolidation phase, different friction angles (ϕ) are used. In the consolidation phase, using $\phi = 0$ densest sample with minimum void ratio (e_{min}) can be prepared. Similarly, if $\phi = \phi_{cyc}$ is used, loosest specimen with maximum void ratio can be prepared. Where ϕ_{cyc} is the friction angle used for cyclic shearing which is assumed to be about 26° in this study. Note that the ϕ used for sample preparation should be less than ϕ_{cyc} in

order to prevent sample stability for cyclic loadings. The consolidation process is stopped when the confining pressure of 100 kPa is reached, and the particles inside the granular assembly are at equilibrium. For the consolidation phase, the friction angle is adjusted to ϕ_{cyc} for later cyclic shearing. To carry out cyclic loading in undrained conditions, the volume of the RVE is maintained constant. Note the gravity is set to be zero and the simulation tests are all carried out in quasi-static conditions with very low strain rate.

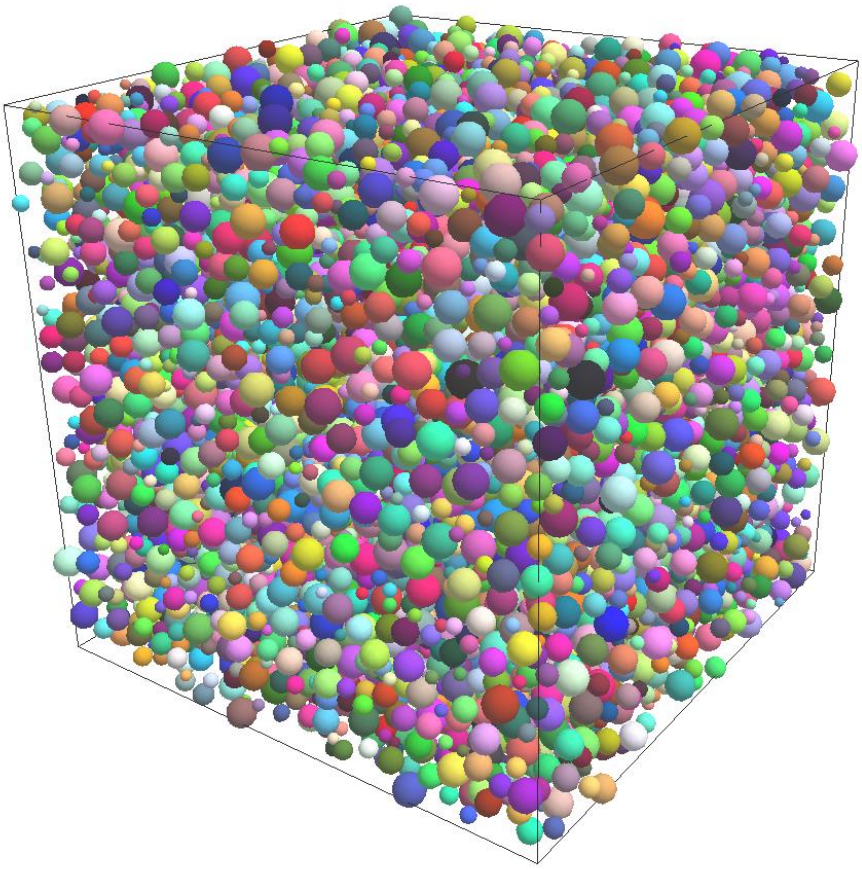


Figure 3.4 3D-DEM granular packing

3.2 Contact-based fabric quantification

Contact-based fabric indicators have been extensively used in many studies to quantify the load-bearing capacity of the soil for different testing conditions (e.g., Guo and Zhao 2013; Kuhn 1999; Ouadfel and Rothenburg 2001; Rothenburg and Bathurst 1989; Thornton 2000).

In this thesis, the coordination number and contact-based fabric tensors are used to understand the internal fabric.

3.2.1 Coordination number

Coordination Number (Z) measures the average number of inter-particle contacts for each particle in the granular assembly and is defined as,

$$Z = \frac{2N_c}{N_p} \tag{3.25}$$

where N_c is the total number of contacts and N_p is the total number of particles.

3.2.2 Contact-based fabric tensor

Fabric tensor is the directional indicator and used to quantify the distribution of numerous space vector. Contact-based fabric tensor ($\boldsymbol{\phi}$) measures the distribution of contact normal vectors ($\boldsymbol{n}$) in an RVE. Following the method proposed by Oda (1982), $\boldsymbol{\phi}$ can be defined as follows:

$$\phi_{ij} = \frac{1}{N_c}\sum_{k=1}^{N_c} n_i^{(k)} n_j^{(k)} = \oint_{\Omega} E(\Omega) n_i n_j d\Omega \tag{3.26}$$

where N_c is the total number of inter-particle contacts and $E(\Omega)$ is the angular distribution function of the normal vectors, such that $\int E_1(\Omega)\, d\Omega = 1$. Following (Sitharam et al. 2009), $E_1(\Omega)$ can be written as follows:

$$E_1(\Omega) = \frac{1}{4\pi}\left(1 + c_{ij} n_i n_j\right) \tag{3.27}$$

$$c_{ij} = \frac{15}{2}\left(\phi'_{ij}\right), \qquad where\ \phi'_{ij} = \phi_{ij} - \left(\frac{\phi_{ii}}{3}\right) \tag{3.28}$$

The following invariant of c_{ij} can be used to quantify the anisotropy degree (a_c) of the fabric tensor and is given by:

$$a_c = \sqrt{\frac{3}{2}(c_{ij} c_{ij})} \tag{3.29}$$

3.3 New fabric descriptors for particle-void distribution

Fabric quantification for liquefied soils is particularly challenging because, at liquefied state, soil loses all its contact and tend to flow like a fluid. Thus, traditional contact-based fabrics depict little to no information about the internal soil fabric at the liquefied state and may not be reliable. Alternatively, a special type of particle-void fabric descriptors was introduced by Wei et al. (2018) for a 2D granular assembly by constructing "particle-void cells" around particles using weighted Voronoi tessellation. No contact information is needed in the construction of these cells; hence, these descriptors were even defined for a liquefied state. In this thesis, the particle-void fabrics introduced by Wei et al. (2018) for 2D assembly is extended to 3D granular assemblies. The flow for the computation of these fabrics goes as follows; first the particle-void cells of 3D granular assembly are computed using Voronoi tessellation, and later the anisotropy of the resulting particle-void cells are computed with the help of Minkowski tensors. Finally, the statistical analysis on the local anisotropy indices from Minkowski tensors gives the final particle void descriptors.

3.3.1 Voronoi tessellation

Frictional granular particles when disorderly packed, they get trapped in a metastable configuration (Pouliquen et al. 2003). When a shear strain is applied on the granular assembly, particles evolves slowly and attains new fabric configurations after every strain increment. However, at grain level, the particle movement for the next strain increment is restricted only to the void space available around it. This void space can be calculated using a popular mathematical technique called Voronoi Tessellation. The Voronoi tessellation for a set of points in a given space is the partitioning of that space into regions such that all locations within any one region are closer to the generating point than to any other points in space. For a regular Voronoi tessellation the partition of the space in between two points is done by drawing a convex plane that bisects the distance between two points. Likewise, the whole space can be divided using these convex planes, w.r.t the points resulting in the formation of convex polyhedron regions generally called Voronoi cells. Such tessellation can be used to divide the space for a mono-dispersed granular assembly.

In general, soils are polydisperse. For a poly-dispersed granular packing weighted Voronoi tessellation can be used to partition the void space around the particle. Compared with the regular Voronoi tessellation instead of bisecting planes, the weighted Voronoi tessellation uses radical planes, hence advancing its applicability for polydisperse assembly of spheres. Radical planes can be generated using the radius of particles as a weight when dividing the void space, such that a single particle will be entirely encased by a polygonal Voronoi cell formed from radical planes, as shown in Figure 3.5. Note that if the regular Voronoi tessellation is used on a polydisperse system, the Voronoi cell may undesirably crosscut particles and might not capture the true particle-void space. Mathematically, the particle and its surrounding void space can be described as $\{P^i, R^i, [V_1^i, V_2^i, \dots V_k^i]\}$, where P^i is the center of the particle 'i', R^i is the radius of the particle 'i', and $[V_1^i, V_2^i, \dots V_k^i]$ are vertexes of the Voronoi cell. As the construction of the Voronoi cells solely depends on the particle positions and radiuses, it can well be used to study the particle void space distribution in liquefied granular assemblies.

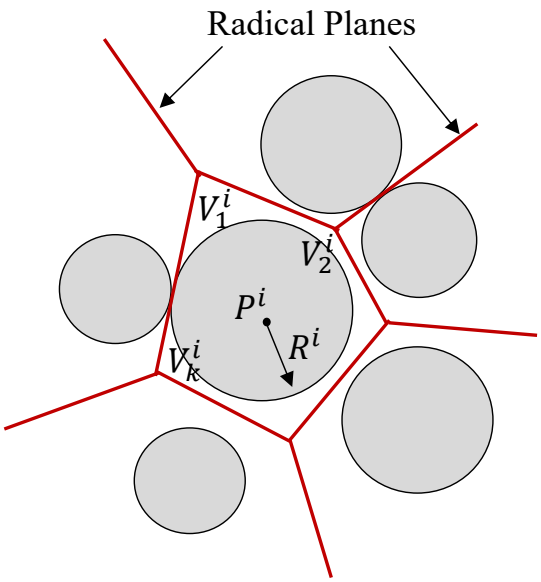


Figure 3.5 Voronoi tessellation

In this thesis, to compute Voronoi tessellations, an open-source software library, Voro++ (Rycroft 2009) is used. Figure 3.6 shows the Voronoi tessellation of a polydisperse packing with periodic boundaries. It is important to note that the number of convex Voronoi cells (N_c) quantified would be same as the number of particles (N_p) in the granular assembly, i.e., $N_c = N_p$.

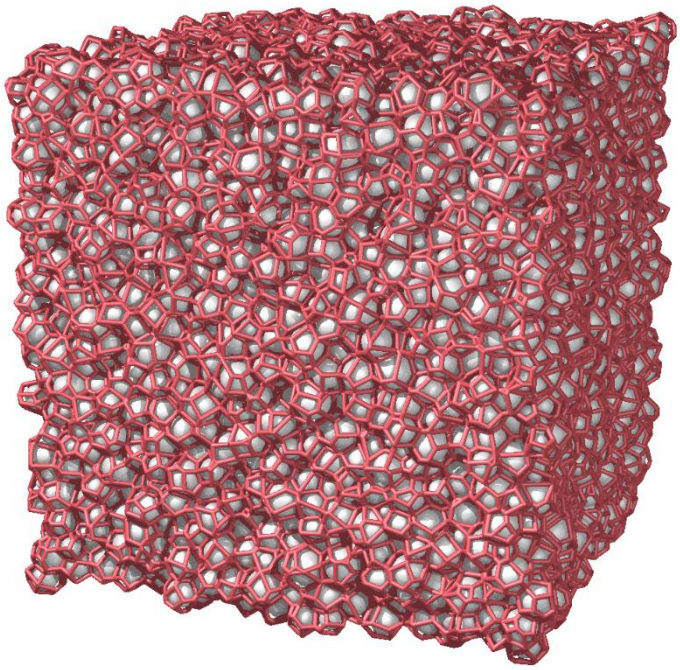


Figure 3.6 Voronoi tessellation on a 3D granular packing

3.3.2 Minkowski tensors

The shape anisotropy of a convex body such as Voronoi cells can be quantified using Minkowski tensors. Minkowski tensors are the family of six independent tensorial shape measures of a convex body, K and are developed in the framework of integral geometry. The eigenvalues of these tensors represent an ellipsoid in space, and the ratio between the minimum and the maximum eigenvalues quantifies the degree of anisotropy of the body K corresponding to the Minkowski tensor (Schröder-Turk et al. 2010). In a Euclidean 3D space for a convex body, K and with bounding surface S the six independent tensors are denoted as $\boldsymbol{W}_0^{20}, \boldsymbol{W}_1^{20}, \boldsymbol{W}_2^{20}, \boldsymbol{W}_3^{20}, \boldsymbol{W}_1^{02}$ and $\boldsymbol{W}_2^{02}$ and are computed as follows,

$$\boldsymbol{W}_0^{20} = \int_K \boldsymbol{r} \otimes \boldsymbol{r}\, dV \tag{3.30}$$

$$\boldsymbol{W}_1^{20} = \frac{1}{3}\int_K \boldsymbol{r} \otimes \boldsymbol{r}\, dA \tag{3.31}$$

$$\boldsymbol{W}_2^{20} = \frac{1}{3}\int_K H(\boldsymbol{r})\, \boldsymbol{r} \otimes \boldsymbol{r}\, dA \tag{3.32}$$

$$\boldsymbol{W}_3^{20} = \frac{1}{3}\int_{\partial K} G(\boldsymbol{r})\, \boldsymbol{r} \otimes \boldsymbol{r}\, dA \tag{3.33}$$

$$\boldsymbol{W}_1^{02} = \frac{1}{3}\int_{\partial K} \boldsymbol{n} \otimes \boldsymbol{n}\, dA \tag{3.34}$$

$$\boldsymbol{W}_2^{02} = \frac{1}{3}\int_{S} H(\boldsymbol{r})\, \boldsymbol{n} \otimes \boldsymbol{n}\, dA \tag{3.35}$$

where $\boldsymbol{r}$ is the position vector, and $\boldsymbol{n}$ is the bounding surface normal. The vector product $\otimes$ is defined as the dyadic product, $\boldsymbol{a} \otimes \boldsymbol{a} = a_i a_j$, which is a second-order symmetric tensor. $H(\boldsymbol{r})$ is the mean curvature, and $G(\boldsymbol{r})$ is the Gaussian curvature of the surface, ∂K at position $\boldsymbol{r}$**.** The list of these six tensors are linearly independent and corresponds to a unique shape measure of the convex body, K.

Schröder-Turk et al. (2011) provided a clear picture of the morphological features that these tensors capture and is shown in Figure 3.7. The first four tensors, i.e., $\boldsymbol{W}_0^{20}, \boldsymbol{W}_1^{20}, \boldsymbol{W}_2^{20}$ and $\boldsymbol{W}_3^{20}$ are called moment tenors because they closely resemble the most common moment of inertia tensor, $\boldsymbol{I}$. For a continuous 3D body with unit density, the moment of inertia tensor can be expressed as $\boldsymbol{I} = \int(|\boldsymbol{r}|^2\, \boldsymbol{E_3} - \boldsymbol{r} \otimes \boldsymbol{r})dV$, where $\boldsymbol{E_3}$ is the identity matrix. The moment tensor, $\boldsymbol{W}_0^{20}$ can be interpreted from the moment of inertia tensor as $\boldsymbol{W}_0^{20} = \frac{tr(\boldsymbol{I})}{2} - \boldsymbol{I}$**,** where tr is the trace of a matrix. From the definition, $\boldsymbol{W}_0^{20}$ is a tensorial measure of the volume mass distribution of the convex body, K around a reference point commonly the centroid. Similarly, $\boldsymbol{W}_1^{20}$ is the moment tensor for measuring the surface mass distribution of the boundary ∂K of K. $\boldsymbol{W}_2^{20}$ and $\boldsymbol{W}_3^{20}$ are the moment tensors for the wireframe of K, assumed with mass situated at vertices but to allow additivity of the tensors, the mass is distributed on to the surface according to the mean curvature and Gaussian curvature respectively. It is important to note that the above four moment tensors depend on the choice of origin at which they are computed. The last two tensors, i.e., $\boldsymbol{W}_1^{02}$ and $\boldsymbol{W}_2^{02}$ are the measure of the orientational distribution of the surfaces and the curvatures of the convex body respectively. Thus, these tensors do not depend on the choice of origin.

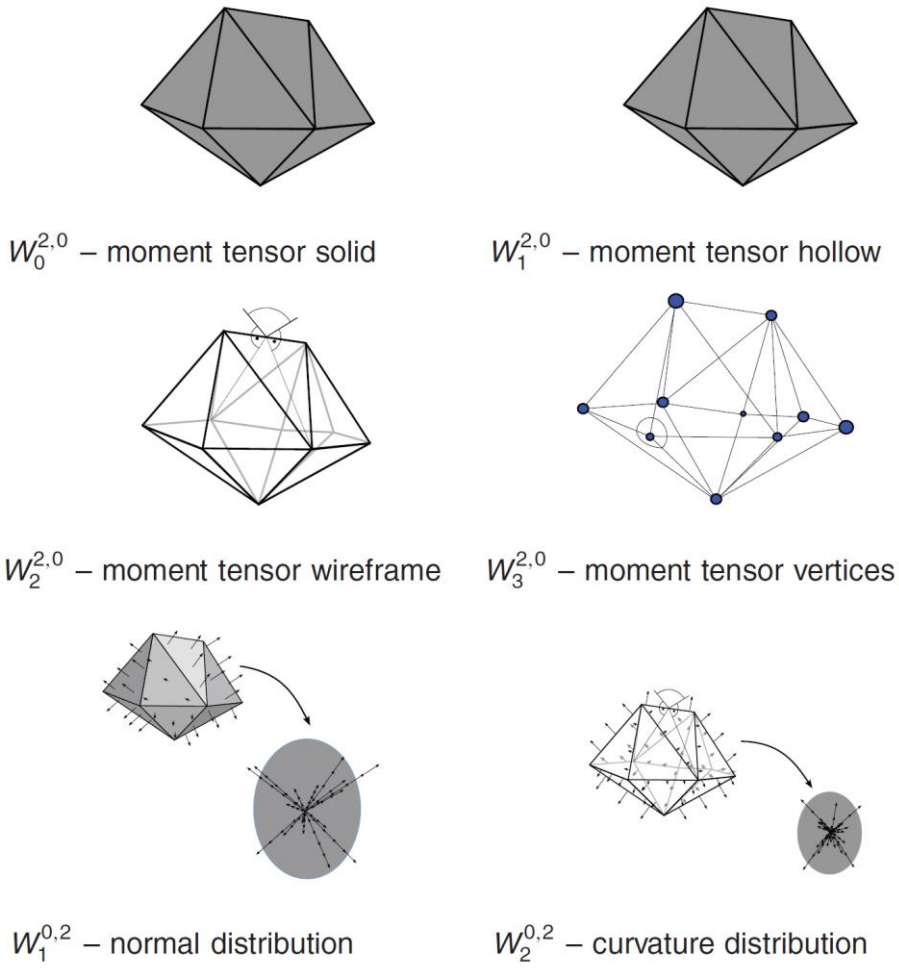


Figure 3.7 Definitions of different Minkowski tensors (after Schröder-Turk et al. 2011)

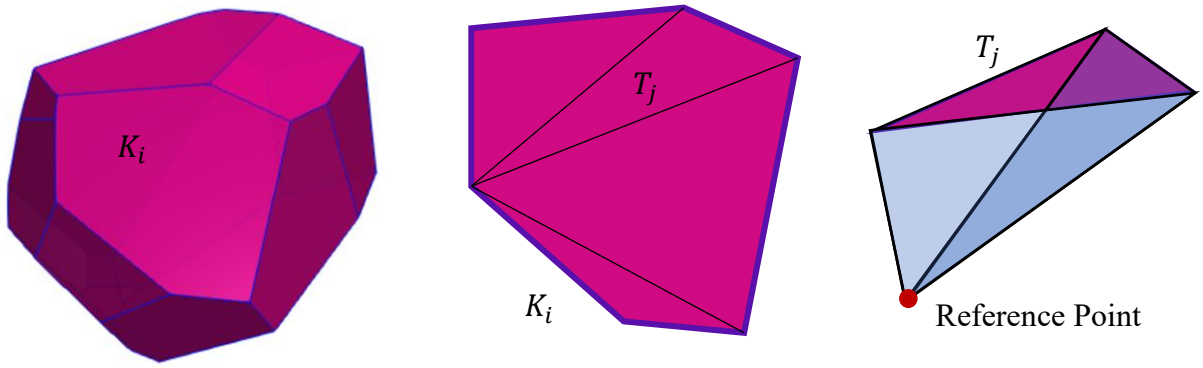
Karambola (Schröder-Turk et al. 2010) is an open-source code written in C++ to compute Minkowski tensors for three-dimensional bodies. This code requires to input the essential data that define a convex polygon, i.e., the coordinates of vertices forming a face and numbering of faces in a special file with ". poly" format. Karambola can be easily used to compute the Minkowski tensors for a couple of Voronoi cells, but when attempted to compute for about 10,000 Voronoi cells, it gets computationally very costly. Thus, a new code is required to handle more number of Voronoi cells effectively.

In the current thesis, the anisotropy of the Voronoi cells are computed using the moment tensor, $\boldsymbol{W}_0^{20}$. A python-based code is written for the computation of Minkowski tensor $\boldsymbol{W}_0^{20}$ which uses parallelization technique, thus decreasing the computational time significantly. It is important to note that the computation of Minkowski tensors of different Voronoi cells is

mutually independent. Hence, they can be easily parallelized on to the different number of computer cores for computation. The algorithm of the python code is inspired from the Karambola and is briefly discussed in the following sub-section.

3.3.2.1 Computation of moment tensor, $\boldsymbol{W}_0^{20}$

The algorithm to calculate $\boldsymbol{W}_0^{20}$ tensor for a convex polyhedron contains 4 main steps: 1) Triangulation of the faces of convex polyhedron 2) Generation of tetrahedrons with respect to reference point 3) Calculation of moment of inertia of the tetrahedron and 4) Computation of $\boldsymbol{W}_0^{20}$.



(a) Convex polyhedron (b) Polygonal face (c) Tetrahedron

Figure 3.8 Computation steps of $\boldsymbol{W}_0^{20}$Minkowski tensor

Figure 3.8(a) shows a general convex polyhedron with convex polygons as its faces. Each polygon is defined by its corresponding vertices and can be easily triangulated. The first vertex is connected to every other vertex of the polygon so that 'n-2' triangles are formed for a polygon with 'n' number of vertices. Likewise, every face of the polyhedron is triangulated. An illustration for triangulation on a random face of a convex polyhedron is shown in Figure 3.8(b).

As discussed in the previous section, $\boldsymbol{W}_0^{20}$ depends on the choice of reference point. The reference point can be anywhere in space about which the tensor $\boldsymbol{W}_0^{20}$ is computed. With a known reference point, the convex polyhedron can be divided into multiple tetrahedrons by joining the reference point to the triangulated faces, as shown in Figure 3.8(c). If the reference point lies inside the polyhedron, the sum of volumes of all the tetrahedrons results to the total

volume of a convex polyhedron. The tensor $\boldsymbol{W}_0^{20}$ of the convex polyhedron about the reference point can be computed from the total moment of inertia tensor, $\boldsymbol{I}$ at the reference point. The total moment of inertia tensor ($\boldsymbol{I}$) is equal to the sum of moment of inertias of individual tetrahedrons at the reference point. Using the formulation proposed by Tonon (2004), the moment of inertia of individual tetrahedrons with known vertices can be easily computed. Suppose, the vertices of a particular triangle 'k' on the triangulated face of a convex polyhedron are $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$. Let the reference point chosen to be the origin $O(0,0,0)$. The moment of inertia tensor, $\boldsymbol{I}_k$ for the tetrahedron '$ABCO$' about 'O' is given as,

$$\boldsymbol{I}_k = \begin{bmatrix} a & -d & -e \\ -d & b & -f \\ -e & -f & c \end{bmatrix} \tag{3.36}$$

where $a, b, c, d, e,$ and f are calculated as follows:

$$a = L.\mu.(y_1^2 + y_2^2 + y_3^2 + y_1y_2 + y_1y_3 + y_2y_3 + z_1^2 + z_2^2 + z_3^2 + z_1z_2 + z_1z_3 + z_2z_3)/60 \tag{3.37}$$

$$b = L.\mu.(x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_1x_3 + x_2x_3 + z_1^2 + z_2^2 + z_3^2 + z_1z_2 + z_1z_3 + z_2z_3)/60 \tag{3.38}$$

$$c = L.\mu.(x_1^2 + x_2^2 + x_3^2 + x_1x_2 + x_1x_3 + x_2x_3 + y_1^2 + y_2^2 + y_3^2 + y_1y_2 + y_1y_3 + y_2y_3)/60 \tag{3.39}$$

$$d = L.\mu.(2x_1z_1 + x_1z_2 + x_1z_3 + x_2z_1 + 2x_2z_2 + x_2z_3 + x_3z_1 + x_3z_2 + 2x_3z_3)/120 \tag{3.40}$$

$$e = L.\mu.(2x_1y_1 + x_1y_2 + x_1y_3 + x_2y_1 + 2x_2y_2 + x_2y_3 + x_3y_1 + x_3y_2 + 2x_3y_3)/120 \tag{3.41}$$

$$f = L.\mu.(2y_1z_1 + y_1z_2 + y_1z_3 + y_2z_1 + 2y_2z_2 + y_2z_3 + y_3z_1 + y_3z_2 + 2y_3z_3)/120 \tag{3.42}$$

where μ is the density of the medium (is considered equal to 1), and L is given as,

$$L = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} \tag{3.43}$$

Thus, the total moment of inertia ($\boldsymbol{I}$) is given by the summation of the moment of inertias of individual tetrahedrons about point 'O' and is mathematically represented as,

$$\boldsymbol{I} = \sum_{k=1}^{n} \boldsymbol{I}_k \tag{3.44}$$

where 'n' is the total number of triangles after triangulation on every face. Note that the formulation is defined for the reference point at the origin. If other than origin is used as a reference point, a local coordinate system needs to be defined where the local origin is fixed at the reference point, and the relative coordinates for the vertices of the triangle are used for the computation.

Finally, the moment tensor $\boldsymbol{W}_0^{20}$ is defined from the moment of inertia tensor as

$$\boldsymbol{W}_0^{20} = \frac{tr(\boldsymbol{I})}{2}\boldsymbol{E}_3 - \boldsymbol{I} \tag{3.45}$$

where $\boldsymbol{E}_3$ is the identity matrix in 3D Euclidean space and $tr()$ is the trace function on a matrix.

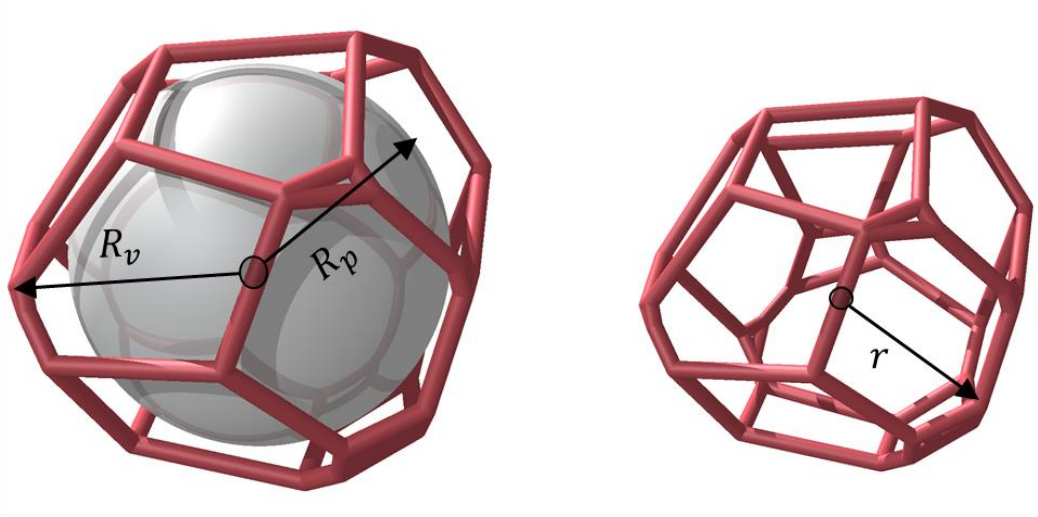
3.3.3 Definition of new particle-void fabric descriptors

3.3.3.1 Particle-void cell

A random Voronoi cell with the particle inside is chosen from the Voronoi tessellation (refer Figure 3.6) is shown in Figure 3.9(a). The potential space for the particle to displace upon loading is the void space contained in the Voronoi cell. To characterize the local void distribution of this void space, the Cell-to-Particle ratio (CP ratio) is defined. If $R_v(\theta, \phi)$ is the radial measure of the Voronoi cell from the particle center and $R_p(\theta, \phi)$ is the radial measure of the particle, then the Cell-to-Particle ratio (CP ratio) is given as follows:

$$r(\theta, \phi) = \frac{R_v(\theta, \phi)}{R_p(\theta, \phi)} \tag{3.46}$$

where $\theta \in (0,2\pi)$ and $\phi \in (0, \pi)$. Thus, $r(\theta, \phi)$ directly measures the local void distribution around the particle. As R_v is the radial measure of the convex Voronoi cell which varies with (θ, ϕ) and R_p is the radius of the spherical particle which is constant for all (θ, ϕ). Thus, $r(\theta, \phi)$ can be assumed as the radial measure of a special convex cell, which purely contains local void distribution information. This special convex cell is defined as particle-void cell and is shown in Figure 3.9(b).



(a) Voronoi cell and Particle (b) Particle-void cell

Figure 3.9 Definition of particle-void cell

3.3.3.2 Local anisotropy of the particle-void cell

The anisotropy of the particle-void cell is be quantified using the Minkowski tensor, $\boldsymbol{W}_0^{20}$. As discussed in the previous section, $\boldsymbol{W}_0^{20}$ is the tensorial shape measure for the volume mass distribution of a homogeneous body. Thus $\boldsymbol{W}_0^{20}$ of a particle-void cell gives the tensorial measure of the void-distribution around the particle center and is computed as follows:

$$\boldsymbol{W}_0^{20} = \int_{K_v} \boldsymbol{r} \otimes \boldsymbol{r}\, dV \tag{3.47}$$

Essentially, the eigenvalues and the eigenvectors of $\boldsymbol{W}_0^{20}$ tensor provide information about best-fitting ellipsoid into the particle-void cell, as shown in Figure 3.10. If λ_{min} and λ_{max} are the minimum and maximum eigenvalues, then the local shape anisotropy (e_d) of the particle-void cell can be defined as follows:

$$e_d = 1 - \frac{\lambda_{min}}{\lambda_{max}} \tag{3.48}$$

If $e_d = 1$, it represents a highly anisotropic shape and $e_d = 0$ indicates an isotropic shape (i.e., spherical) of the particle-void cell. The unit eigenvector, $\boldsymbol{v}$, corresponding to the maximum eigenvalue, provides the orientation of the particle-void cell.

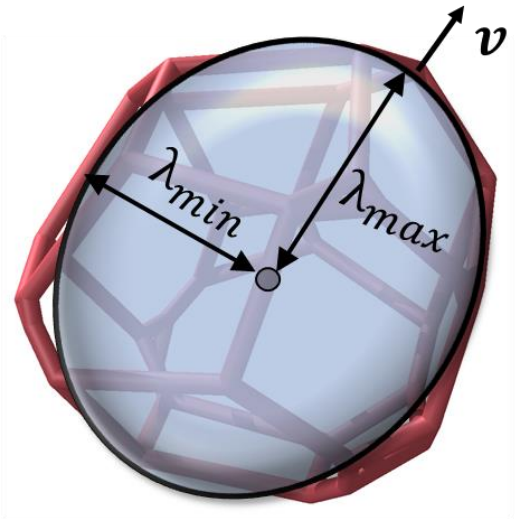


Figure 3.10 Ellipsoidal fitting to a particle-void cell

3.3.3.3 Quantification of particle-void descriptors (E_d,A_d)

The e_d and $\boldsymbol{v}$ described in the above sub-section can quantify the local void anisotropy of single particle-void cell. Where e_d measures the shape elongation and $\boldsymbol{v}$ provides the principal orientation of the particle-void cell. By statistical analysis of $\{e_d, \boldsymbol{v}\}$ associated with all the particle-void cells in the granular assembly, the anisotropy of particle-void distribution for the entire packing can be quantified.

The first descriptor, E_d, is the global shape (elongation) descriptor defined as the mean of $\{e_d\}$ for all the N_v Voronoi cells, mathematically represented as:

$$E_d = \frac{1}{N_v}\sum_{i=1}^{N_v} e_d^{(i)} \tag{3.49}$$

Ideally, the probability density function of e_d should provide an idea for distribution of local shape factor but interestingly the density function e_d/E_d follows a gamma distribution and is the same for different samples at various loading stages. The detailed explanation would be

provided in the next chapter while discussing the evolution of the fabric. Hence, E_d can solely be used as a descriptor for global shape factor.

The second descriptor, A_d, is a global descriptor for anisotropy of particle-void orientation, derived from the unit eigenvectors $\{\boldsymbol{v}\}$. First, a particle-void fabric tensor (V_{ij}) is defined as:

$$V_{ij} = \frac{1}{N_v}\sum_{k=1}^{N_v} v_i^{(k)} v_j^{(k)} = \oint_{\Omega} F(\Omega) v_i v_j d\Omega \tag{3.50}$$

where $E(\Omega)$ is the angular distribution of particle-voids, such that $\oint_{\Omega} E(\Omega) d\Omega = 1$. It is important to note that, each void orientation can be represented as a pair of unit vectors with opposite directions thus making the function $F(\Omega)$ symmetric with respect to the origin. Using Fourier series expansion and ignoring higher-order terms, $F(\Omega)$ can be written as:

$$F(\Omega) = \frac{1}{4\pi}(D + D_{ij} v_i v_j) \tag{3.51}$$

where D and D_{ij} are deviator tensors of order zero and two respectively. Following (Ken-Ichi 1984), they are given as:

$$D = 1 \tag{3.52}$$

$$D_{ij} = \frac{15}{2}\left[V_{ij} - \frac{1}{3}\delta_{ij}\right] \tag{3.53}$$

We use the following invariant of D_{ij} to quantify the anisotropy degree of the fabric tensor,

$$A_d = \sqrt{\frac{3}{2}(D_{ij} D_{ij})} \tag{3.54}$$

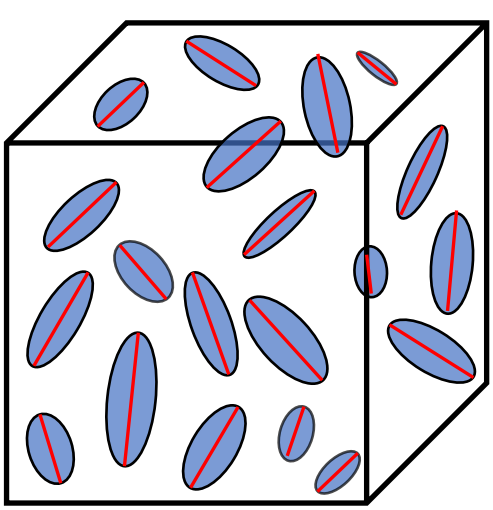
Thus, global shape elongation descriptor E_d and global anisotropy descriptor A_d are adopted to quantify the particle-void fabric evolution.

3.3.4 Physical meaning of E_d and A_d

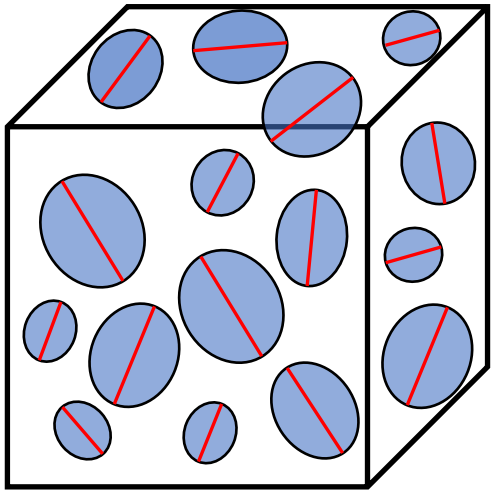
The particle-void cells corresponding to each particle in the granular assembly has a local shape elongation and local orientation. From the statistical analysis on the local particle-void descriptors, a set of global descriptors E_d and A_d are defined in the above section. E_d and A_d,

can physically be understood from Figure 3.11. Granular packing with large E_d and small A_d as shown in Figure 3.11(a) represents that the local particle-voids are highly anisotropic (elongated) in shape but are oriented in different directions in space. Figure 3.11(b) shows a packing with low E_d and low A_d where the particle-void cells are almost isotropic in both local shape and void-orientations. Figure 3.11(c) and (d) shows the packing instances at which A_d is large. It resembles most of the particle-void cells has a preferred orientation in one direction and making the assembly highly anisotropic in terms of particle-void orientations. Note when the E_d value is very low, i.e., local particle-void cells are almost spherical thus principal orientation can be any direction in space making anisotropy descriptor A_d of no significance.

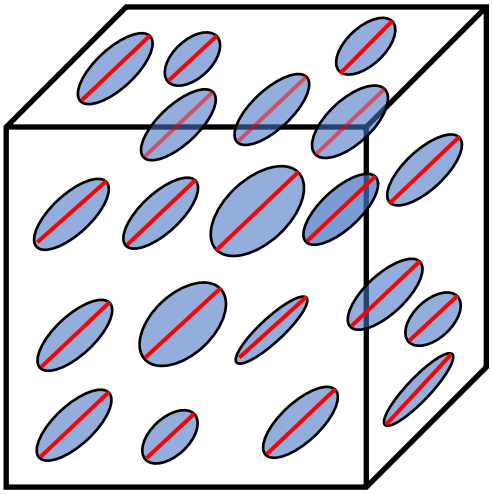
The global descriptor, E_d also signifies the relative density of the granular assembly. It is observed from numerical simulations that the denser packings have lower E_d values and correlation seem to be linear. Similar observations are also discussed in Turk et al. (Schröder-Turk et al. 2010).



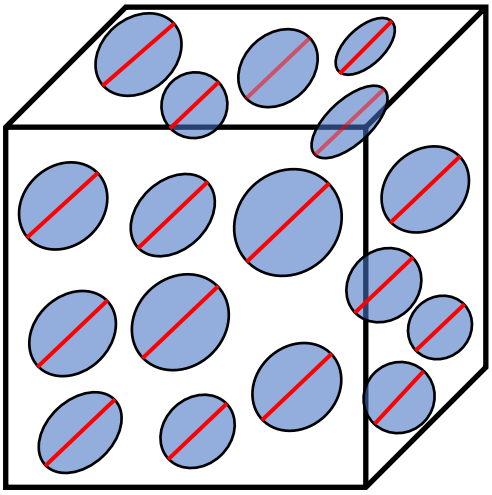
(a) High E_d and low A_d



(b) low E_d and low A_d



(c) High E_d and high A_d



(d) low E_d and high A_d

Figure 3.11 Physical meaning of E_d and A_d

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CHAPTER 4

Fabric Evolution in Liquefaction and Re-liquefaction of Granular Soils

Undrained cyclic simple shear response of a medium to dense sands includes different features such as the decrease in effective stress in the pre-liquefaction stage, change of response from contractive to dilative near initial liquefaction, and shear-strain accumulation in the post-liquefaction stage. Even the robust constitutive models in the literature fail to capture all these features accurately. It is essential to get a microscopic understanding of the above macroscopic features which can facilitate physical quantities that could refine the current constitutive models for better predictions.

In this chapter, different features of undrained cyclic shear response are examined from the microscopic level. The evolution in contact-based fabric descriptors and the newly defined particle-void based fabric descriptors are found to have clear correlations with different features in the undrained cyclic shear response. It is found that the variation in stress-path is due to evolution in contact fabric and variation in stress-strain is due to evolution in particle-void fabric. A unique Hardening State Line (HSL) is observed in the void-fabric space that delineates the flow state and the hardening state in the post-liquefaction stage. The shear strain accumulation is found to have a clear correlation with the evolution of void fabric in the post-liquefaction stage. Also, the compressibility of samples in the post-liquefaction stage has a clear correlation with the particle-void fabrics developed. DEM simulations have also been conducted to identify critical factors that influence the granular soil's resistance to multiple

liquefaction. It is observed that completely different soil fabric will be formed if the liquefied soil is consolidated at different states in the post-liquefaction stage, resulting in vastly different resistance to re-liquefaction.

4.1 Sample preparation and loading conditions

Numerical simulations are conducted using an open-source DEM code, Yade (Šmilauer et al. 2011). A three-dimensional (3D) cubic representative volume element (RVE) consisting of 10000 spheres is generated. The radius of the particles ranges from 0.15 mm to 0.45 mm with the mean radius (R_{50}) being 0.3 mm. Contact interactions are calculated using the Hertz-Mindlin contact model with Young's modulus of 70 GPa and Poisson's ratio of 0.3 assigned to all the particles. Rolling resistance proposed by Belheine et al. (2009) is used at the contact points of the spherical particles to account for surface roughness and particle shapes (Bagi and Kuhn 2004). The dimensionless coefficients for rolling resistance (β), and for plastic moment (η) as defined in Belheine et al. (2009) is set as 0.1 and 0.05 respectively in the simulation. Note that periodic boundary condition is prescribed to the RVE in order to minimize the boundary effects.

The sample preparation method discussed in section 3.1.5 of this thesis is used to prepare three samples Dr40, Dr50, and Dr60 with relative densities 40%, 50%, and 60% respectively. The e_{min} and e_{max} are found to be 0.52 and 0.82 at 100kPa confining pressure, respectively. After initial consolidation, ϕ is set as 26° for the later cyclic shearing. Samples are subjected to different cyclic simple shear loadings, i.e., to different Cyclic stress ratios (CSR= $\tau_x/\sigma'_{z,o}$) as shown in Table 4.1. Note that z-direction direction is considered as the vertical direction and cyclic shearing is applied in the x-direction of the granular assembly. Undrained conditions are maintained by keeping the volume of RVE constant. The rate of shear strain loading is maintained to be 0.01/s in order to ensure a quasi-static condition.

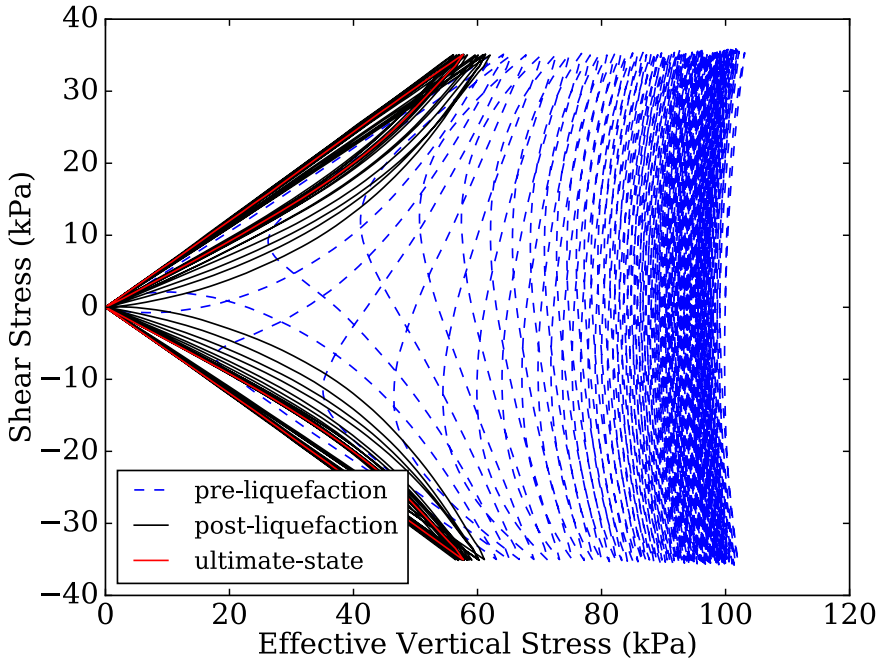
Table 4.1 Samples and respective loading conditions in DEM simulations

Sample	**Relative Density**	**CSR=$\tau_x/\sigma'_{z,o}$**
Dr40	40%	0.16, 0.18, 0.20
Dr50	50%	0.22, 0.25, 0.28
Dr60	60%	0.30, 0.35, 0.40

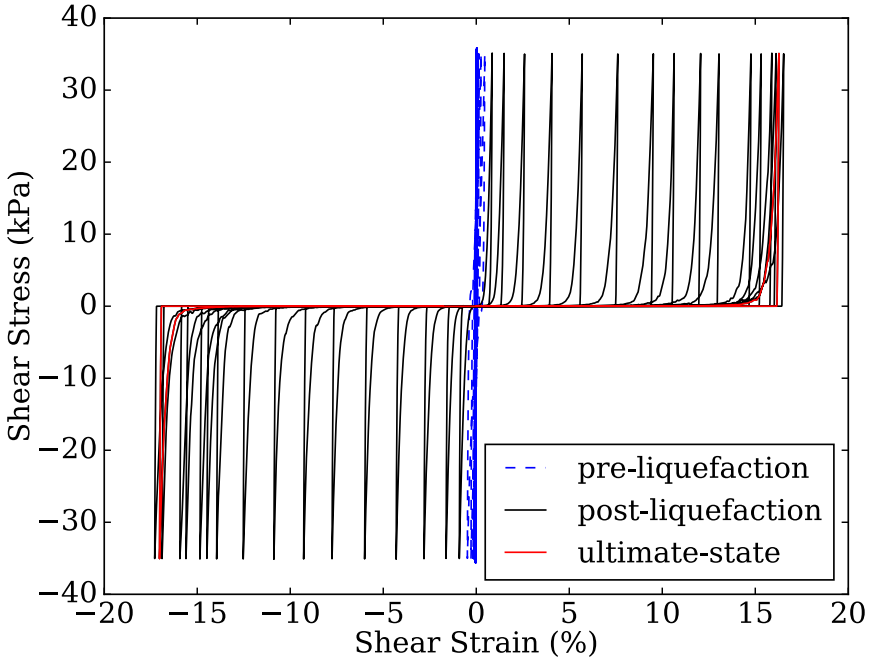
4.2 Cyclic liquefaction simulation using DEM

It is observed that DEM is capable to qualitatively capture the typical undrained cyclic simple shear response in all the loading cases. Figure 4.1 shows the undrained response of sample Dr60 for a CSR of 0.35. The soil behavior is divided into two stages, pre-liquefaction stage and post-liquefaction stage. Initial liquefaction separates the two stages and is defined as the point at which the effective vertical stress (σ_z') is zero or numerically when $\sigma_z' < 0.5$ kPa for the very first time.

In the pre-liquefaction stage, it is observed that upon repeated cyclic loading, the effective vertical stress gradually reduces and after a certain number of cycles it reaches to initial liquefaction, as shown in Figure 4.1a. During this stage, the shear strain does not evolve; instead, the behavior is similar in every cycle, as shown in Figure 4.1b. In other words, the load-bearing structure degrades with every cycle during pre-liquefaction without generating any considerable deformation. The number of cycles taken to reach initial liquefaction for different loading conditions on different samples is shown in Figure 4.2.



(a) Stress path



(b) Stress-strain

Figure 4.1 Macroscopic behavior of 3D granular assembly in cyclic liquefaction

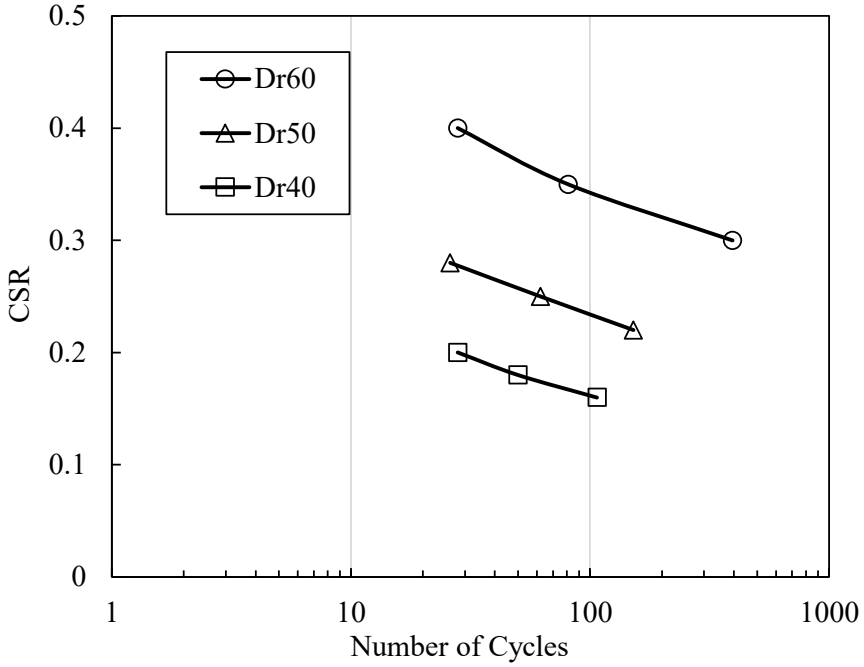


Figure 4.2 Liquefaction resistance of samples with different relative density

In the post-liquefaction stage, i.e., once after initial liquefaction, the stress paths follow similar butterfly loops every cycle while the shear strain amplitude increases cycle after cycle as in Figure 4.1(a,b). Within a loading cycle in the post-liquefaction stage, the granular assembly attains a flow state and the hardening state, accordingly, the shear deformation is divided into flow strain (γ_o) and hardening strain (γ_d) as shown in Figure 4.3. Flow strain is the double amplitude strain component where the shear stiffness is very low, and the effective stress is almost zero. Hardening strain is the strain component where both shear stiffness and effective stress have a sharp increase from zero. In other words, during the flow strain, the particle assembly behaves almost like a fluid whereas in hardening stage it behaves like a solid. It is worth noting that the hardening strain in each cycle is almost the same, whereas the flow strain increases gradually with every cycle and is responsible for the strain accumulation in the post-liquefaction stage. After few cycles in the post-liquefaction stage, the stress-strain response gets stabilized, i.e., no further accumulation in shear strain is observed with increase in the number of cycles and is referred to as ultimate state as shown in Figure 4.1(a,b). Qualitatively similar behavior was observed in the post-liquefaction stage for all the test cases. The maximum double amplitude shear strain at ultimate states are found to be dependent on the relative density of the sample but are independent of loading paths. For the three samples Dr40, Dr50, and Dr60 the maximum double amplitude strains at ultimate state are found as 54%,46%, and 34% respectively. Note that at ultimate state, both stress path and stress-strain response are stabilized and do not evolve further with increasing cycle number.

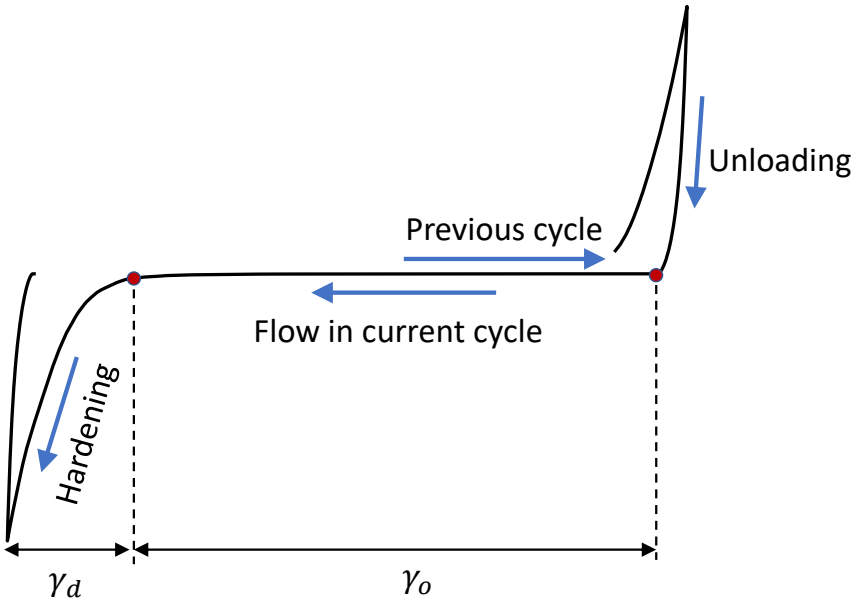


Figure 4.3 Typical stress-strain response in the post-liquefaction stage

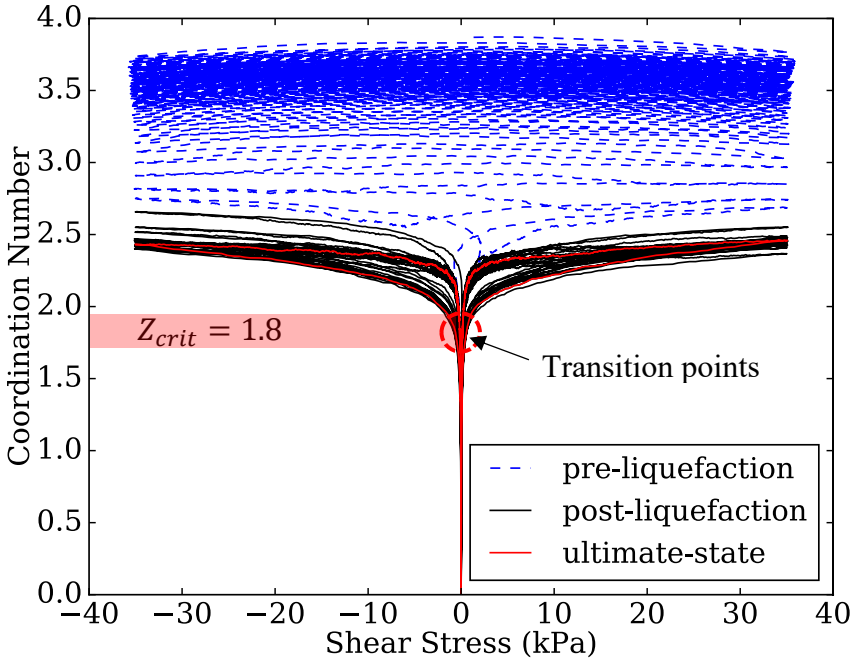
4.3 Evolution of contact-based fabric descriptors in cyclic liquefaction

In Section 3.2 of this thesis two fabric indicators, Z and a_c, are introduced that are based on particle-contact information. Z is the average coordination number and a_c is the degree of anisotropy in the orientation of contact normals. The evolution of these descriptors at different stages of cyclic liquefaction are discussed here.

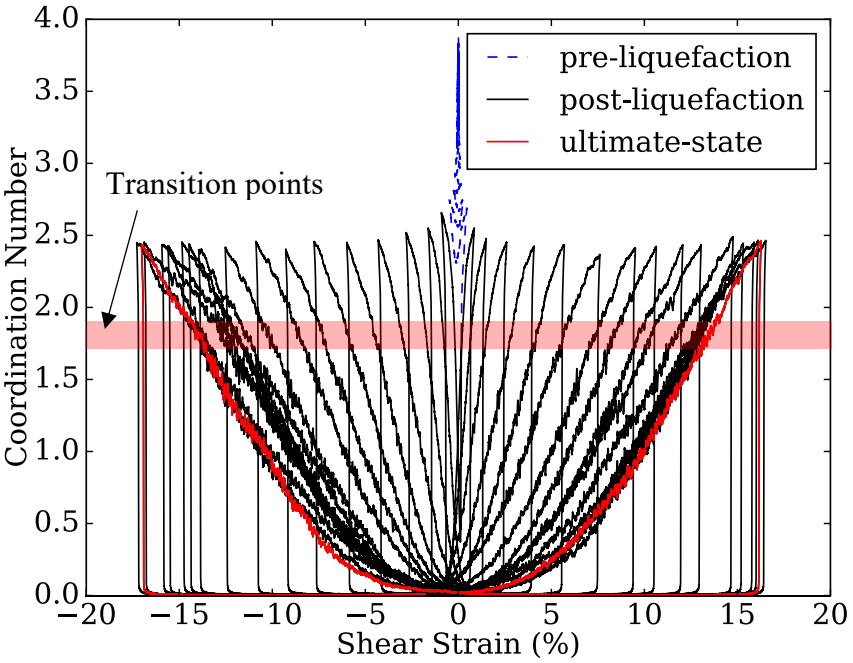
4.3.1 Evolution of coordination number (Z)

The evolution of the average coordination number (Z) during cyclic liquefaction of Dr60 sample for CSR 0.35 loading is shown in Figure 4.4(a,b). The initial sample before the cyclic loading has Z equals to about 3.8. In the pre-liquefaction stage, the coordination number gradually decreases with each loading cycle and reaches to about 1.8, after which the load-bearing capacity of granular assembly is entirely lost, and the coordination number immediately falls to 0, resulting in initial liquefaction condition.

In the post-liquefaction stage, the evolution of coordination number against shear stress is similar in every cycle of loading, as shown in Figure 4.4a. Starting from the flow state, with an increase in the amplitude of shear strain, the Z slowly evolves from 0 and reaches to about 2.4 by the end of the hardening stage. The transition from flow stage to hardening stage is considered if the shear stress of the packing is more significant than 0.1 kPa. It is observed that the transition from flow state to the hardening state always occurred at a range of $Z = (1.7 - 1.9)$, i.e. an average value of about 1.8 in every cycle of the post-liquefaction stage. Only after the transition point (i.e., Z =1.8), the granular assembly is able to resist the shear stress by developing a stable load-bearing structure. Note that the shear strain required to reach the transition point is different in different cycles of loading in the post-liquefaction stage, as shown in Figure 4.4b. It is interesting to observe that the transition point (i.e., Z=1.8) is independent of both the loading path and the relative density of the sample. For all the test cases, it is observed the transition from flow state to hardening state occurred at coordination number equals to 1.8. It is postulated that particle size distribution (PSD) can be a possible parameter that might affect the transition point, as it may change the jamming characteristics of the granular assembly. Thus, for a given PSD the transition in the post-liquefaction stage happens only after a critical coordination number (Z_{crit}). It is found, Z_{crit}=1.8 for this study.



(a) Coordination number Vs Shear stress



(b) Coordination number Vs Shear strain

Figure 4.4 Evolution of coordination number

4.3.2 Evolution of degree of anisotropy in contacts (a_c)

Degree of anisotropy in contacts is the statistical measure of average orientation in contact normals of a granular assembly. The evolution of a_c within the first cycle of loading of the sample Dr60 with CSR 0.35 loading is shown in Figure 4.5. The variation of anisotropy degree is analogous to the variation of shear stress in one loading cycle, i.e., with the increase in shear stress the anisotropy degree also increases and vice versa. Within a cycle, the anisotropy degree is maximum when the shear stress amplitude is maximum. If $a_{c,max}$ is considered as the maximum anisotropy degree in one loading cycle, the evolution of $a_{c,max}$ during the cyclic liquefaction is shown in Figure 4.6. In the pre-liquefaction stage, the $a_{c,max}$ increases gradually with the loading cycles, i.e., $a_{c,max}$ increases from 0.16 to about 0.6 in 81 cycles. Whereas, after initial liquefaction the $a_{c,max}$ increases quickly from 0.6 to around 1.5 in just 10 post-liquefaction cycles. Later $a_{c,max}$ gets stabilized when it reaches to the ultimate state and doesn't evolve further. The evolution of $a_{c,max}$ is qualitatively similar for all the loading cases. Note that within a post-liquefaction cycle during the flow state, the number of contacts in the granular assembly is very low. As a_c is an average quantity over number of contacts, for such a low number of contacts in flow state the value a_c might shoot up to unreasonable numbers and are ignored.

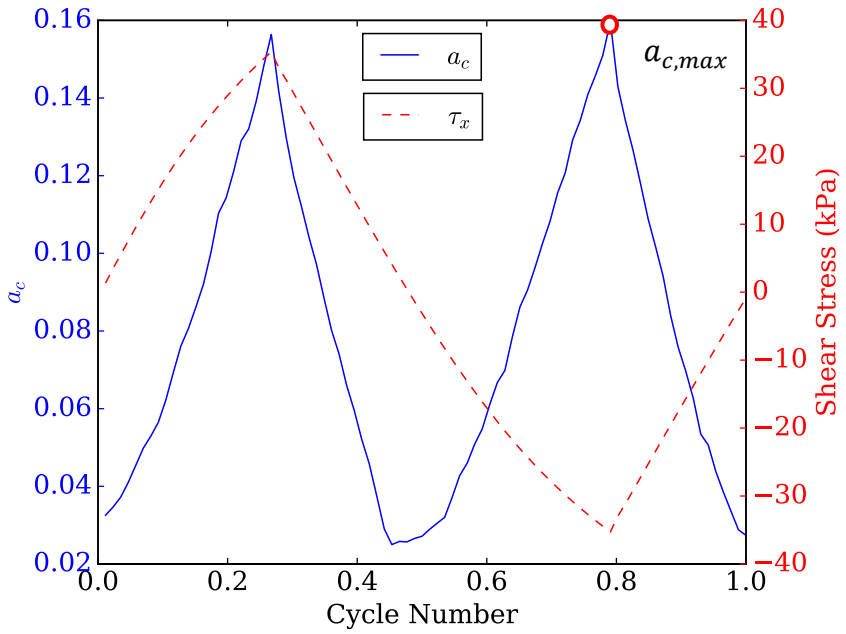


Figure 4.5 Evolution of anisotropy degree in the first cycle of loading

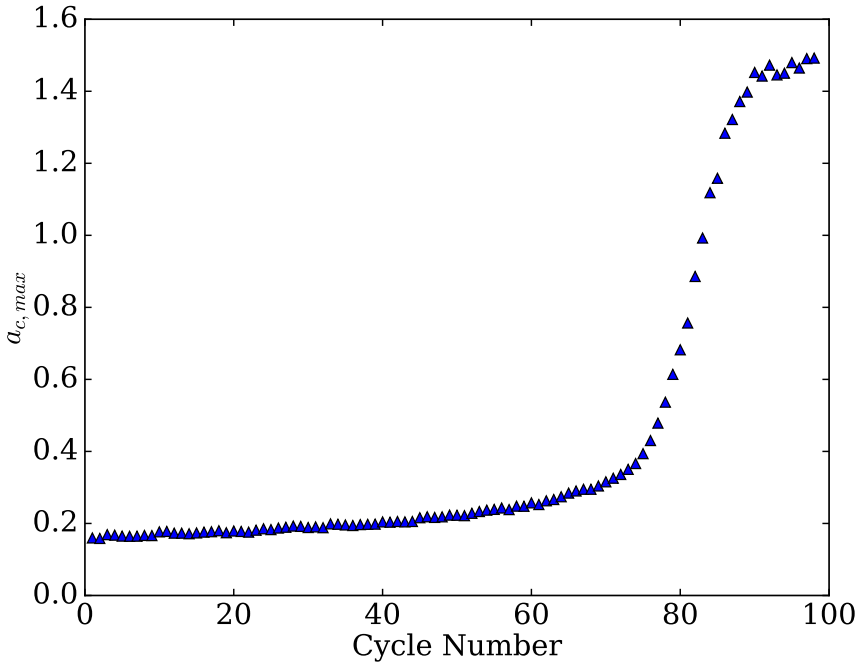
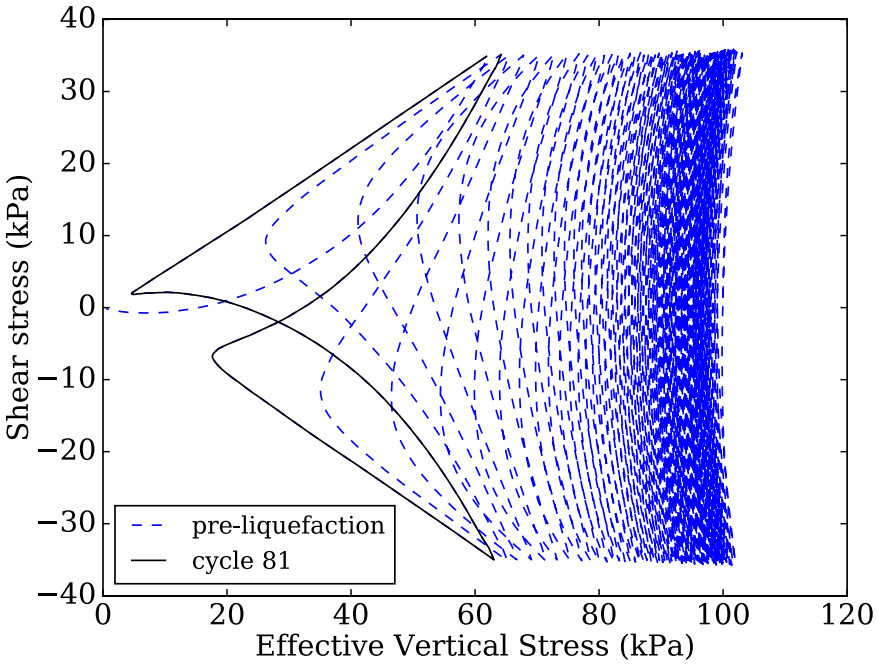


Figure 4.6 Evolution of maximum anisotropy degree

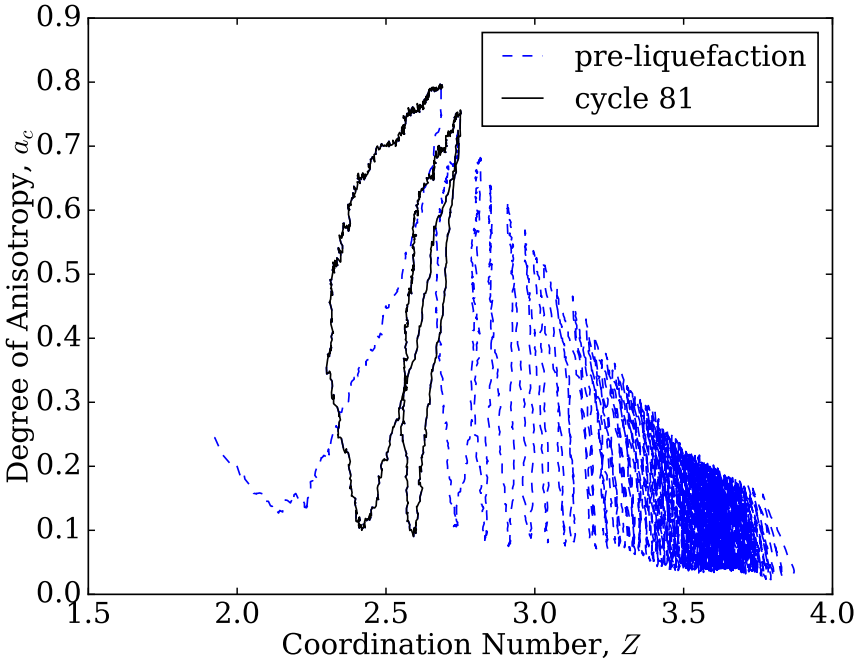
4.3.3 Stress path and $Z - a_c$ space

In the pre-liquefaction stage, the effective stress path ($\sigma_z' - \tau_x$) is comparable to the evolution of microstructure in $Z - a_c$ space. Figure 4.7a shows the effective stress path of sample Dr60 for CSR 0.35 loading in the pre-liquefaction stage and Figure 4.7b shows the respective microstructure evolution in $Z - a_c$ space. In the pre-liquefaction stage, the gradual decrease in effective vertical stress with every cycle of loading is similar to the gradual decrease in coordination number with each cycle. As the number of cycles progress, the coordination number reduces. To accommodate shear stress of 35 kPa, the average orientation in contact normals should increase for these reduced number of contacts. Thus, the degree of anisotropy increases gradually with the decrease in coordination number, as shown in Figure 4.7b. Within a loading cycle in the pre-liquefaction stage near to initial liquefaction, both the contractive and dilative type of behaviors is observed. Figure 4.8a shows the stress path response in cycle 81, i.e., one cycle before initial liquefaction, where starting from Point A to Point C, the behavior is contractive and from Point C to Point D, it is dilative. Similarly, from Point D to Point F, the behavior is contractive and from Point F to Point G, its dilative again. Point C and

Point F are phase transformation points where the soil behavior changes from contractive to dilative. Figure 4.8b shows the microstructure evolution in $Z - a_c$ space for cycle 81. It is observed that at phase transformation points, i.e., Point C and Point F, the trends in the evolution of coordination number changes, i.e., from decreasing to increasing. The trend change in coordination number at the micro-level is attributed to the phase transformation behavior at the macro level. At Point B and Point E, the degree of anisotropy attains its local minimum value because after the stress reversal at Point A and Point D, the contacts in the granular assembly which were previously oriented to support the shear stress tends to lose. Gradually the orientation in contacts reach to a near isotropic state at point B and Point E. Upon further loading from these points; new contacts are formed in the other direction to support the shear stress in other direction, thus increasing the degree of anisotropy from Point B and Point E.

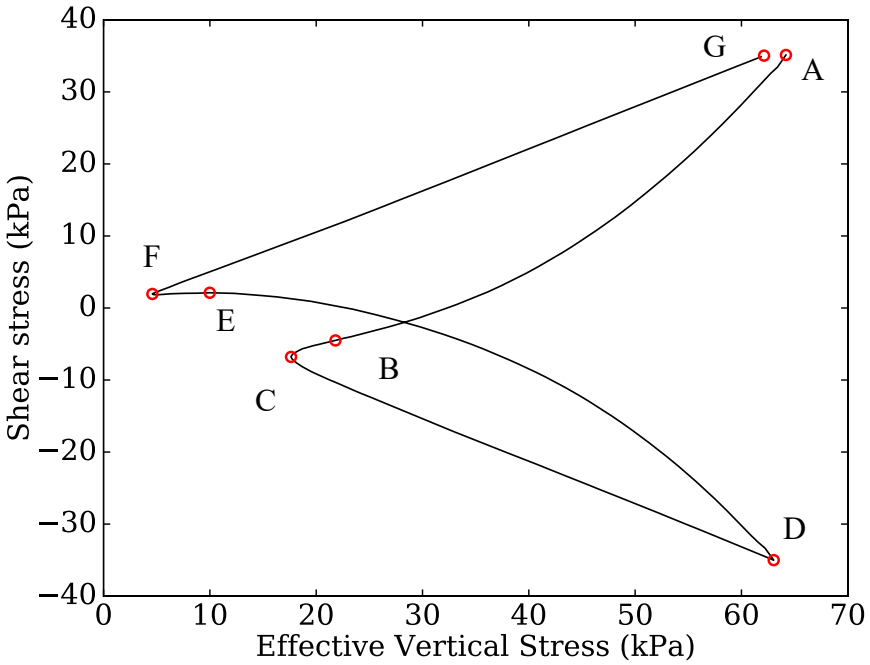


(a) Stress path in the pre-liquefaction stage

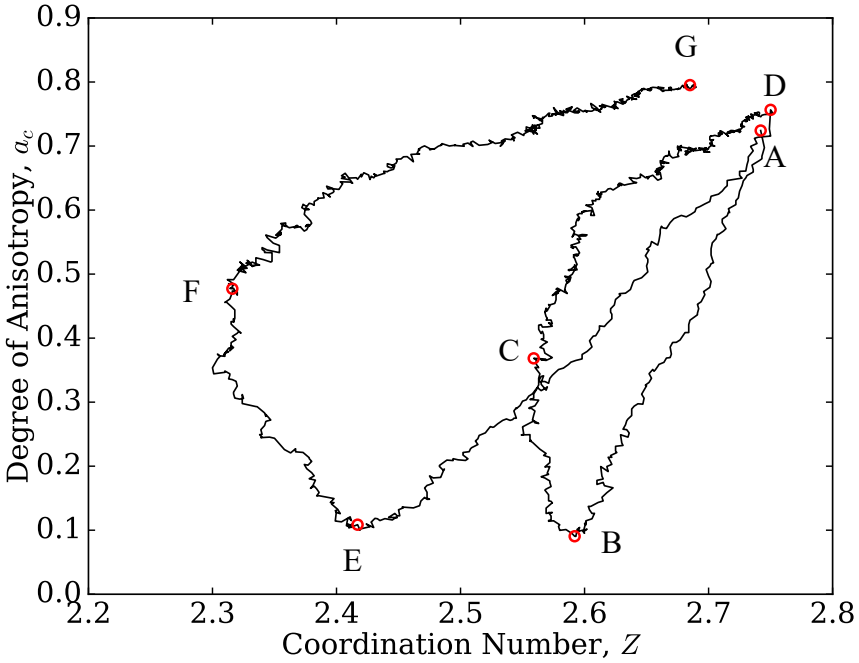


(b) $Z - a_c$ space in the pre-liquefaction stage

Figure 4.7 Comparing stress path to $Z - a_c$ space



(a) Stress path in one pre-liquefaction cycle



(b) Variation in $Z - a_c$ space

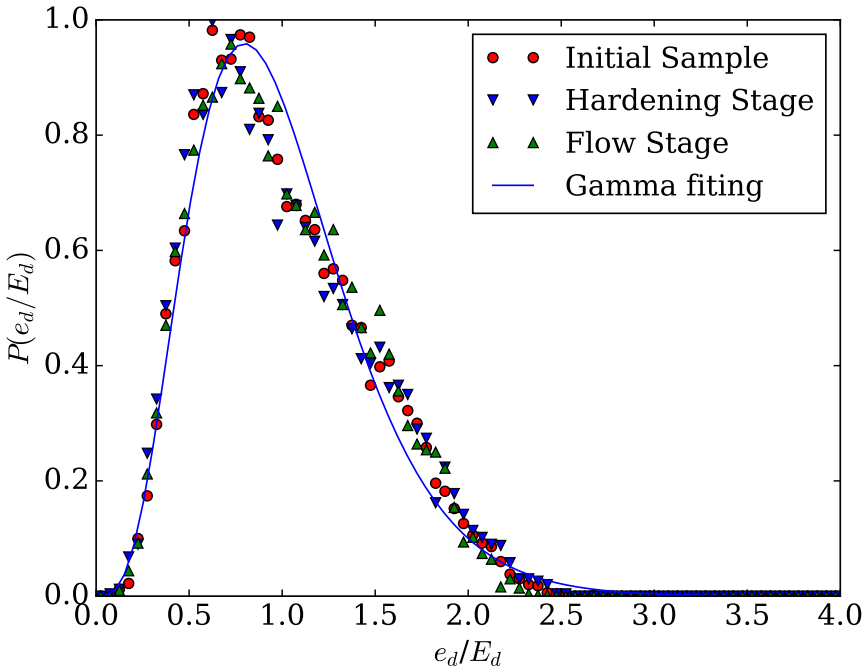
Figure 4.8 Variation of stress path and $Z - a_c$ space in one cycle of pre-liquefaction stage

4.4 Evolution of void-based fabric descriptors in cyclic liquefaction

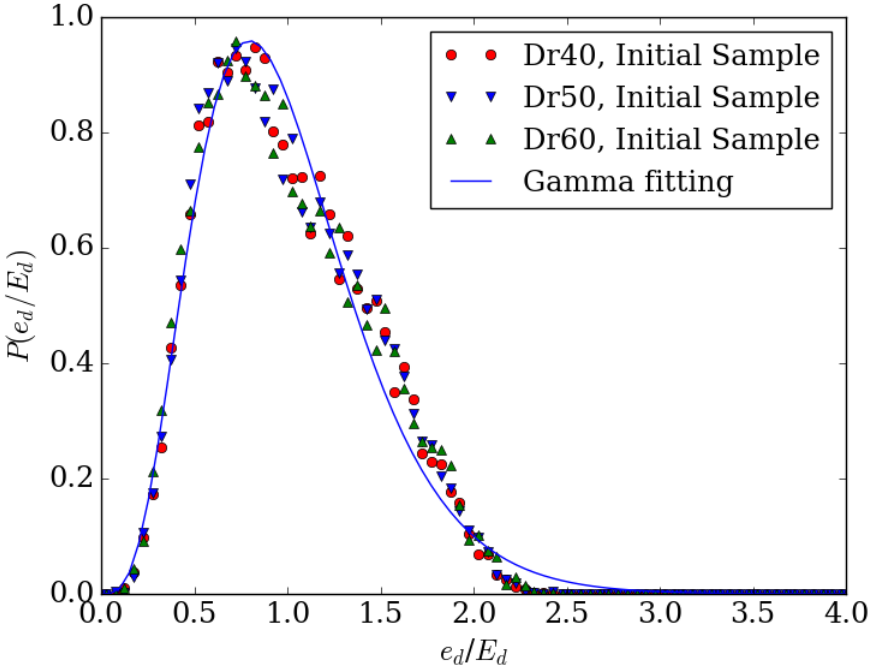
The concept of particle-void cells and the definitions for particle void descriptors, E_d and A_d are introduced in section 3.3 of this thesis. E_d is the global shape (elongation) descriptor and A_d is the degree of anisotropy in the particle-void cell orientations. E_d and A_d are obtained from the statistical analysis of e_d and $\boldsymbol{v}$, where e_d measures the local shape elongation of a particle-void cell and $\boldsymbol{v}$ provides the principal orientation of the same. In a granular packing, e_d and $\boldsymbol{v}$ are associated with every particle of the assembly, thus forming an extensive data set of population N_p (total number of particles). E_d is computed as the mean value of e_d. It is interesting to observe that the probability density function (PDF) of the normalized value, i.e., e_d/E_d, remains unchanged at different loading stages. Figure 4.9a shows the PDF of e_d/E_d at three different stages, i.e., at Initial sample, at hardening state, and at flow state in the cyclic loading test of Dr60 sample. The PDF of e_d/E_d can be well fit using a gamma distribution function as follows:

$$P(x|a,b) = \frac{1}{b^a \Gamma(a)}(x)^{a-1} e^{-x/b} \tag{4.1}$$

where $x = e_d/E_d$; $\Gamma(a)$ is the gamma function; $a = 4.83$ and $b = 0.208$. Moreover, the PDF of e_d/E_d is found to be independent of the relative density of the sample. Figure 4.9b shows the PDF of e_d/E_d for the initial samples of Dr40, Dr50, and Dr60. The same gamma distribution function can fit well for the different relative density samples. Thus, the mean value of e_d, i.e., E_d is sufficient as a descriptor to quantify the shape elongation of the whole particle-void distribution.



(a) PDF of e_d/E_d at three different stages of cyclic liquefaction



(b) PDF of e_d/E_d for different relative density samples

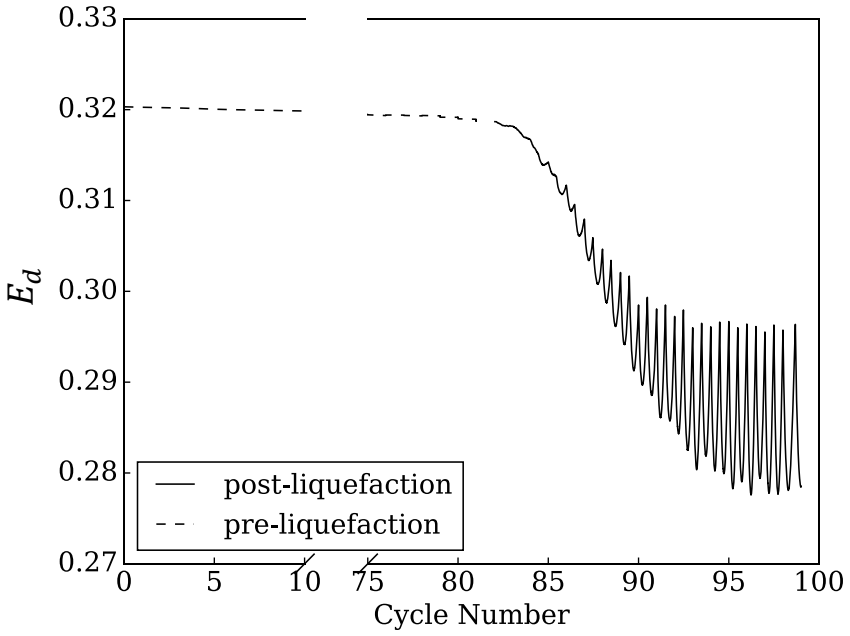
Figure 4.9 Probability distribution function of e_d/E_d

E_d for the initial consolidation samples of Dr40, Dr50 and Dr60 are 0.335,0.329, and 0.320 respectively. It is found that the isotropically consolidated samples have different E_d values for different relative densities, where higher relative density samples have lower E_d values. Though the samples are isotropically consolidated, the particle void cells are still anisotropic in terms of their shape. E_d =0 represents a perfectly isotropic shape of particle void cells. A_d for the initial samples of Dr40, Dr50 and Dr60 are 0.116,0.118, and 0.079 respectively. It is found that isotropically consolidated samples have very low A_d values resembling particle-void cells are randomly oriented, making it near isotropic in terms of particle-void cells orientation.

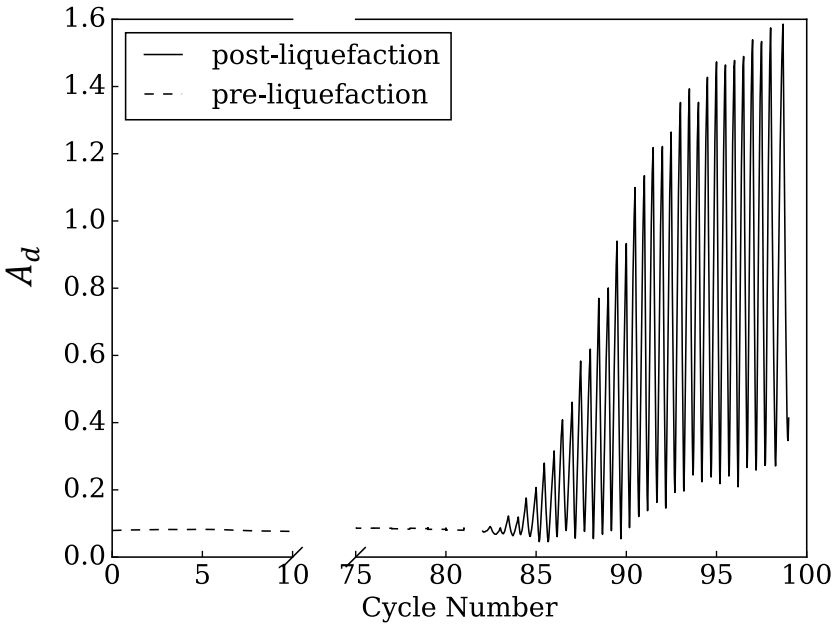
4.4.1 Evolution of E_d and A_d

The evolution of E_d and A_d during cyclic liquefaction of sample Dr60 for CSR 0.35 loading is shown in Figure 4.10. It is observed that in the pre-liquefaction stage, there is no significant variation in both E_d and A_d, i.e., E_d evolves from 0.329 to 0.32 and A_d evolves from 0.079 to

0.085 in the whole pre-liquefaction stage of 81 loading cycles. But significant variations in the evolution of these descriptors are observed during the post-liquefaction stage. After initial liquefaction within a few cycles of loading the E_d value drops down from 0.32 to a minimum value of about 0.278. Similarly, the A_d values rise from 0.085 to a maximum value of about 1.59. This variation happens in just 14 cycles after initial liquefaction. At ultimate state the evolution in both E_d and A_d got stabilized, where E_d keeps varying in between 0.295 to 0.278 and A_d vary in between 0.22 to 1.59, as shown in Figure 4.10(a,b). Qualitatively similar behavior in the evolution of $E_d$and A_d is observed for different loading paths and for different relative densities.



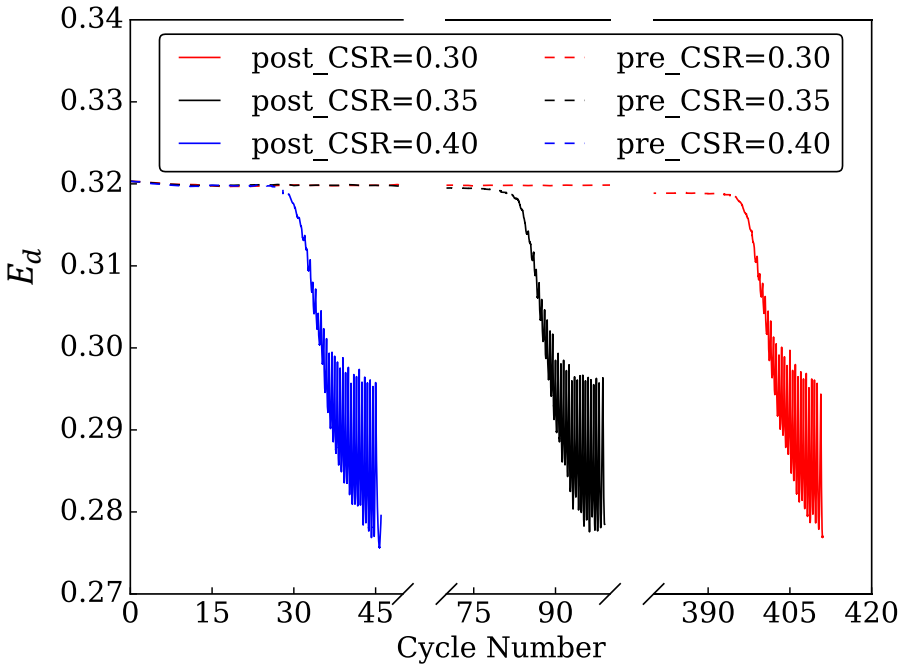
(a) Evolution of E_d



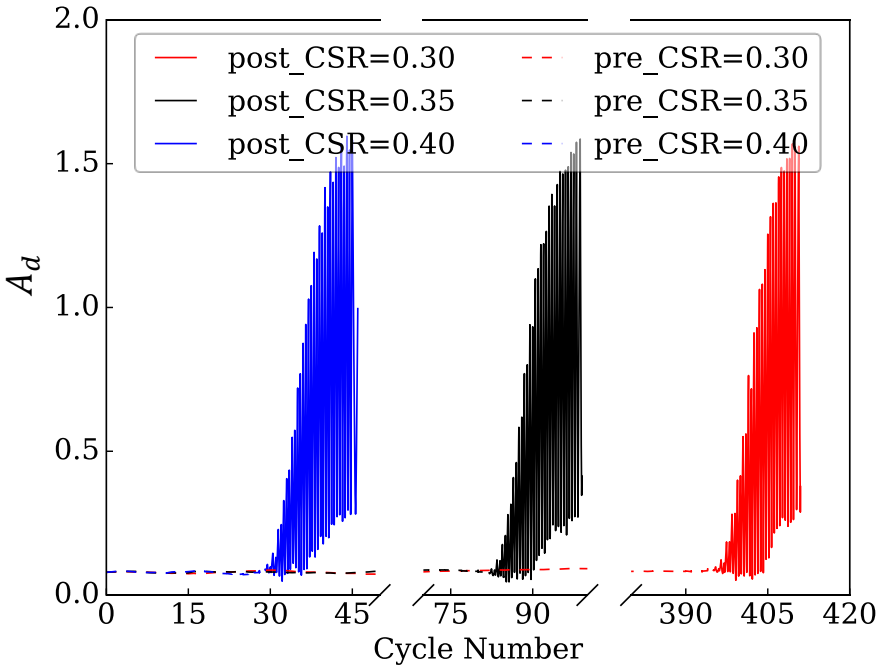
(b) Evolution of A_d

Figure 4.10 Evolution of E_d and A_d in cyclic liquefaction

Figure 4.11(a,b) shows the evolution of E_d and A_d for sample Dr60 in different loading paths (i.e., CSR=0.3,0.35 and 0.4). Overall from all the test cases, it is found that in the pre-liquefaction stage, very negligible change in E_d and A_d is observed and the evolution mostly happens in the post-liquefaction stage. It is also important to note that in all the cases E_d value tends to decrease and A_d value tends to increase after initial liquefaction. Decreasing in E_d resembles that the local particle-void cells are tending to become more isotropic in shape (i.e., more spherical), in other words, the void space around a particle is becoming more isotropic with every cycle of loading in the post-liquefaction stage. Thus, continuous cyclic loadings will drive the particle configuration towards the packing in which local void is more isotropically distributed around a particle. In such a configuration most of the particle-void cells tend to elongate and orient in the direction of loading, i.e., high A_d value before the jamming takes place in the granular assembly, thus allowing high shear strains. The accumulation of shear strains in the post-liquefaction stage is because with every cycle E_d decreases (more isotropic) thus A_d increases to allow jamming, thereby higher shear strain is observed in the current cycle compared to the previous cycle.



(a) Evolution of E_d

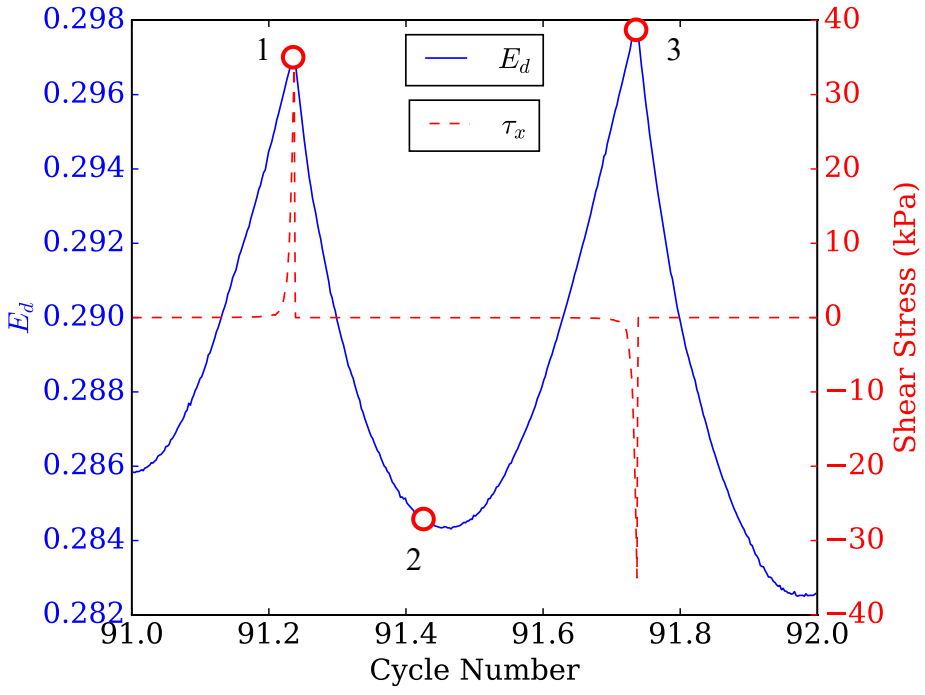


(b) Evolution of A_d

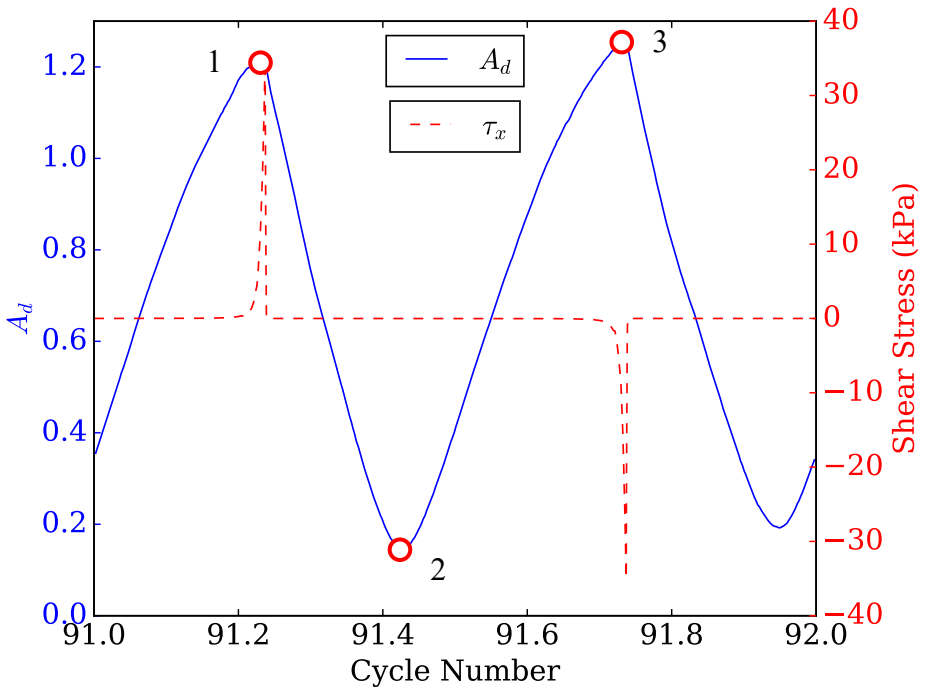
Figure 4.11 Evolution of E_d and A_d in different loading paths

4.4.2 Evolution of E_d and A_d within a loading cycle

The evolution of E_d and A_d in the 9th cycle of the post-liquefaction stage, i.e., the 91st cycle of the cyclic loading test on sample Dr60 for CSR 0.35 loading is shown in Figure 4.12(a,b). Within a loading cycle, it is observed that when the shear stress reaches its maximum amplitude, both E_d and A_d reaches their local maximum values. At the maximum shear stress, i.e., at Point 1 and Point 3, the granular assembly has a load-bearing structure where local particle-void cells are relatively more elongated and are oriented in a preferred direction. When the shear stress is reversed at Point 1 or Point 3, the load-bearing structure collapses and the value E_d and A_d slowly decreases. When the shear stress is varying from one peak (i.e., from Point 1) to another peak (i.e., to Point 3), the granular assembly passes through Point 2, where the degree of anisotropy (A_d) reaches its local minimum value (refer to Figure 4.12b). At this point, the E_d value also approximately reaches to a local minimum value (refer to Figure 4.12a). Thus, at Point 2 the particle-void cells are more isotropic both in terms of its shape (E_d) and the orientation (A_d).



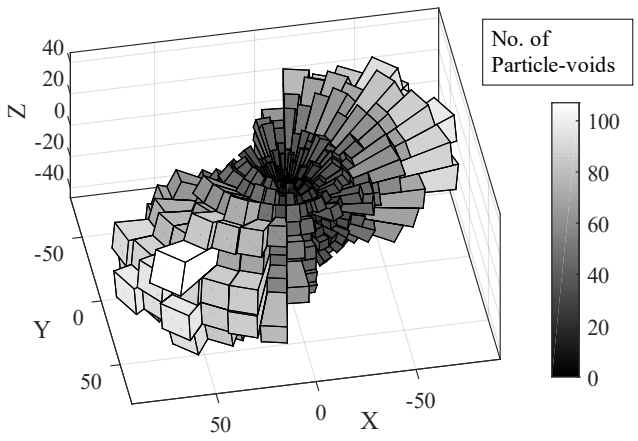
(a) Evolution of E_d



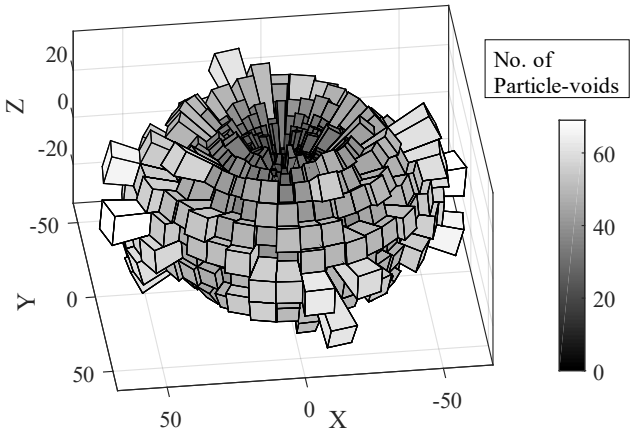
(b) Evolution of A_d

Figure 4.12 Evolution of E_d *and* A_d in one cycle

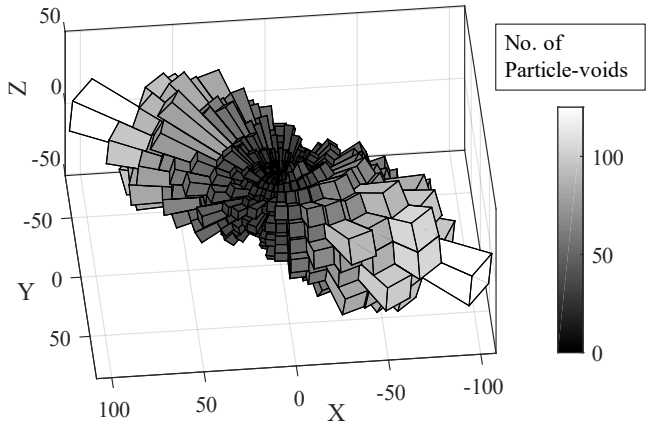
Figure 4.13(a,b,c) shows the spatial distribution of the orientations in particle-void cells at Point 2, Point 2, and Point 3, respectively, in the form of spherical histograms. Each bin in the spherical histogram is separated at an angle of 10° in both azimuthal and polar directions. The radius of the spherical histogram corresponding to any bin represents the number of particle-void cell orientations falling in that bin. Highly anisotropic distributions in the orientation of particle void cells are observed at Point 1 and 3, thus resulting in a high A_d values. At Point 2 near to isotropic shape in the distribution is observed, thus resulting in a low A_d value.



(a) Point-1



(b) Point-2



(c) Point-3

Figure 4.13 Spherical histograms of particle-void orientations at different points in the post-liquefaction stage

4.4.3 Hardening state line in $E_d - A_d$ space

From the discussions mentioned in section 4.2 and 4.3, it was made clear that in the post-liquefaction stage the granular assembly will transform from a flow state to a hardening state under cyclic shearing (refer to Figure 4.3). The point at which such a state transformation happens was referred to as transition points. Also, it was found that there exists a critical coordination number, Z_{crit}, where the transition from flow state to the hardening state always happens. Z_{crit} is independent of loading conditions and the relative density of the sample. In the current study, for the chosen particle size distribution Z_{crit} was observed to be 1.8. Figure 4.14 shows the evolution of E_d and A_d during cyclic loading of sample Dr60 for CSR 0.35 loading. The evolution is shown in $E_d - A_d$ space and it can be observed that E_d decreases with increasing A_d. At ultimate state the value of E_d and A_d gets stabilized and oscillates around the same path. The plot also highlights the transition points, i.e., Z =1.8 along with Z =1 points for reference. It is interesting to notice that the transition points clearly define loci of a Hardening state line (HSL) that separates flow state and the hardening state in $E_d - A_d$ space. As can be seen in Figure 4.14, the HSL can be fit using a curved line which delineates the boundary to flow state and the hardening state in $E_d - A_d$ space. Below the HSL, the granular assembly behaves like a fluid, and above the HSL the assembly has a stable load-bearing structure, thus behaves like a solid. The fact that the HSL is a concave upward curve, the load-bearing structure in the post-liquefaction stage is formed only when the sufficiently large A_d values is reached for the decreasing E_d values.

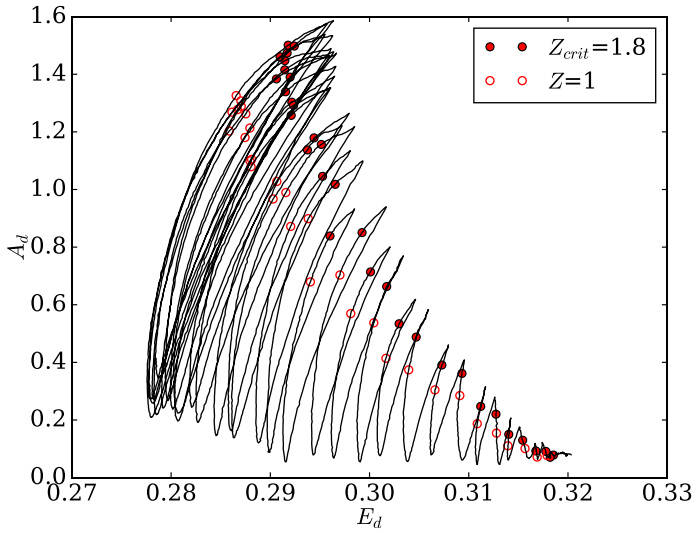
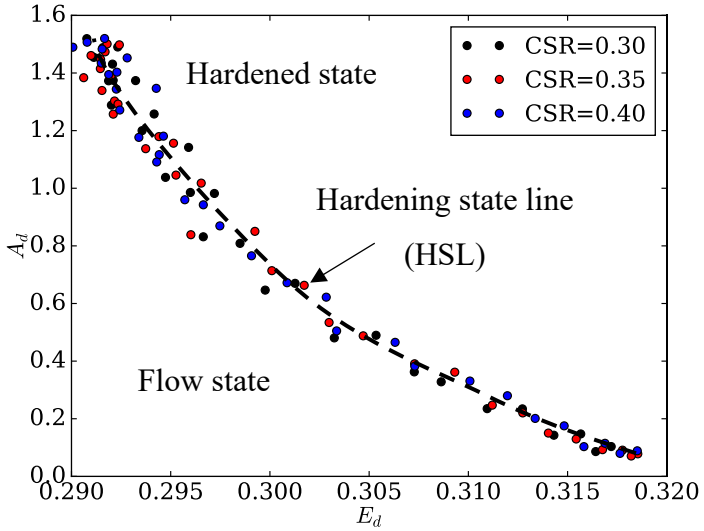
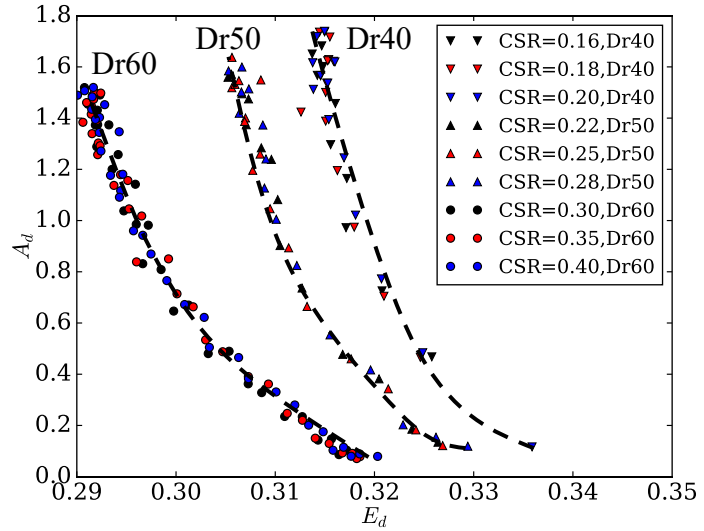


Figure 4.14 Evolution of particle-void fabric in $E_d - A_d$ space

Interestingly, the HSL is found to be unique for a sample and is independent of the loading paths. Figure 4.15a shows the transition points in $E_d - A_d$ space for sample Dr60 in different loading paths (i.e., CSR = 0.30, 0.35 and 0.40). Transition points in these loading paths are all aligned on a single curved line indicating the uniqueness of HSL line. It is found that the relative density of the sample has an influence on the HSL line, as shown in Figure 4.15b. As the relative density decreases the corresponding HSL line shifts to the right with increased slopes.



(a) $E_d - A_d$ (effect of loading path on HSL)

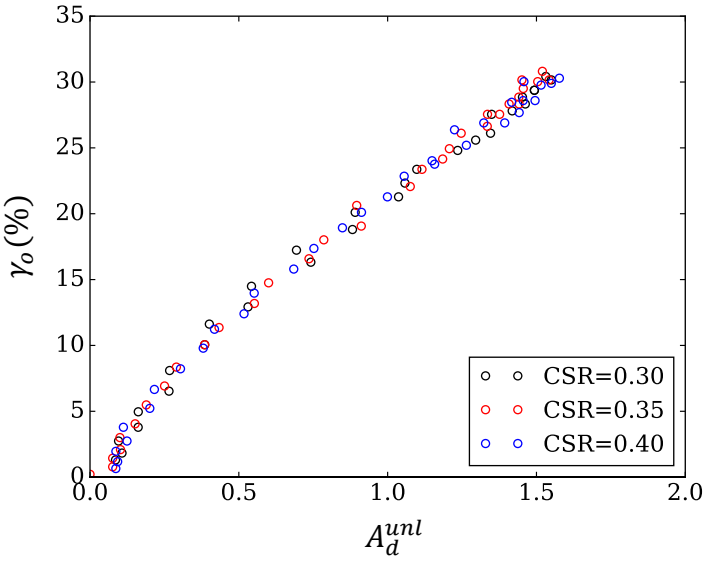


(b) $E_d - A_d$ (effect of relative density on HSL)

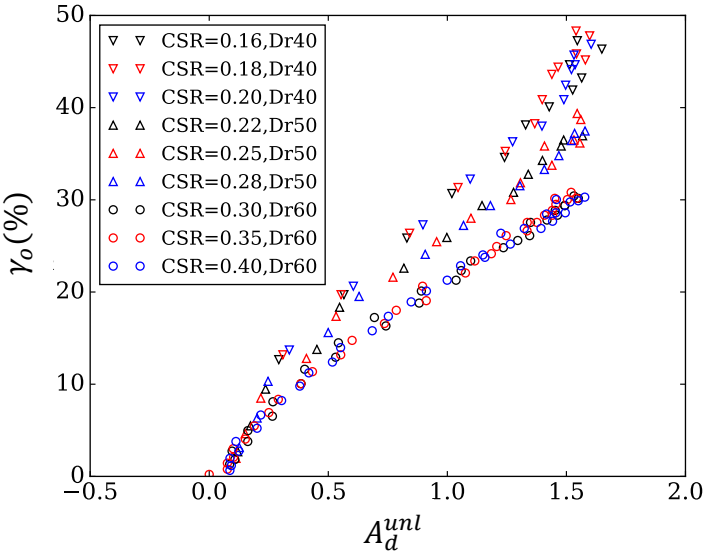
Figure 4.15 Particle-void fabric space ($E_d - A_d$)

4.4.4 Influence of particle-void distribution on shear strain accumulation

The critical feature in the post-liquefaction stage of cyclic mobility behavior is the flow strain accumulation. Within a cycle in the post-liquefaction stage, flow strain amplitude, γ_o, is defined as the shear strain experienced by the sample from an unloading point to the next transition point (refer to Figure 4.3). A_d^{unl} is defined as the degree of anisotropy in the particle void distribution right at the unloading point. Figure 4.16a shows the $A_d^{unl} - \gamma_o$ relation for the sample Dr60 in different loading paths. It is found that A_d^{unl} and γ_o has a strong correlation, i.e., with higher A_d value at the unloading point (A_d^{unl}) will be followed by a larger amplitude of flow strain, γ_o to reach the next transition point . Also, the correlation seems to be unaffected with the change in loading paths for a given sample as shown in Figure 4.16a. The relative density of a sample seems to have little influence on the slope of the $A_d^{unl} - \gamma_o$ correlation. Figure 4.16b shows the $A_d^{unl} - \gamma_o$ curve with different relative densities. It is observed that the slope of the correlation becomes steeper as the relative density decreases indicating higher flow strains for the same A_d, higher flow strains are observed for looser samples. It is also interesting to note that at the ultimate state, the flow strain amplitude seems to increase with the decrease in relative density. For Dr60 sample, the flow strain amplitude (γ_o) at ultimate state is about 30% whereas for Dr40 sample is about 50%.



(a) $A_d - \gamma_o$ Space (effect of loading path)



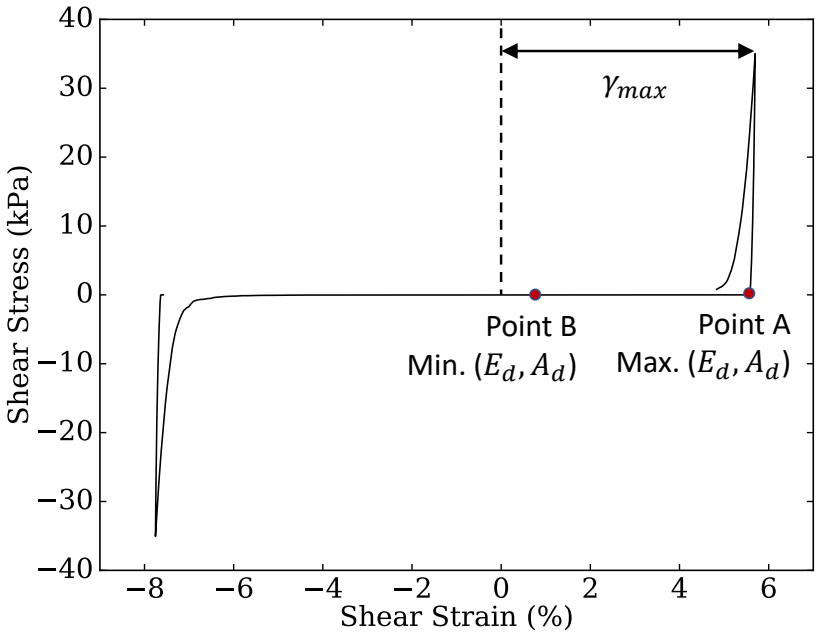
(b) $A_d - \gamma_o$ space (effect of relative density)

Figure 4.16 $A_d - \gamma_o$ Space

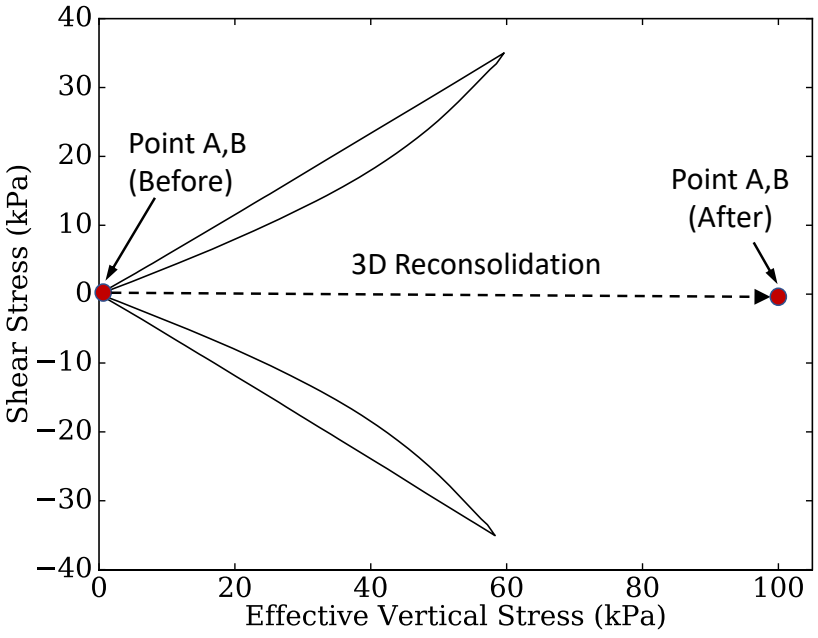
4.5 Volumetric compression in post-liquefaction consolidation

After initial liquefaction, the dissipation of excess pore pressure results in an increase of effective confining stress and the volumetric compression of the granular soil. In DEM, the post-liquefaction consolidation is simulated by moving the boundaries of 3D RVE to increase effective stress, until the effective stress reaches its initial hydrostatic state of 100 kPa. A stepwise consolidation algorithm is used to move the boundaries in order to compress the packing to a hydrostatic stress state. Within a cycle in the post-liquefaction stage, two points, i.e., point A and point B are chosen in the flow state for consolidation, as shown in Figure 4.17a. Point A corresponds to the just unloading point from the positive peak shear stress (refer to Fig. 4.17a) where the (E_d, A_d) are at their local maximum values. Point B corresponds to the instant where (E_d, A_d) are at their local minimum values. Different samples are saved corresponding to these points in different post-liquefaction cycles. Within a post-liquefaction cycle, samples at point A and B would have experienced the same maximum shear strain (γ_{max}) and is equal to the single amplitude strain in the previous post-liquefaction cycle (refer to Fig. 4.17a). The stress state before and after reconsolidation is shown in Figure 4.17b. In total, 18

consolidated samples are prepared corresponding to the point A and B from 9 post-liquefaction cycles of Dr60 and CSR 0.35 loading response.



(a) Unloading points

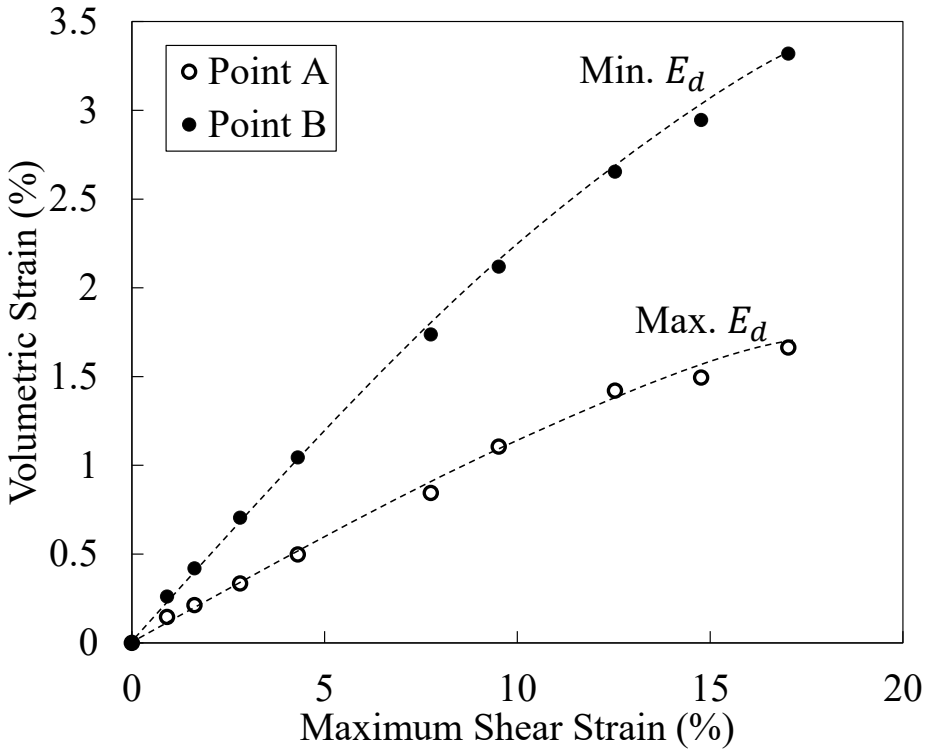


(b) Reconsolidation path

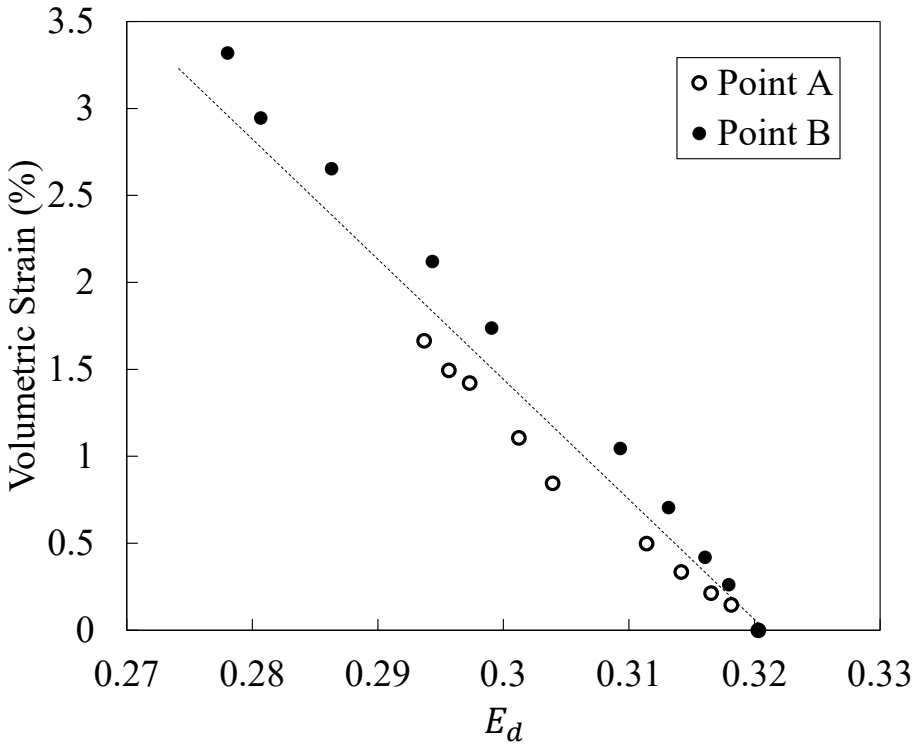
Figure 4.17 Unloading points in a post-liquefaction cycle and reconsolidation path

The volumetric strain (ϵ_v) resulting from the consolidated samples (both at point A and point B) have a clear correlation with the maximum shear strain (γ_{max}) experienced by the sample before consolidation as shown in Figure 4.18a. Samples that have experienced higher shear strain before consolidation tends to have higher volumetric compression. A similar phenomenon is observed in experimental studies, as reported in Ishihara et al. (2016). From Figure 4.18a, it is also observed that samples at point A have relatively lower compressibility compared to samples at point B. This highlights a range of possible volumetric compressions of samples corresponding to different points of unloading in the post-liquefaction cycle.

The volumetric strain (ϵ_v) is well correlated with the particle-void descriptor, E_d and is shown in Figure 4.18b, where E_d is recorded just before the consolidation at both point A and point B. It can be observed that the samples with relatively lower E_d value has higher volumetric compression. A granular packing with relatively low E_d value means that the particle voids are more isotropic (more spherical) and when consolidated hydrostatically results in more uniform compression locally. Hence, giving rise to a higher volumetric compression overall. In other words, E_d is a good indicator for the compressibility of the granular assembly.



(a) Maximum shear strain and Volumetric strain



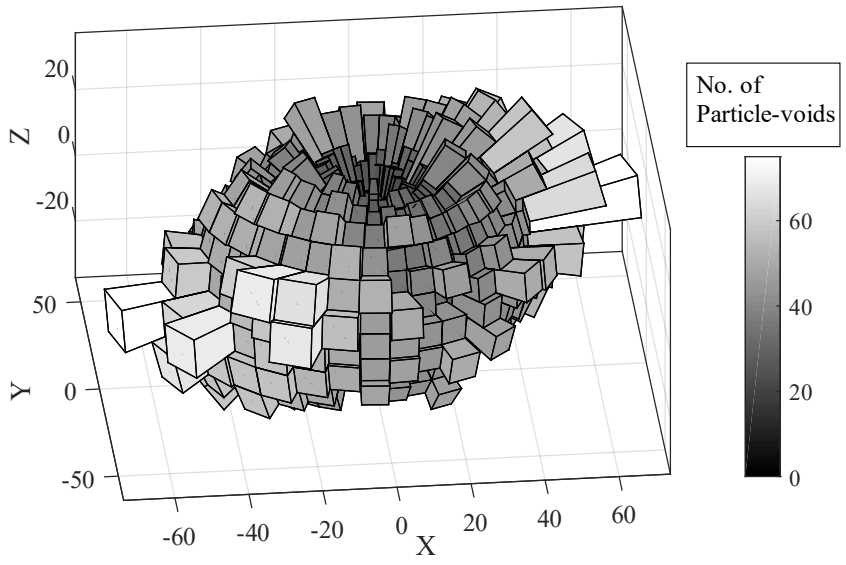
(b) E_d and Volumetric strain

Figure 4.18 Correlation of Volumetric strain with macro and micro quantities

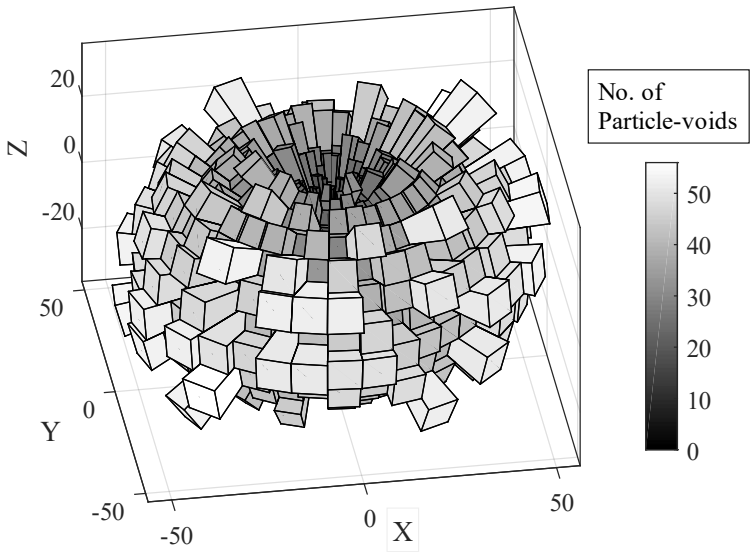
4.6 Re-liquefaction behavior

In this study, our primary focus is to study the effect of unloading point in a post-liquefaction cycle to the re-liquefaction resistance. The samples at different unloading points give rise to different anisotropic fabrics when reconsolidated. Recent studies argue that anisotropy is the most influential factor and surpasses the effect of relative density on liquefaction resistance (Yamada et al. 2010). The samples saved at point A and B corresponding to 5th cycle in the post-liquefaction stage of Dr60-CSR 0.35 loading response are reconsolidated and examined for re-liquefaction strengths (refer to Fig. 4.17a). The post-liquefaction consolidation procedure mentioned in the previous section is used to reconsolidate the samples at point A and B. Note that during reconsolidation the interparticle friction angle is slightly reduced in order to generate more contacts internally and later it is adjusted back to 26° during shearing. Hydrostatic stress state of 100 kPa is reached upon reconsolidation in both the samples (refer to Fig. 4.17b). Before reconsolidation, the distribution of orientation in particle-void cells at

point A and B in the form 3D spherical histograms is shown in Figure 4.19(a,b) respectively. Sample at point A has highly anisotropic distribution in the orientation of particle-void cells, whereas, at point B, it is nearly isotropic distribution. This is because point A and B are chosen at local maximum and minimum A_d values. It is interesting to observe that the anisotropy in the orientation of particle-voids before reconsolidation determines the anisotropy in the orientation of contact normals after reconsolidation. After reconsolidation, the distribution of contact normals at point A and B in the form 3D spherical histograms is shown in Figure 4.20(a,b) respectively. Reconsolidated sample at point A has high anisotropy in contact normals distribution, whereas reconsolidated sample at point B has an isotropic distribution. Based on these observations, it can be concluded that, during reconsolidation, particles in the granular packing tends to form new contacts in the direction perpendicular to the orientations of their particle void cells. Therefore, high anisotropy in the orientation of particle void cells before reconsolidation generate high anisotropy in contact normals after reconsolidation.

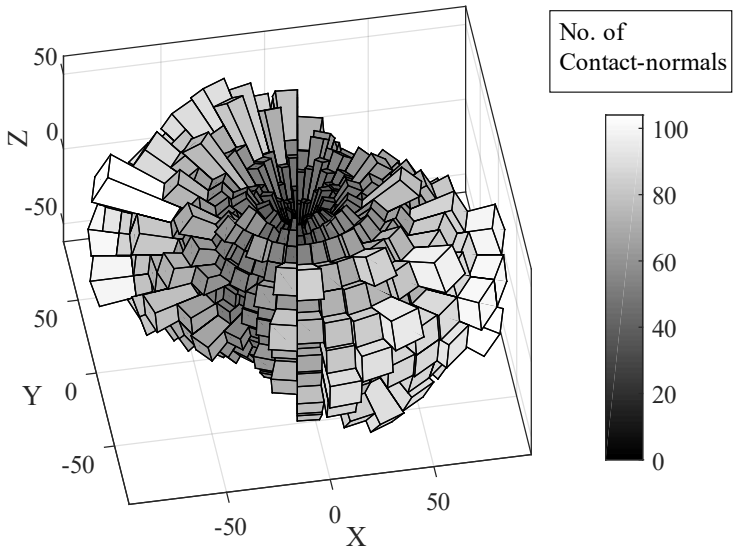


(a) Point A

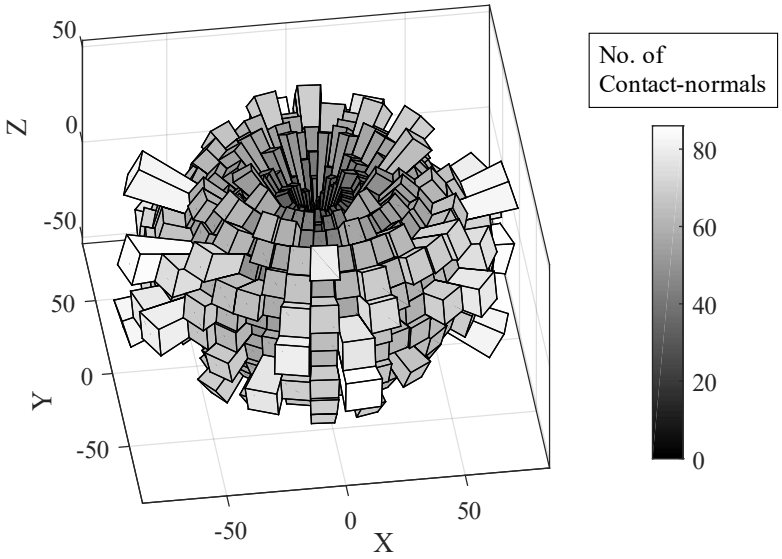


(b) Point B

Figure 4.19 Spherical histogram of particle-void orientation at point A and point B



(a) Point A

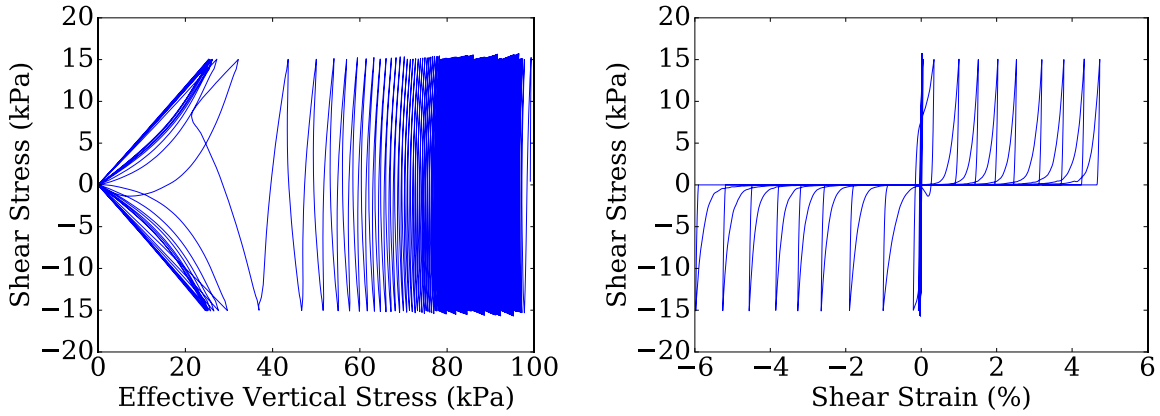


(b) Point B

Figure 4.20 Spherical histogram of contact-normal orientation in reconsolidated samples at point A and point B

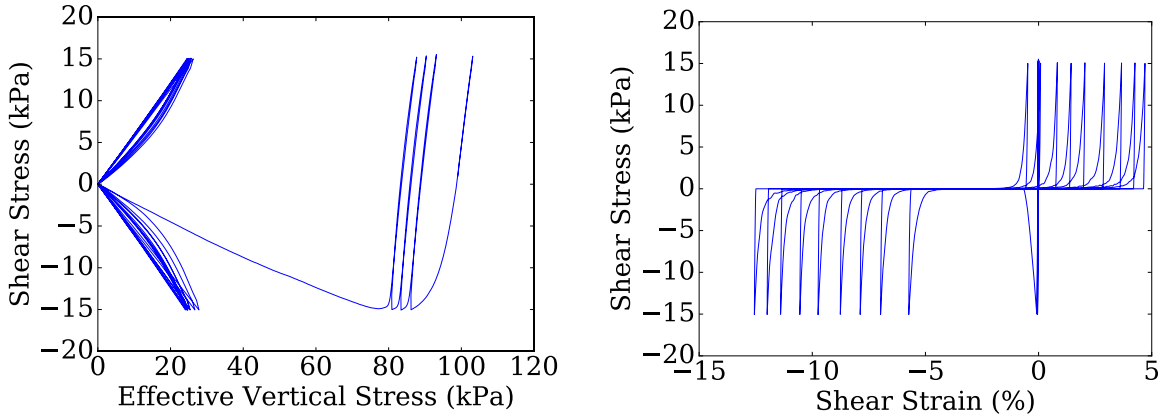
The re-liquefaction behavior for reconsolidated samples at point A and B for a CSR of 0.15 is shown in Figure 4.21(a,b) and 4.22(a,b) respectively. It is observed that the reconsolidated sample at point A requires just three cycles to liquefy, whereas reconsolidated sample at point B takes about 236 cycles to reach initial liquefaction. The relative densities of samples at A and B after reconsolidation are 64% and 66%, which are almost the same, but the variation in liquefaction resistance is quite significant. Clearly, contact fabric anisotropy has a significant role in determining the liquefaction resistance. As shown in Figure 4.20a, Point A has a high anisotropic distribution in contact normals. Majority of the contact normal vectors are distributed in only a few bins and in a particular orientation. Due to such preferential orientation of contact normals in the reconsolidated sample at point A, it happened to be very stiff in one direction and very loose in the other, resulting in the overall degradation of liquefaction resistance. At point B, the contact normal distribution is almost the isotropic in all the directions as shown in Figure 4.20b, so the reconsolidated sample behaves the same for both positive shear stress and negative shear stress thus resulting in many cycles for complete degradation

of stiffness. Hence, different unloading points in the post-liquefaction stage results in different fabric after consolidation, which dramatically affects the re-liquefaction strength.



(a) Stress path (b) Stress - Strain

Figure 4.21 Re-liquefaction behavior of reconsolidated sample at point A



(a) Stress path (b) Stress - Strain

Figure 4.22 Re-liquefaction behavior of reconsolidated sample at point B

4.7 Conclusions

In summary, the evolution of fabric indicators in the cyclic liquefaction phenomenon is explored. Two new particle-void based descriptors, E_d and A_d are developed based on the shape and the anisotropy of the particle-void cells which are generated from the Voronoi tessellation of the granular packing. Characteristics of the fabric evolution can be summarized as below:

1) The contact-based fabric indicators are found to evolve primarily in the pre-liquefaction stage. Coordination number (Z) reduces gradually with every cycle of loading before initial liquefaction, which is analogous to the decrease in effective stress. As the Z decreases, the degree of anisotropy in contact normal (a_c) is observed to increase.

2) Before initial liquefaction, within a cycle, the change in behavior from contractive to dilative, i.e., phase transformation, is attributed to the change in trend from decrease to increase in the coordination number.

3) It is observed that the evolution of the new particle-void descriptors primarily occurs in the post-liquefaction stage, accompanied by increasing post-liquefaction deformation. After initial liquefaction, the value of E_d decreases and A_d increases gradually cycle after cycle. For a given relative density, the post-liquefaction deformation of the granular packing is strongly correlated to the evolution of the particle-void fabric descriptors, so these fabric indices can be used to quantify the amplitude of flow deformation. Finally, a stable state of particle-void distribution can be achieved in post-liquefaction after a sufficient number of loading cycles.

4) In the post-liquefaction stage, the jamming transition could occur in the granular flow, which trans-forms the granular assemblage from a “fluid-like” state to a “solid-like” state to arrest the flow deformation. The coordination number (Z_{crit}) at which this transition occurs is found to be unique and independent of both loading path and relative density. Given the relative density, unique hardening state line (HSL) is defined in the void-fabric space, that delineates the boundary between a flow stage and a jamming/hardening stage when the soil is fully liquefied.

5) It is observed that the post-liquefaction volumetric strain depends on the magnitude of the maximum shear strain that is experienced by the granular packing. For a given magnitude of maximum shear strain, a range of volumetric deformations is possible based on the point of unloading in a post-liquefaction cycle. In either case, it is observed with the decrease in particle-void descriptor the volumetric strain has increased. This suggests E_d can be a good indicator for the compressibility of a granular packing.

6) In this study, one of the key important factors, i.e., the point of unloading in the post-liquefaction stage and its effect on re-liquefaction resistance is explored. It is observed that completely different soil fabrics are formed if the liquefied soil is consolidated at different unloading stages, resulting in vastly different resistance to the re-liquefaction.

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CHAPTER 5

Fabric Evolution in Multi-directional Loading of Granular Assembly

Seismic loads are essentially irregular with varying loading magnitude and direction. In this chapter, a medium dense sample in a DEM simulation is subjected to several regular multi-directional loading paths (Circular/Oval, Figure-8, and Uni-directional). DEM well simulates the qualitative behavior in these different multi-directional loading paths.

In this chapter, the evolution of both contact-based and particle-void fabric indicators in multi-directional loading tests is quantitatively studied. It is found that contact-based fabric evolution is strongly correlated with the loading path, whereas the void-based fabric evolution is related to the generation of shear strains. The trajectory of evolution in the contact-based fabric tensor components in the initial few cycles is very similar to the loading path. For all circular/oval loading path, the trajectory of evolution eventually ends up being the same, indicating the formation of the same contact-fabric eventually in all the loading paths. The trajectory of evolution in the void-based fabric tensor components is similar to the trajectory of generation in shear strain, indicating a strong correlation to the deformation of the granular system. Also, a new criterion to indicate soil liquefaction in case of multi-directional loading paths is proposed.

5.1 Sample generation and loading method

5.1.1 Sample generation

The DEM simulations are conducted using open-source code, Yade (Šmilauer et al. 2011). A three dimensional (3D) cubic representative volume element (RVE) consisting of 10000 randomly spaced spheres is generated. The radius of the spheres ranges from 0.2 mm to 0.6mm with a mean radius (R_{50}) being 0.4mm. Hertz-Mindlin nonlinear contact model is used to calculate forces at inter-particle contacts with Young's modulus of 70 GPa and Poisson's ratio of 0.3 assigned to all the particles. Note that periodic boundary condition is prescribed to the RVE to minimize the boundary effect. Also, no rotational stiffness is considered at the contact allowing particles for free rotation. The particle assembly is randomly generated in space with no initial contacts and later is isotropically consolidated to a confining stress of 100 kPa, thus the initial effective stress ($\sigma'_{v,o}$) is also 100 kPa. To attain different void ratios, the friction angle (ϕ) can be adjusted during the consolidation. In the current study, a medium dense sample with a void ratio equal to 0.64 is generated in the initial consolidation process. After initial consolidation, ϕ is set to 26° for the later cyclic shearing. Undrained conditions are maintained by keeping the volume of RVE constant. It is important to note that the rate of shear strain loading is maintained to be 0.01/s in order to ensure quasi-static test conditions.

5.1.2 Multi-directional loading in DEM

In a multi-directional undrained simple shear, the total shear stress can be interpreted as two independent components in mutually perpendicular directions varying harmonically and can be denoted as τ_x and τ_y. The harmonic functions of τ_x and τ_y can be represented as:

$$\tau_x = \tau_1 \cos(\omega_1 t) \tag{5.1}$$

$$\tau_y = \tau_2 \cos(\omega_2 t + \phi) \tag{5.2}$$

where (ω_1, ω_2) are frequencies of loading and (τ_1, τ_2) are the amplitudes of the shear stress in x and y direction respectively. ϕ is the phase difference/ phase lag. In the current study, three typical loading paths are considered: Circular/Oval, Figure-8, and Uni-directional following Kammerer (2002). To generate an oval loading path: $\omega_1 = \omega_2$ and $\phi = -\frac{\pi}{2}$ are assigned. If

$\tau_1 = \tau_2$, the loading path is circular. In case of uni-directional loading path, $\tau_2 = 0$. To generate a figure-8 type of loading path, $\omega_2 = 2\omega_1$ and $\phi = -\frac{\pi}{2}$ are assigned.

In the current DEM simulations, the strains in the respective direction are controlled in order to generate stresses in that direction. The strain-controlled algorithm to generate a multi-directional loading path is little complex. The difficulty lies in determining the correct shear strain rate since the strain increment may not be coaxial with the stress increment. To prescribe the multi-directional stress path, a five-step procedure is followed:

(1) Divide the multi-axial loading path into target values $(\tau_x, \tau_y)_{tar}$ with an increment of $\Delta(\omega_1 t) = 1^o$.

(2) Calculate the direction of shear stress increment based on the target stress and current stress, $\boldsymbol{n}_{\Delta\tau} = \frac{\Delta\tau_x \boldsymbol{e}_x + \Delta\tau_y \boldsymbol{e}_y}{\sqrt{(\Delta\tau_x)^2 + (\Delta\tau_y)^2}}$, where $\Delta\tau_x = (\tau_x)_{tar} - (\tau_x)_{cur}$ and $\Delta\tau_y = (\tau_y)_{tar} - (\tau_y)_{cur}$. $\boldsymbol{e_x}$ and $\boldsymbol{e_y}$ are the unit vectors along x and y directions.

(3) Assume the direction of shear strain increment is same as the direction of shear stress increment, $\boldsymbol{n}_{\Delta\gamma} = \boldsymbol{n}_{\Delta\tau}$. Thus the direction of shear strain can be determined as $\dot{\boldsymbol{\gamma}} = \dot{\gamma}_o \boldsymbol{n}_{\Delta\gamma}$, where $\dot{\boldsymbol{\gamma}} = (\dot{\gamma}_x, \dot{\gamma}_y)$ and $\dot{\gamma}_o$ is the magnitude of strain rate and is always 0.01/sec.

(4) Move the boundary of the packing based on the calculated shear-strain rate in step (3) and update the current shear stress $(\tau_x, \tau_y)_{cur}$.

(5) Calculate the new shear stress increment $\Delta\tau_x$ and $\Delta\tau_y$. If the magnitude of $\Delta\tau_x$ and $\Delta\tau_y$ are both smaller than a preset tolerance, the target shear stress is reached. Otherwise, go back step (2) and restart the loop. If the target shear stress is reached update the new target shear stress based on the target loading path. For current simulations, the preset tolerance assigned is $\|\Delta\tau_x\|$ <0.3kPa and $\|\Delta\tau_y\|$ <0.3kPa.

For an inelastic medium like granular material, it takes quite a few iterations in order to reach the target stress values. As the directions of stress increment and strain increment are not coaxial, it takes multiple iterations to adjust the strain increments direction to reach the target stress values. Overall, with the help of above procedure, the target multi-directional loading paths can be achieved. Figure 5.1 shows the stepwise flow chart for the above stress-controlled algorithm to generate multi-directional loading path.

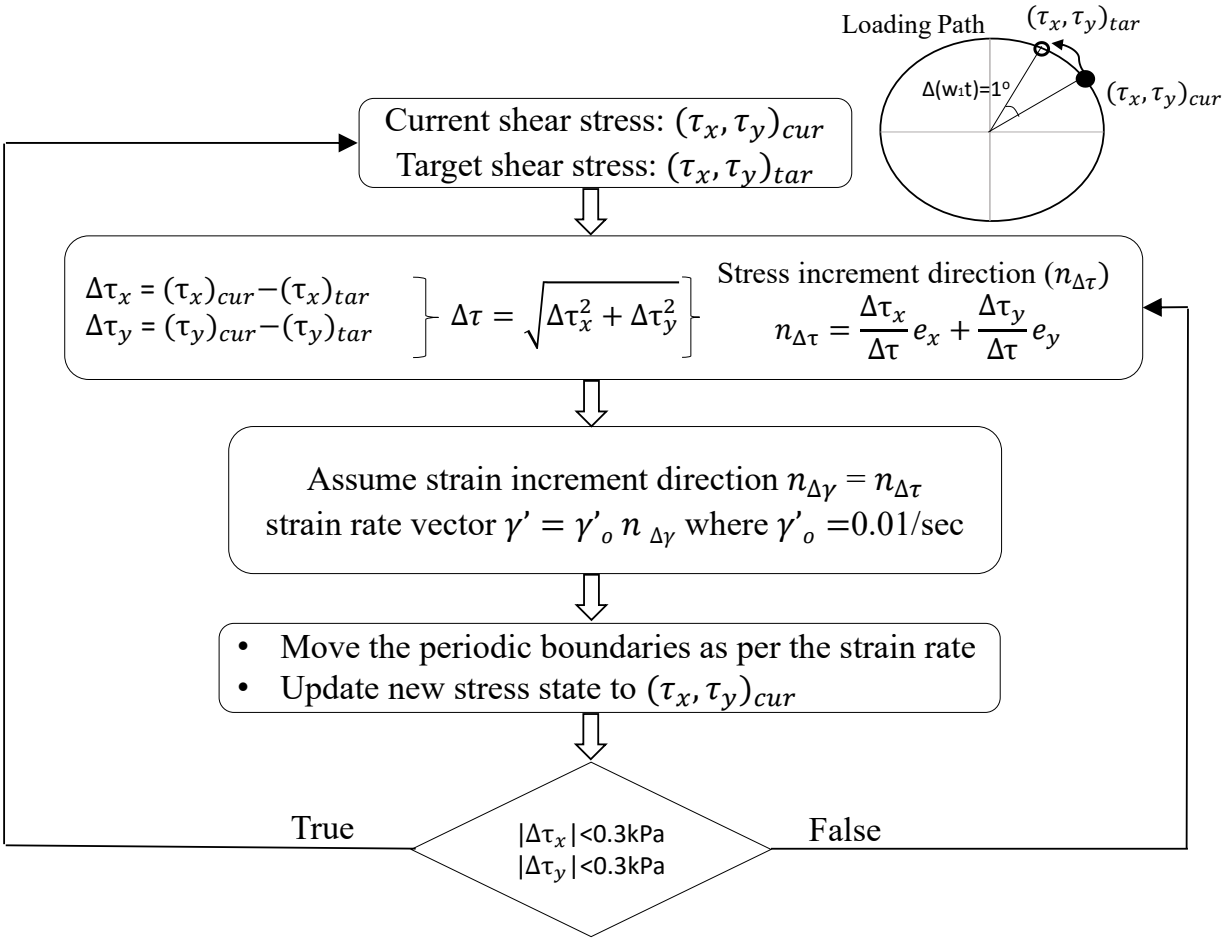


Figure 5.1 Algorithm to generate stress-controlled multi-directional loading in DEM

Cyclic stress ratio (CSR) is defined as the ratio between the shear stress in the horizontal direction to the initial effective stress. For a multi-directional test, $CSR_x = \tau_1/\sigma'_{v,o}$ and $CSR_y = \tau_2/\sigma'_{v,o}$ can be individually defined. Also, the aperture ratio (AR) which defines the extent of multi directionality can be quantified as the ratio of CSRs in orthogonal directions, i.e., $AR = CSR_x/CSR_y$. For an circular/oval loading path, $AR = 0$ represents a uni-directional loading case while $AR = 1$ represents a circular loading path. In simulations, we assume that $\tau_1 \geq \tau_2$without the loss of generality. Accordingly, CSR_x is always higher than CSR_y. Also, in the present study, τ_1 is fixed to be 25 kPa (i.e., $CSR_x = 0.25$) whereas τ_2 is varied between 0 and 25 kPa to generate loading paths of different apertures. Table 5.1 summarizes all the loading paths that are generated for the current study. Figure 5.2 shows the loading paths that are generated based on the above stress-controlled multi-directional algorithm. It is important to note that the initial isotropically consolidated sample is first subjected to monotonic loading in

the x-direction with very low shear strain to reach $\tau_x = 25\ kPa$. All the tests start from this current stress state, i.e., $(\tau_x, \tau_y)_{cur} = (25,0)\ kPa$ and later target stress values $(\tau_x, \tau_y)_{tar}$ are calculated according to the loading path that is followed.

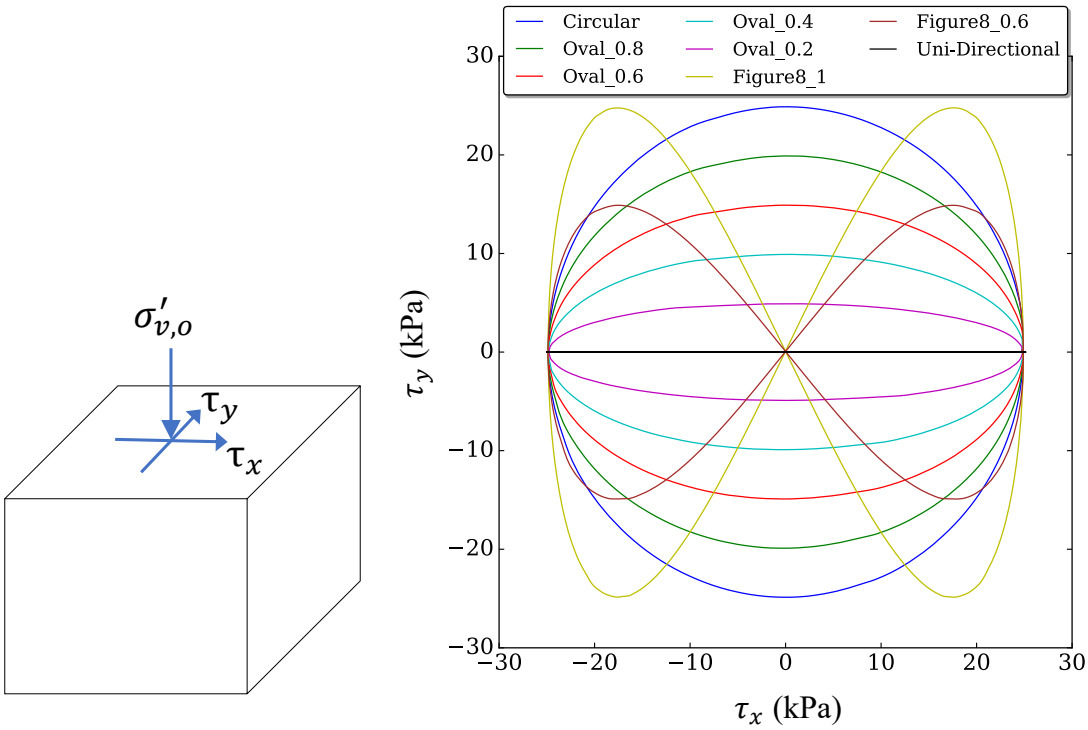


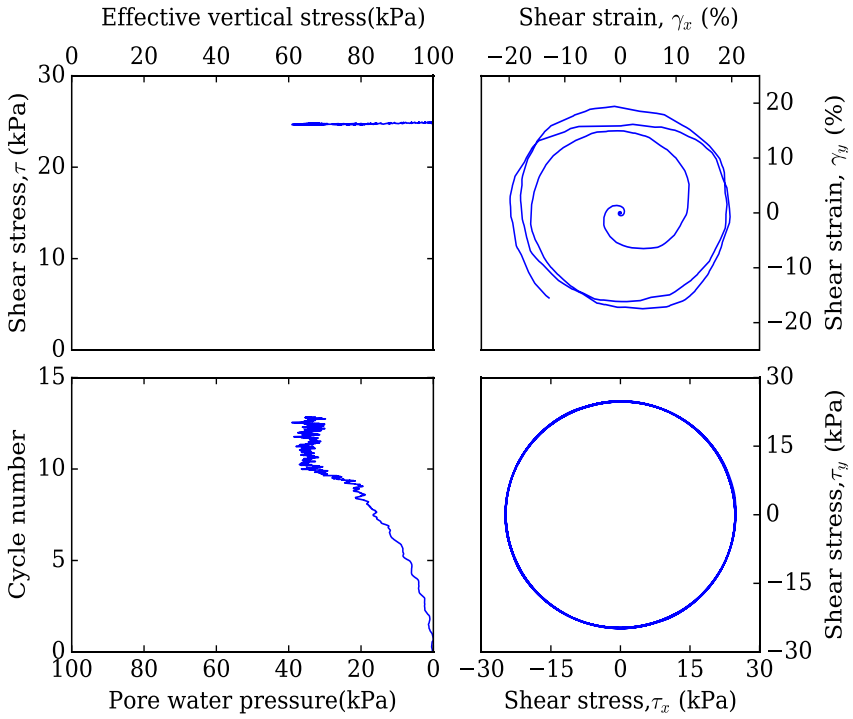
Figure 5.2 Different loading paths considered for this study

Table 5.1 Summary of Loading Paths

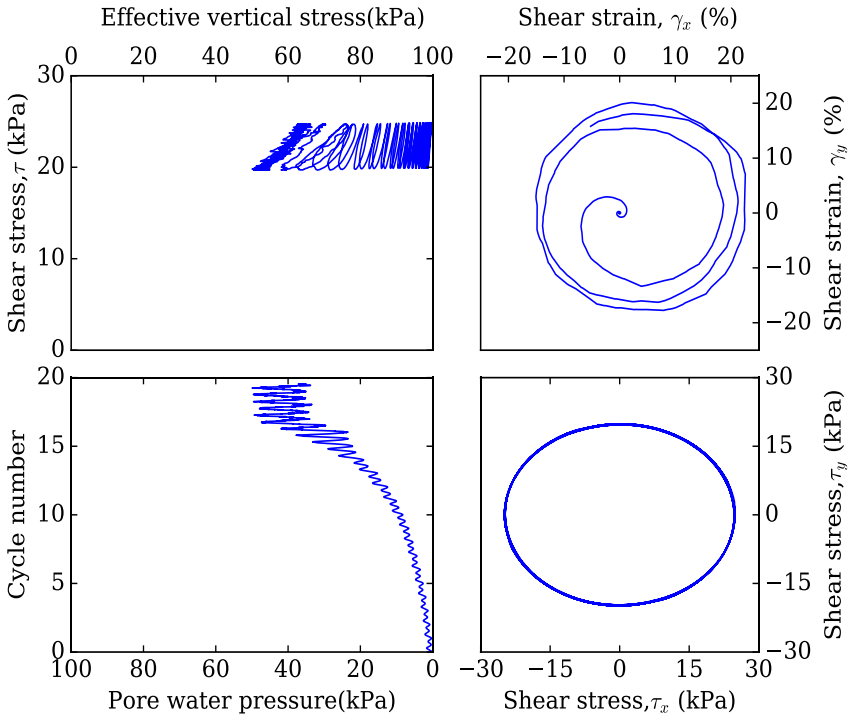
Loading path	$\boldsymbol{CSR_x}$	$\boldsymbol{CSR_y}$	$\boldsymbol{AR}$
Uni-directional loading	0.25	0	0
Oval loading		0.05	0.2
		0.10	0.4
		0.15	0.6
		0.20	0.8
Circular loading		0.25	1
Figure-8 loading		0.15	0.6
		0.25	1

5.2 Macroscopic soil behavior in multi-directional loading

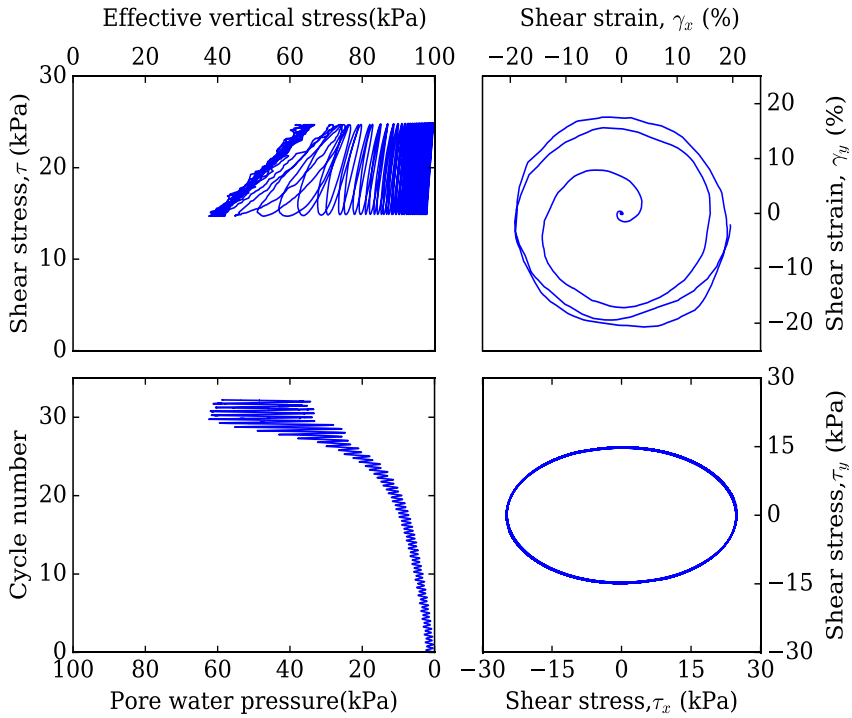
The multi-directional soil response under different loading paths is represented in the form of "4-way plots", as shown in Figure 5.3. In a 4-way plot, the bottom right plot is the loading path and the upper right plot is the development of shear strain in the XY plane corresponding to the loading path. The upper left plot is the variation in effective vertical stress (σ_v') plotted against the total shear stress ($\tau = \sqrt{\tau_x^2 + \tau_y^2}$) and the bottom left plot represents the development of excess porewater pressure against the loading cycle number. Only for the uni-directional response in Figure 5.3(h), the upper right plot shows the stress-strain plot in x-direction, i.e., $\tau_x - \gamma_x$, and the upper left plot represents the stress path in $\sigma_v' - \tau_x$ space. Overall, it is observed that the multi-directional response of soil in numerical simulations are qualitatively very similar to the experimental results reported by Kammerer (2002).



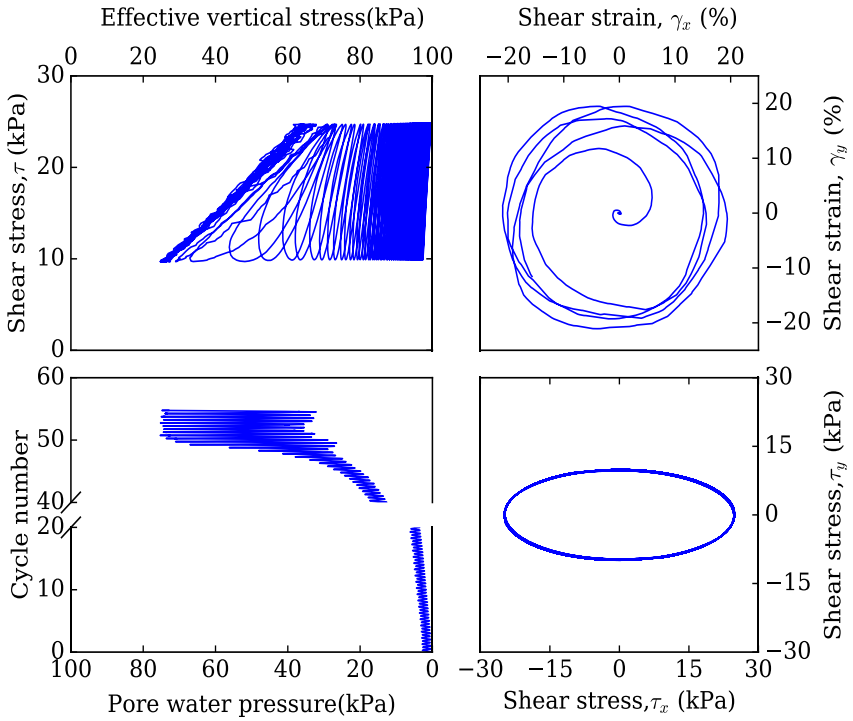
(a) Circular loading path



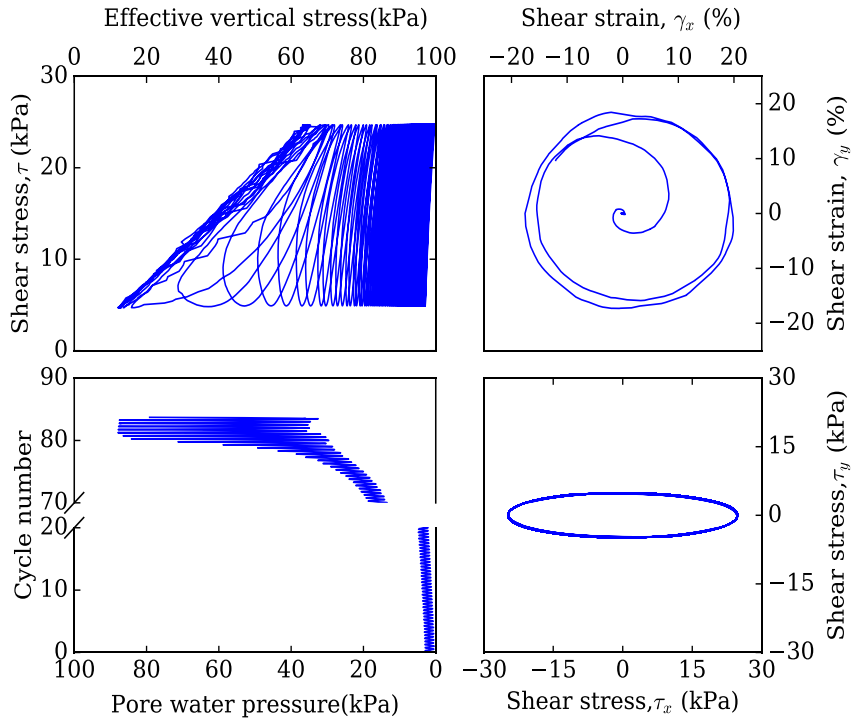
(b) Oval_0.8 loading path



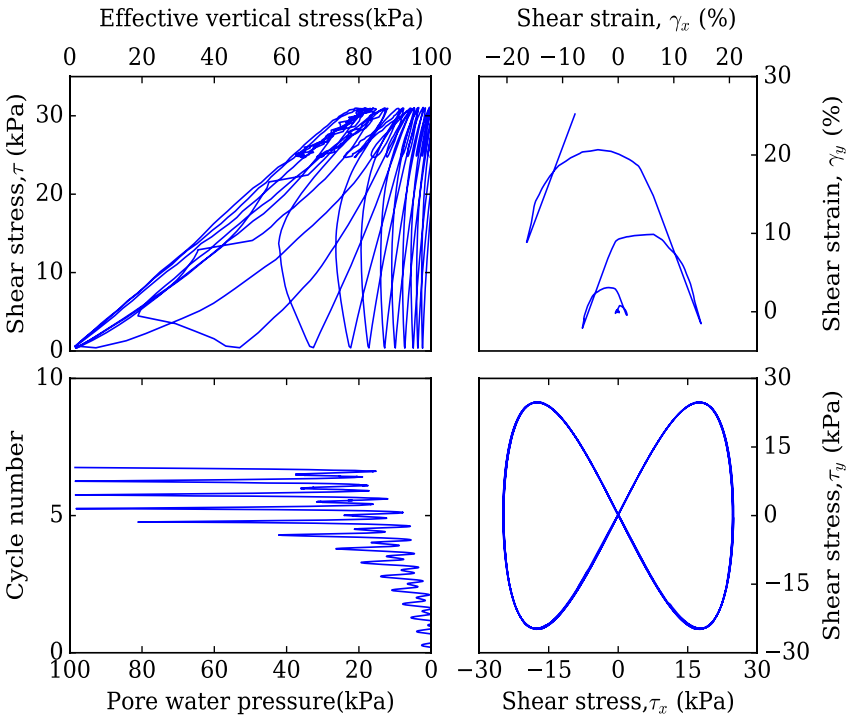
(c) Oval_0.6 loading path



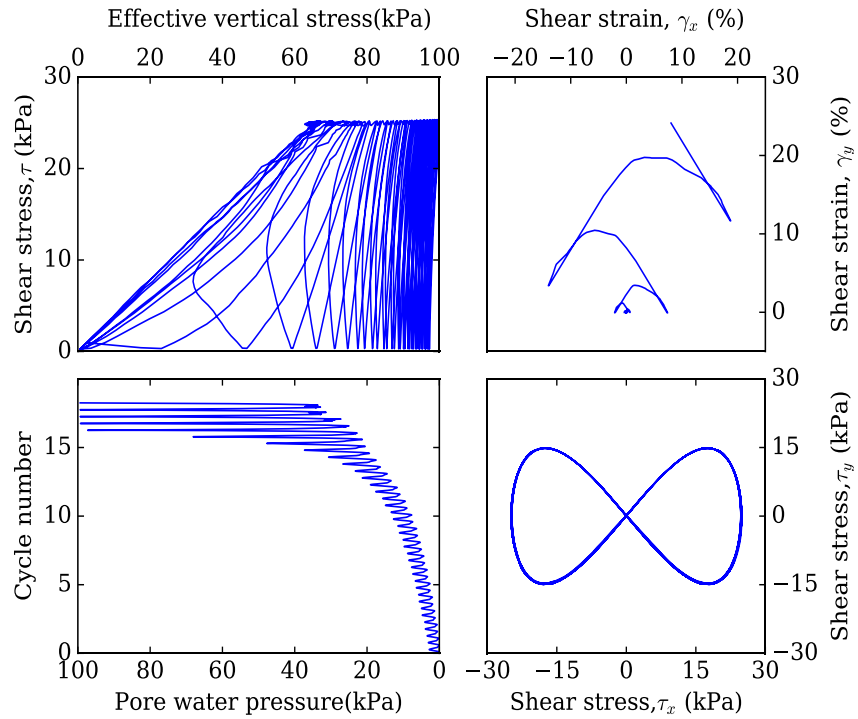
(d) Oval_0.4 loading path



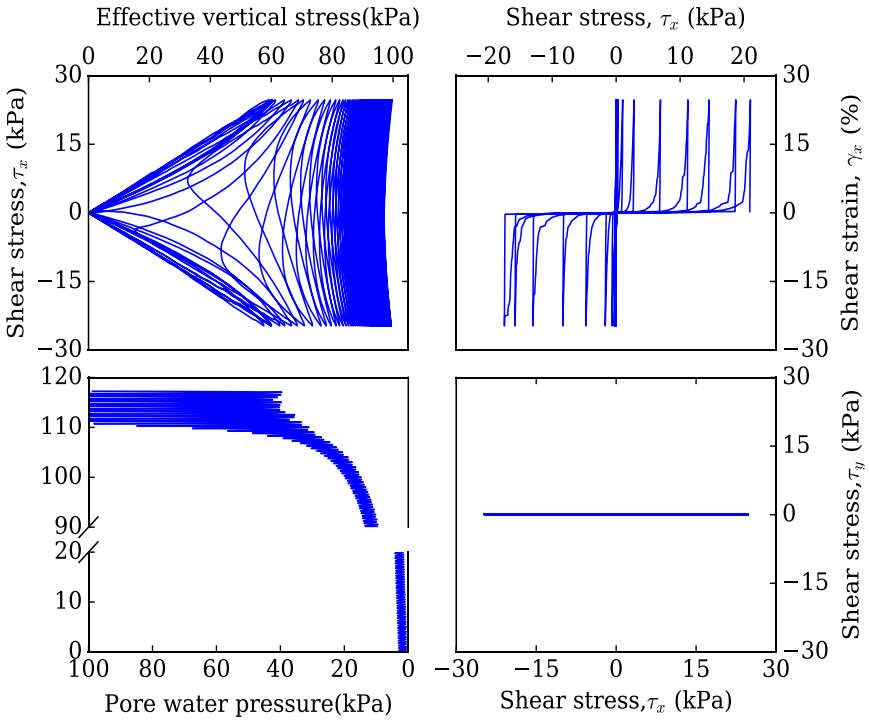
(e) Oval_0.2 loading path



(f) Figure-8_1 loading path



(g) Figure-8_0.6 loading path



(h) Uni-directional loading path

Figure 5.3 Macroscopic cyclic response in multi-directional loading paths

5.2.1 Development of shear strain

The top right plot of the 4-way plots shows the development of shear strain for different loading paths. In all the loading paths, the shear strains are observed to accumulate in both X and Y directions with every cycle of loading. The qualitative way these shear strains accumulate is found to depend on the type of loading path. For a figure-8 loading path, it is observed that shear strains in X-direction accumulate both in positive and negative directions, but in the Y-direction, the accumulation of shear strain is observed only in the positive Y-direction (refer to Figure 5.3(f,g)). With every cycle of figure-8 loading the granular packing tends to incline more towards the positive y-direction. For a circular/oval loading path, the accumulation of shear strain is observed in all the directions, i.e., in both positive and negative directions of both X and Y axes, resulting in the generation of strains in a circular fashion. Interestingly, after a certain number of loading cycles, the accumulation of shear strain in XY-plane is observed to cease, i.e., strains after reaching to about 20% in both X- and Y-directions ceases to develop further. In other words, an ultimate state in the development of shear strain in XY-plane (≈20% strain circle) is observed. Even in loading paths with different aspect ratio, the

ultimate state in strain levels is observed to be almost the same, i.e., the aspect ratio of loading path has no effect on the determination of ultimate state (refer to Figure 5.3(a-e)). For a uni-directional loading path, the shear strains are observed to accumulate only in positive and negative directions of the X-axis. An ultimate state in shear strain development is also observed in uni-directional loading, i.e., about 20% shear strain in positive and negative direction of the x-axis (refer to Figure 5.3h).

Figure 5.4 shows the development of the total shear strain (γ=$\sqrt{\gamma_x^2 + \gamma_y^2}$) against the number of cycles of loading in different loading paths. In all the loading paths, the total shear strain generated is insignificant for most of the initial cycles, whereas, after a certain number of loading cycles, the total shear strain starts to increase tremendously, i.e., within a few cycles of further loading, the shear strain reached to about 20-25%. This resembles a microstructure breakdown and sudden failure in the granular assembly.

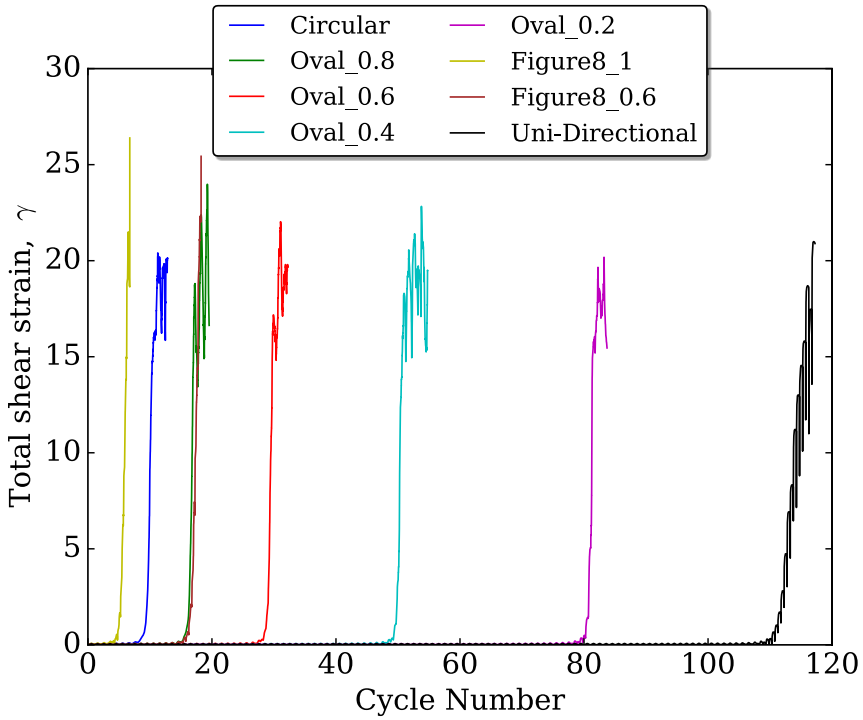


Figure 5.4 Total shear strain generation in multi-directional loading paths

5.2.2 Development of excess pore water pressure

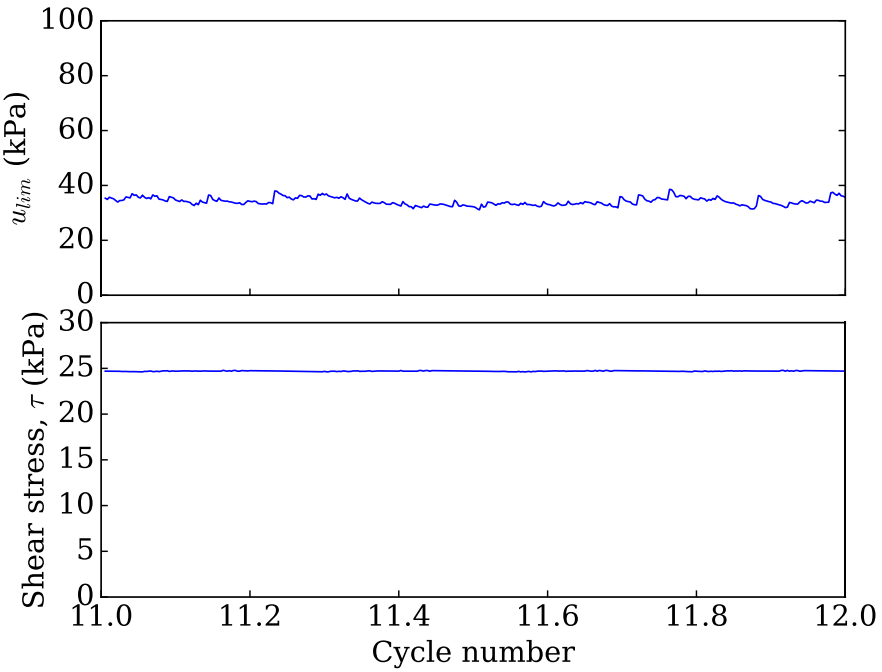
The left bottom plot of the 4-way plots shows the development of excess pore water pressure for different loading paths. With the increase in the number of loading cycles, the effective vertical stress is observed to decrease by accompanying the increase in excess pore water pressure for all the loading paths (refer to Figure 5.3). When the packing is cyclically sheared for long enough, the stress path reaches a limiting state where the pattern of development in excess pore water pressure remains the same even on further loading. It is observed that at limiting state with the increase in total shear stress ($\tau = \sqrt{\tau_x^2 + \tau_y^2}$) the vertical effective stress has also increased, thereby decreasing the excess pore water pressure. Figure 5.5 shows the variation of excess pore water pressure and the total shear stress in one cycle of loading in limiting state of different loading paths. For all the loading cases, the u_{lim} varies inversely with the total shear stress. Also, it is observed that only for the uni-directional and figure-8 loading test cases the maximum value of u_{lim} i.e., $u_{lim,max}$ reaches to initial confining pressure of 100 kPa. Whereas for the other loading cases, the $u_{lim,max}$ is always smaller than the initial confining pressure. Figure 5.6 shows the relationship between $u_{lim,max}$ and AR, the equation for their correlation is given as:

$$u_{lim,max} = -61.7AR + 100 \tag{5.3}$$

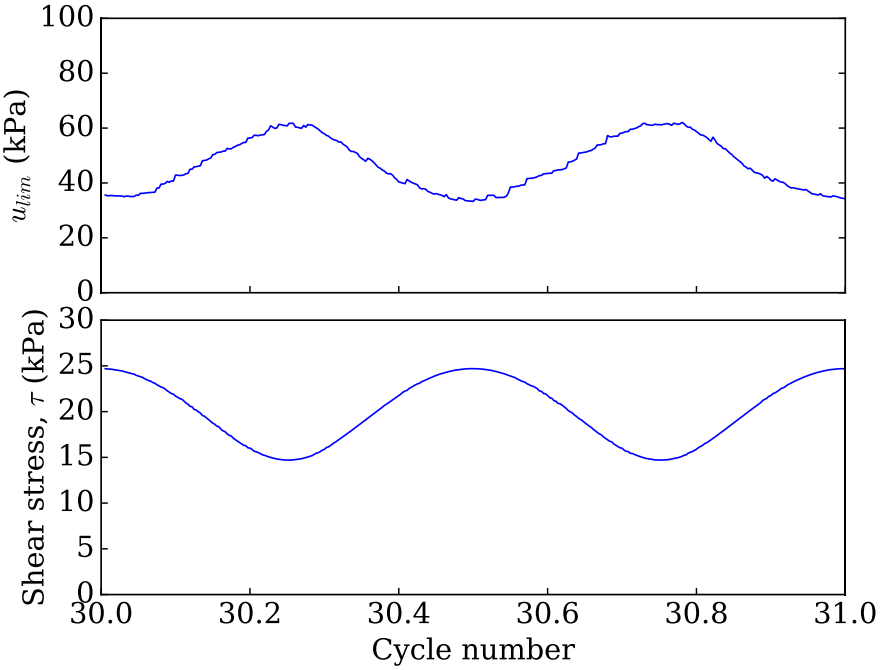
When normalized with the initial confining pressure, the maximum excess pore pressure ratio $r_{u,max}$ is given as:

$$r_{u,max} = -0.62AR + 1 \tag{5.4}$$

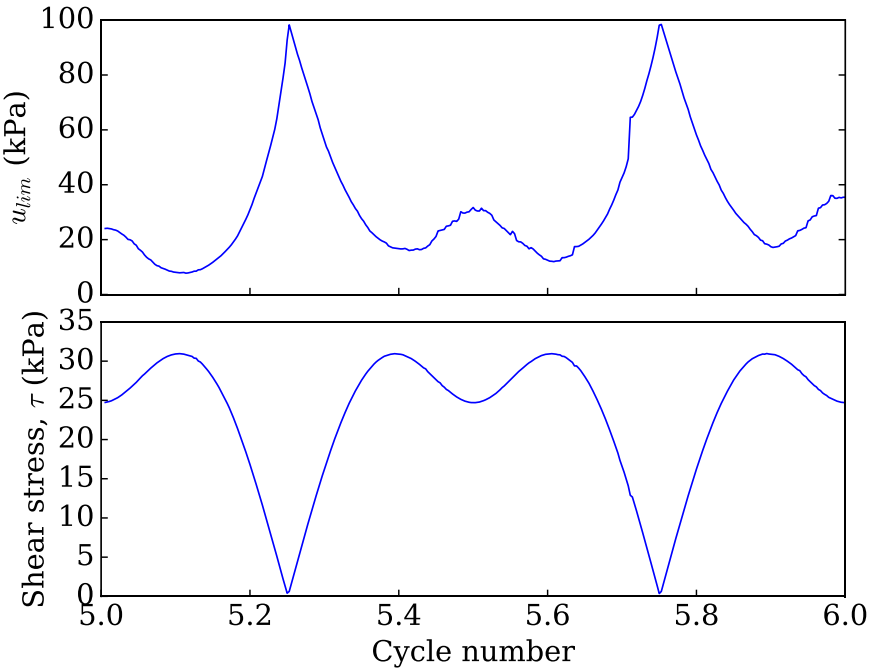
In general, $r_{u,max}$ is used as an indicator for the degree of liquefaction, where $r_{u,max} = 1$ represents full liquefaction. As per this definition, liquefaction can only occur in case of uni-directional and figure-8 loading tests. But, for other loading tests, though the $r_{u,max}$ values never reaches 1, the shear strains developed in the granular assembly are tremendous. This shows that large strains can be developed even when the $r_{u,max} < 1$. Thus, a new criterion to decide liquefaction failure in cyclic loading should be adopted, which is consistent for all the loading tests.



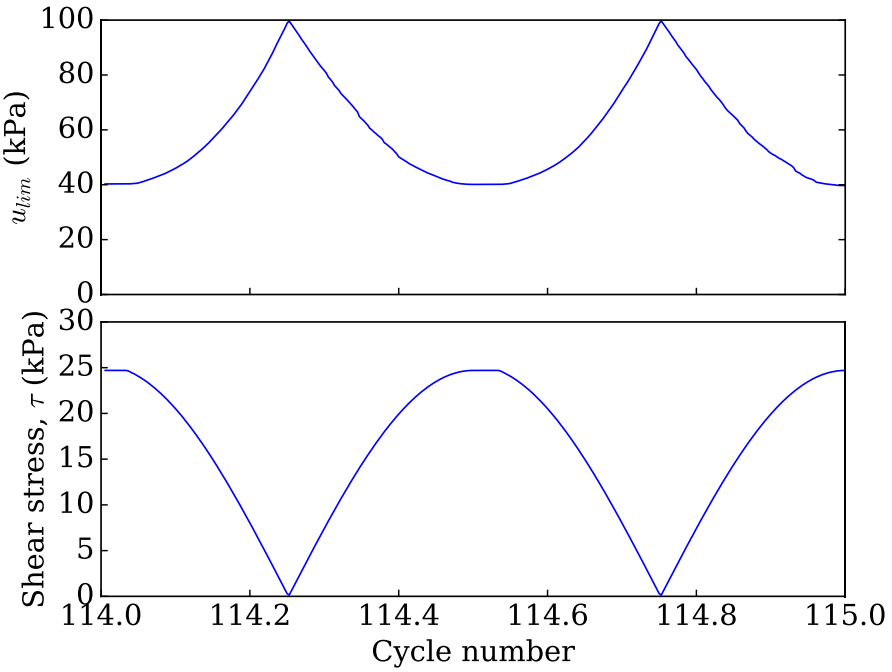
(a) Circular loading path



(b) Oval_0.6 loading path



(c) Figure-8_1 loading path



(d) Uni-directional loading

Figure 5.5 Variation of excess pore water pressure in one cycle of limiting state

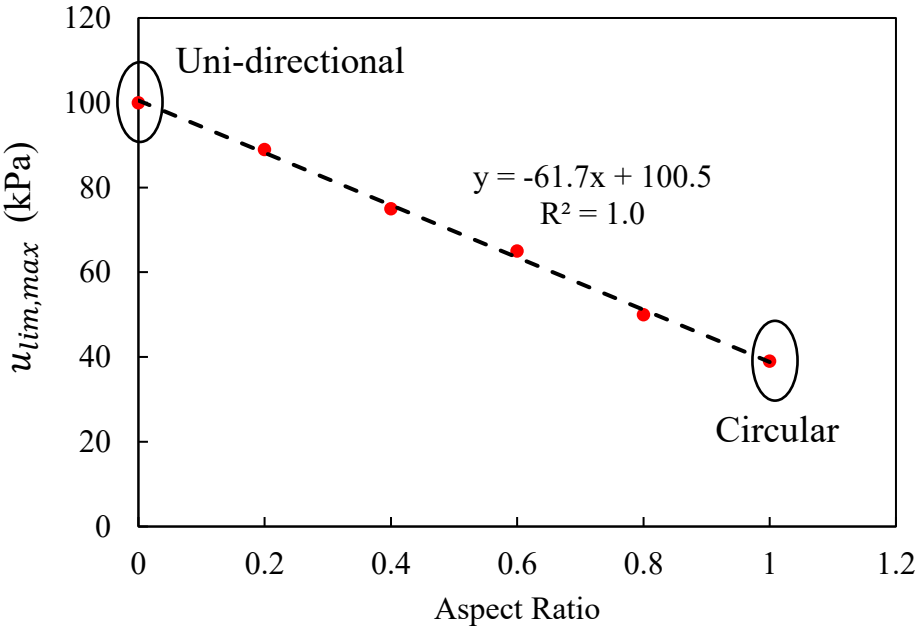


Figure 5.6 Relation between $u_{lim,max}$ and AR of circular/oval loading paths

5.2.3 Liquefaction resistance

In this study, the failure criterion adopted by Ishihara and Yamazaki (1980) is used to determine the resistance to cyclic liquefaction. The liquefaction resistance is defined as the number of loading cycles required to generate a total shear strain ($\gamma = \sqrt{\gamma_x^2 + \gamma_y^2}$) greater than 3%. Figure 5.7 shows the number of cycles taken by different loading paths to reach 3% total shear strain, against the AR values of the loading paths. It is observed that the cyclic resistance for the uni-directional loading path is highest compared to other loading paths. For the oval/circular loading paths the number of cycles to reach initial liquefaction decreases with the increase in AR value thus circular loading path having the lowest cyclic resistance. Also, for the same CSR_x and the CSR_y values, figure-8 loading path has lower cyclic resistance compare to the oval/circular loading paths. Overall it can be observed that with the addition of cyclic stress component in the second direction, the soil packing is more prone to failure.

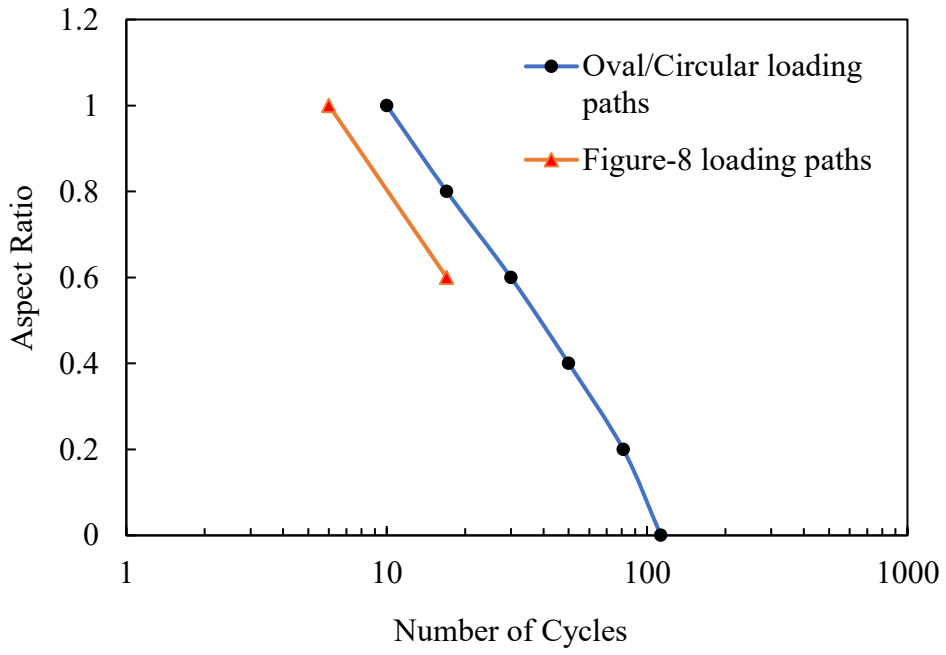


Figure 5.7 Liquefaction resistance in different loading paths

5.3 Contact-fabric evolution in multi-directional Loading

In Section 3.2 of this thesis two fabric indicators, Z and a_c, are introduced that are based on particle-contact information. Z is the average coordination number and a_c is the degree of

anisotropy in the orientation of contact normals. a_c is computed as an invariant of the deviatoric part of the fabric tensor, $\boldsymbol{\phi}$. The contact-normal fabric tensor ($\boldsymbol{\phi}$) is a second-order tensor and is the directional indicator of contact normals. The evolution of the fabric tensor and the fabric indicators in the multi-directional loading is discussed in this section.

5.3.1 Evolution of coordination number (Z)

Figure 5.8 shows the evolution of coordination number against the number of cycles for different loading paths. The initial coordination number of the RVE is about 4.1. When subjected to different multi-directional loading paths, it is observed that Z drops from 4.1 and gradually evolves to about 3.2 when it reaches the limiting state. The rate of evolution in Z is different for different loading paths. The maximum coordination number in limiting state is about 3.2 in all the loading paths.

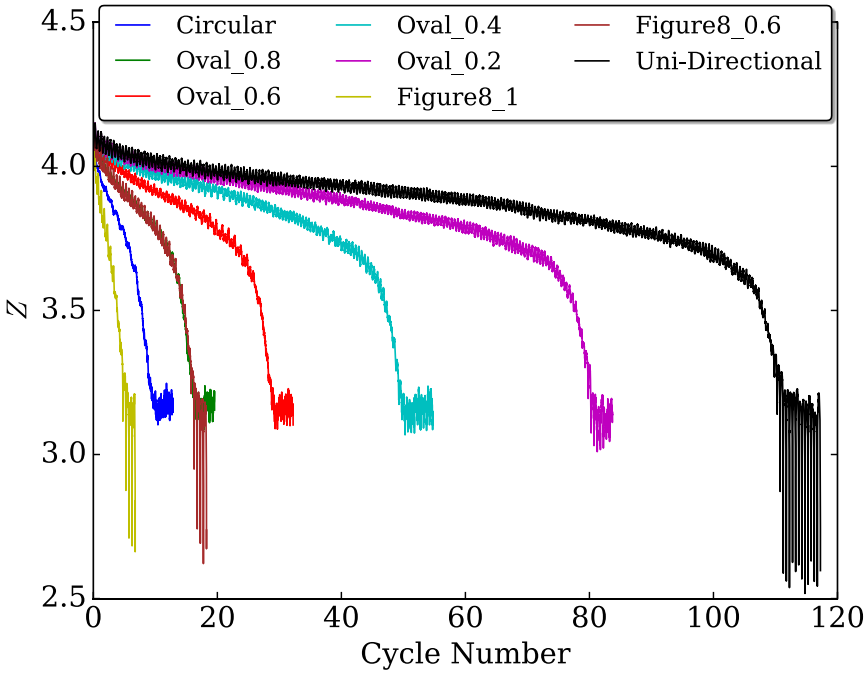
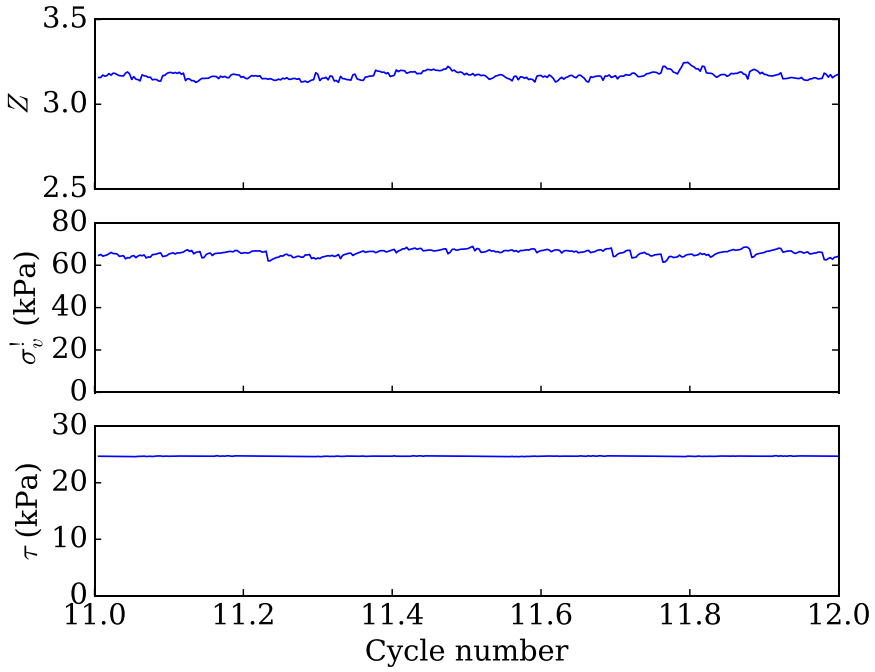
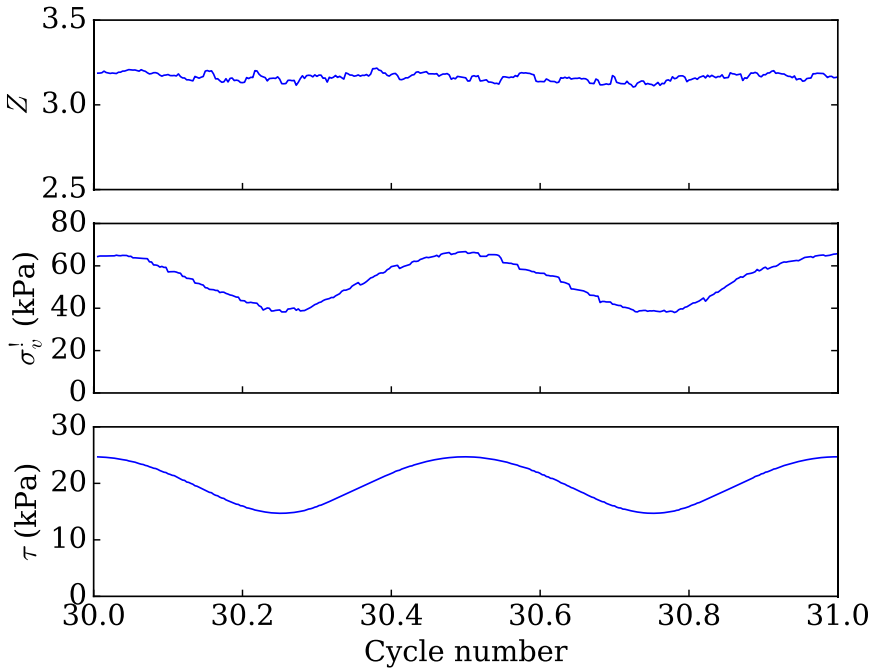


Figure 5.8 Variation of coordination number in different loading paths

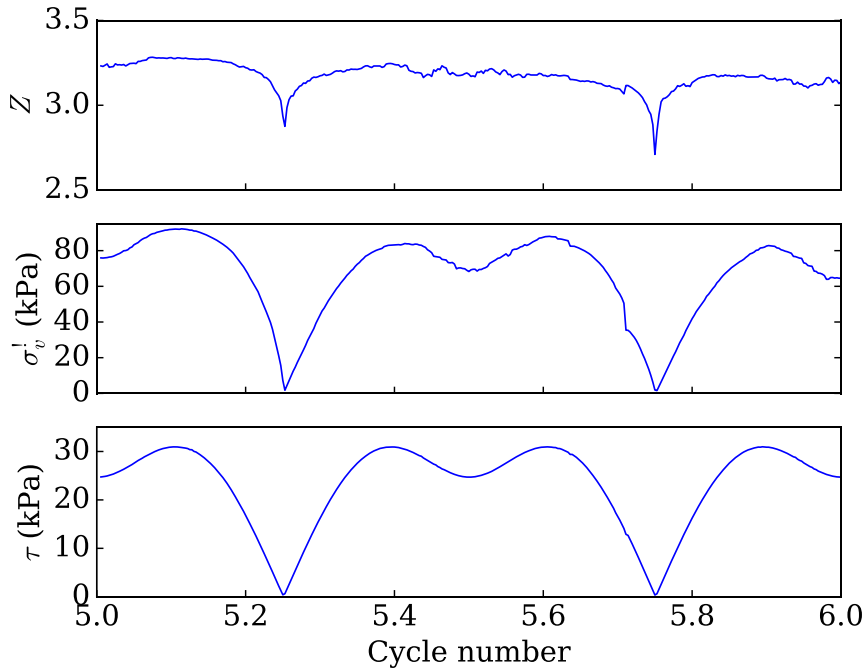
Figure 5.9 shows the evolution of coordination number, variation of effective vertical stress, and total shear stress in one cycle of loading in the limiting state for different loading paths. In the case of circular/oval loading path, in the limiting state, the coordination number in the sample is always around 3.2 (refer to Figure 5.9 (a,b)). There are no significant fluctuations in the evolution of coordination number, even for a significant variation in shear stress and effective vertical stress (refer to Figure 5.9b). However, for uni-directional and figure-8 loading paths, in the limiting state, the coordination number further drops from 3.2, as shown in Figure 5.9 (c,d). Both the uni-directional and the figure-8 loading paths go through the zero total shear stress twice in one cycle of loading. It is observed that when the shear stress approaches to zero, the effective vertical stress also approaches to zero, resulting in a sudden dip in coordination number. The dip in coordination number corresponds to the flow in granular assembly resulting in the generation of high shear strains. With continuous shearing, the interparticle contacts re-establish and form a load-bearing structure. During which the coordination number evolves back to 3.2 as the effective vertical stress increase.



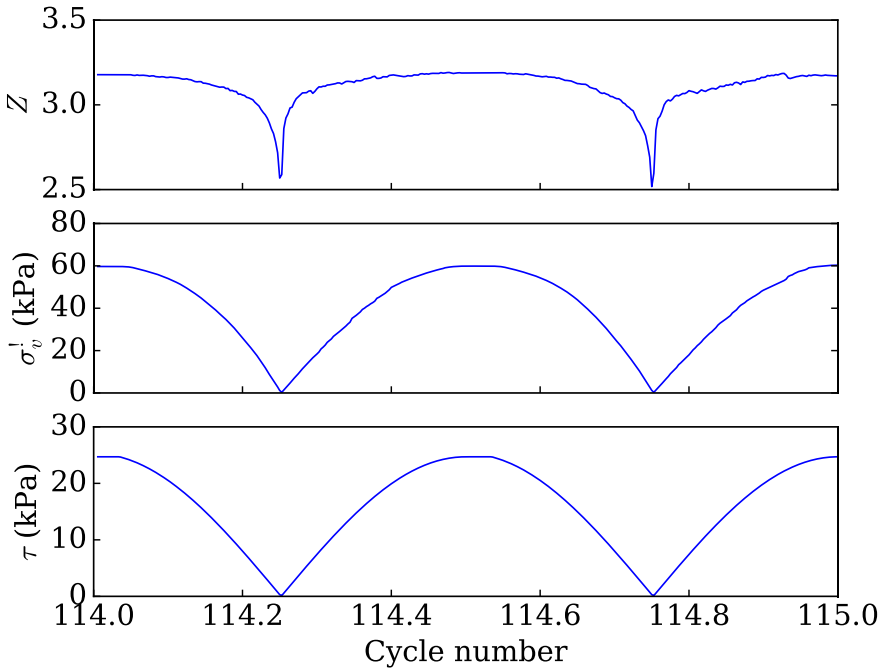
(a) Circular loading path



(b) Oval_0.6 loading path



(c) Figure-8_1 loading path



(d) Uni-directional loading path

Figure 5.9 Variation of coordination number and vertical effective stress in the limiting state

Overall, in the limiting state, the load-bearing structure formed would be very unstable. To sustain any shear stress, the granular assembly would develop a coordination number of 3.2 and collapses immediately if the shear stress is zero.

5.3.2 Evolution of fabric tensor ($\boldsymbol{\phi}$) and degree of anisotropy in contacts (a_c)

As mentioned in section 3.2.2, the contact-based fabric tensor ($\boldsymbol{\phi}$) is defined as:

$$\phi_{ij} = \frac{1}{N_c}\sum_{k=1}^{N_c} n_i^{(k)} n_j^{(k)} \tag{5.5}$$

where $\boldsymbol{n}^k$ is the kth contact normal and N_c is the total number of contacts. $\boldsymbol{\phi}$ is a second-order symmetric ($\phi_{ij} = \phi_{ji}$) fabric tensor and is the directional indicator of the contact normal. The initial sample is isotropically consolidated using spherical particles. Thus, no inherent anisotropy in contact normal is developed. Therefore, non-diagonal terms are approximately

zero, i.e., $\phi_{xy} = \phi_{zy} = \phi_{zx} \approx 0$, and diagonal terms are equal to 1/3, i.e., $\phi_{xx} = \phi_{yy} = \phi_{zz} \approx 1/3$.

Figure 5.10 shows the evolution of each component of fabric tensor (ϕ_{ij}) in a circular loading test. It is observed that the diagonal components of the fabric tensors have no significant evolution and almost remains constant (≈ 0.33) during the whole loading process. Whereas the off-diagonal terms, ϕ_{xz} and ϕ_{yz} have significant variation during the whole loading process. It is due to the fact that shear stress is applied in the XY-plane. The evolution in these components also contributes to the development of anisotropy in the granular assembly.

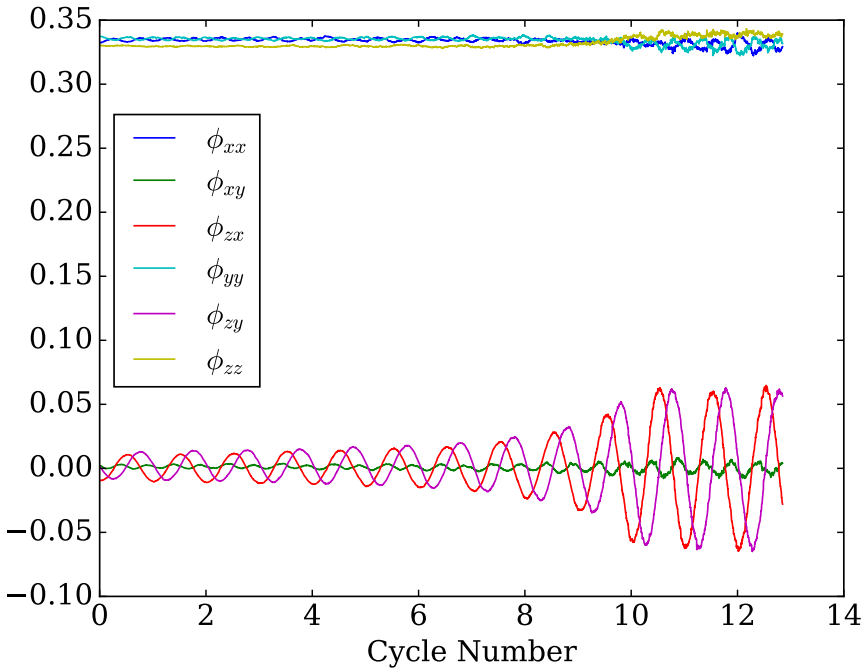
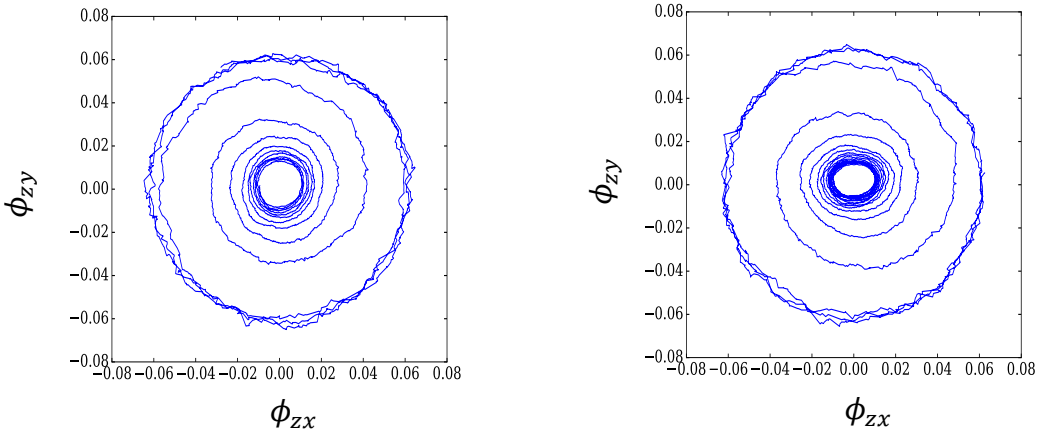


Figure 5.10 Evolution of fabric tensor components in the circular loading path

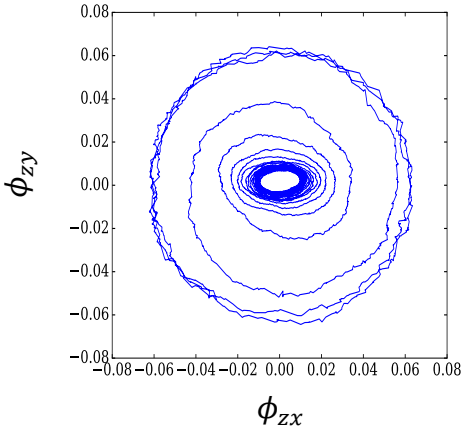
Figure 5.11 shows the evolution in the fabric tensor components ϕ_{zx} versus ϕ_{zy} for circular/oval loading paths. The initial evolution in the fabric components is observed to take the trajectory similar to the loading path, i.e., circular/oval. With the increase in cycle number,

the trajectory of fabric evolution gradually expands to a bigger circle and gets stabilized when it approaches the limiting state.

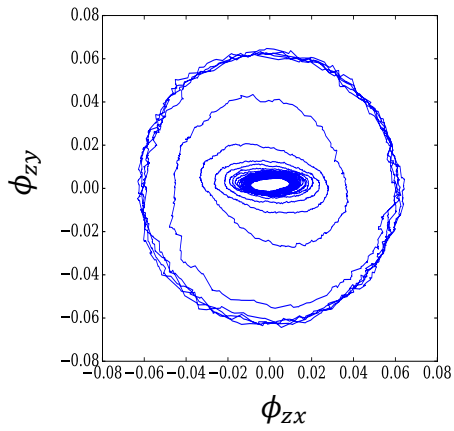


(a) Circular

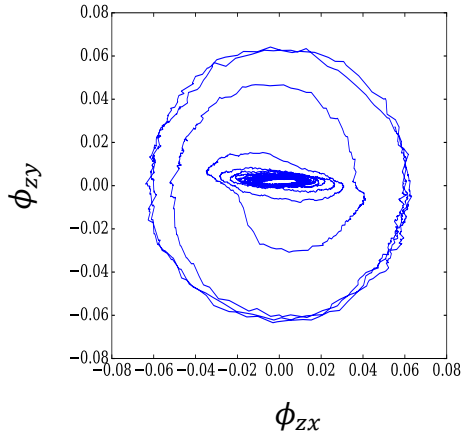
(b) Oval-0.8



(c) Oval-0.6



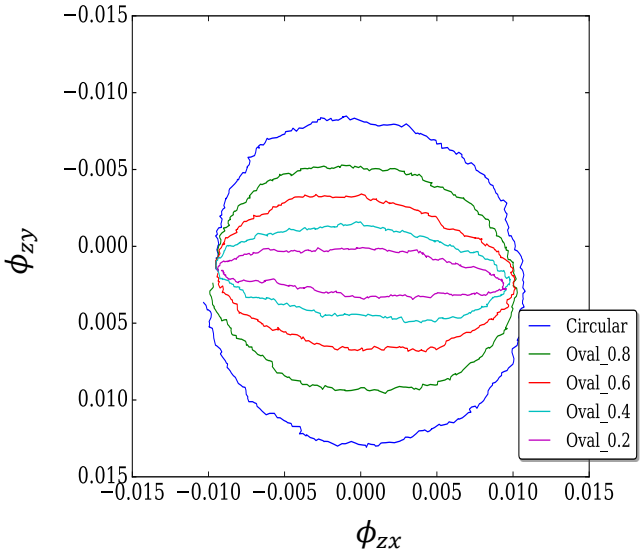
(d) Oval-0.4



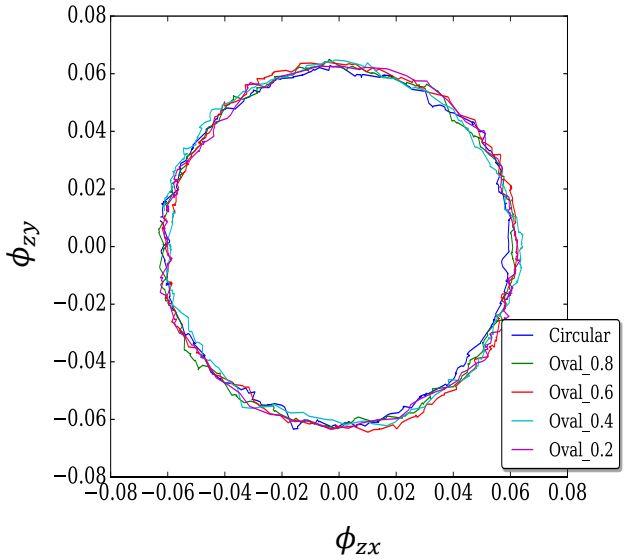
(e) Oval-0.2

Figure 5.11 Evolution of ϕ_{zx} and ϕ_{zy} in circular/oval loading paths

Figure 5.12a shows the evolution of the fabric components in the first cycle of loading and the cycle corresponding to the limiting state for different loading paths. It is evident in the first cycle of loading, the trajectory of fabric evolution is analogous to the loading path whereas in the limiting state the trajectory of fabric evolution is independent of the loading path and it follows a big circle of radius 0.06 as shown in Figure 5.12b.



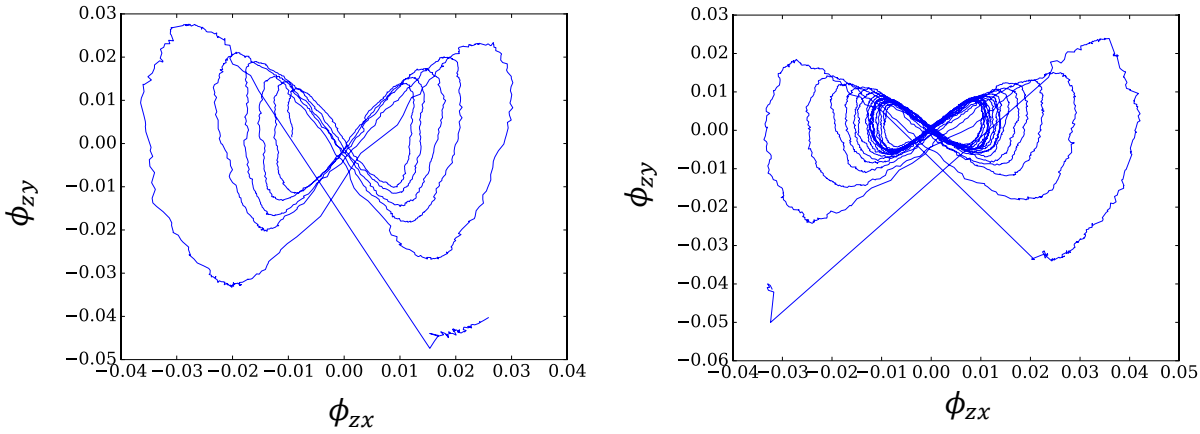
(a) First cycle



(b) Last cycle

Figure 5.12 Evolution ϕ_{zx} and ϕ_{zy} in first and last cycle of circular/oval loading paths

Unlike the behavior in circular/oval loading paths, granular packing subjected to unidirectional and figure-8 loading path undergoes full liquefaction, resulting in flow strains. In the flow stage, the average number of contacts hits a low number with no load-carrying ability between the contacts. At such a low coordination number, the fabric components, ϕ_{zx} and ϕ_{zy} may reach to unreasonable values. The evolution of fabric tensor components in figure-8 loading paths (AR= 1 and 0.6) before full liquefaction is shown in Figure 5.13. The trajectory of fabric components evolution in figure-8 loading paths is also a figure-8 that evolves to a bigger size gradually with increase in the number of cycles in loading (refer to Figure 5.13(a,b)). For the uni-directional loading as the shear stress vary only in the x-direction, the evolution of ϕ_{zy} is insignificant compared to the evolution of ϕ_{zx}. Figure 5.14 shows the evolution of the fabric tensor component ϕ_{zx} in uni-directional loading and the variation in shear stress is also shown for reference. As the number of cycles increases the ϕ_{zx} gradually increases.



(a) Figure-8_1

(b) Figure-8_0.6

Figure 5.13 Evolution of ϕ_{zx} and ϕ_{zy} in figure-8 loading paths

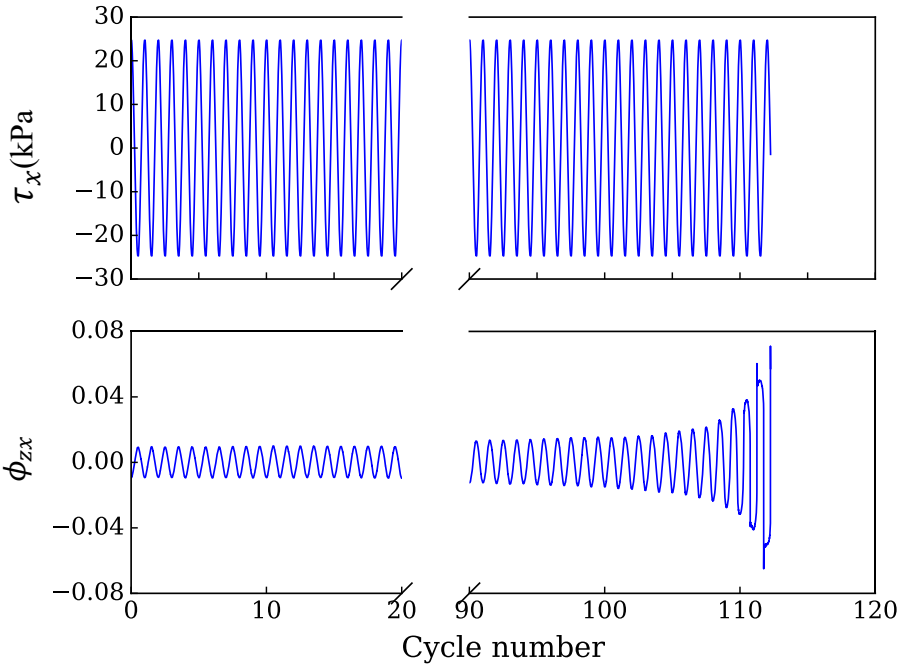


Figure 5.14 Evolution of ϕ_{zx} in uni-directional loading

The fabric components ϕ_{zx} and ϕ_{zy} have a maximum contribution to the quantification of the degree of anisotropy (a_c). The evolution in contact anisotropy degree (a_c) against the cycle number for different loading paths is shown in Figure 5.15. As the initial sample has no inherent anisotropy the a_c value is very low, i.e., about 0.14. The rate of evolution of a_c is different for different loading paths but when stress path reaches to the limit state, the anisotropy degree increases gradually to about 0.8. Figure-8 loading path with AR equals to 1 has the fastest evolution in a_c and uni-directional loading has the slowest evolution.

5.4 Evolution of particle-void descriptors in multi-directional loading

The concept of particle-void cells and the definitions for particle void descriptors, E_d and A_d are introduced in section 3.3 of this thesis. E_d is the global shape (elongation) descriptor and A_d is the degree of anisotropy in the particle-void cell orientations. E_d and A_d are obtained from the statistical analysis of e_d and $\boldsymbol{v}$, where e_d measures the local shape elongation of

particle-void cells and $\boldsymbol{v}$ provides the principal orientation of the same. A fabric tensor ($\boldsymbol{V}$) is also defined based on the orientation of particle-void cells.

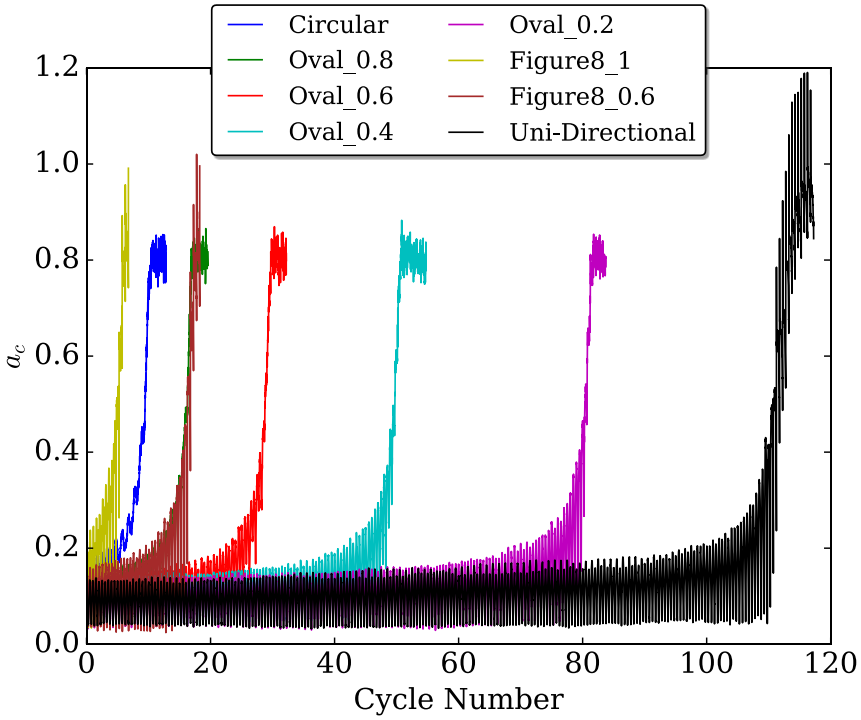
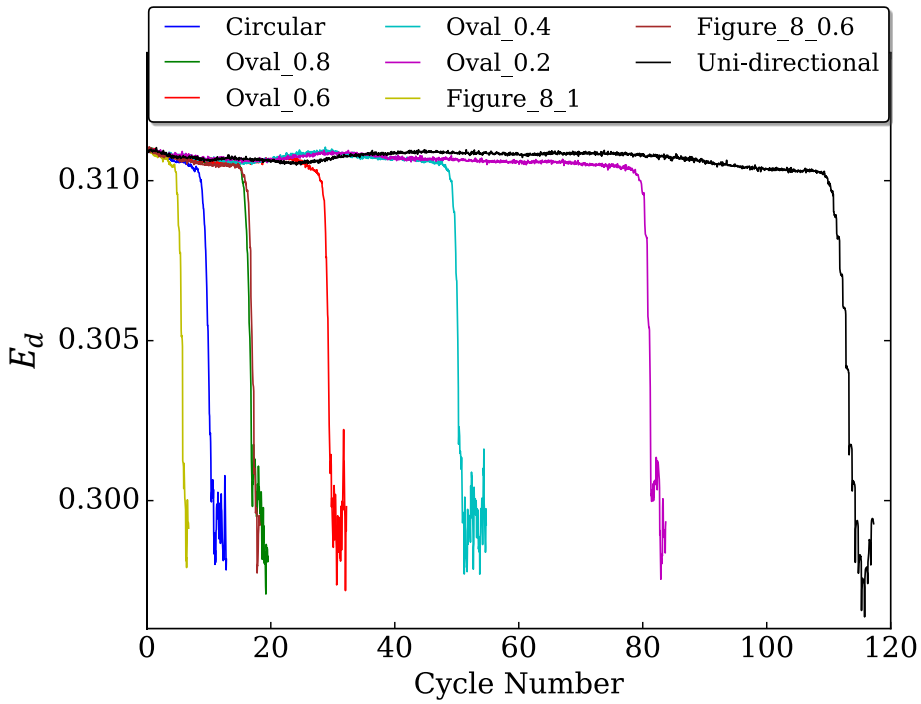


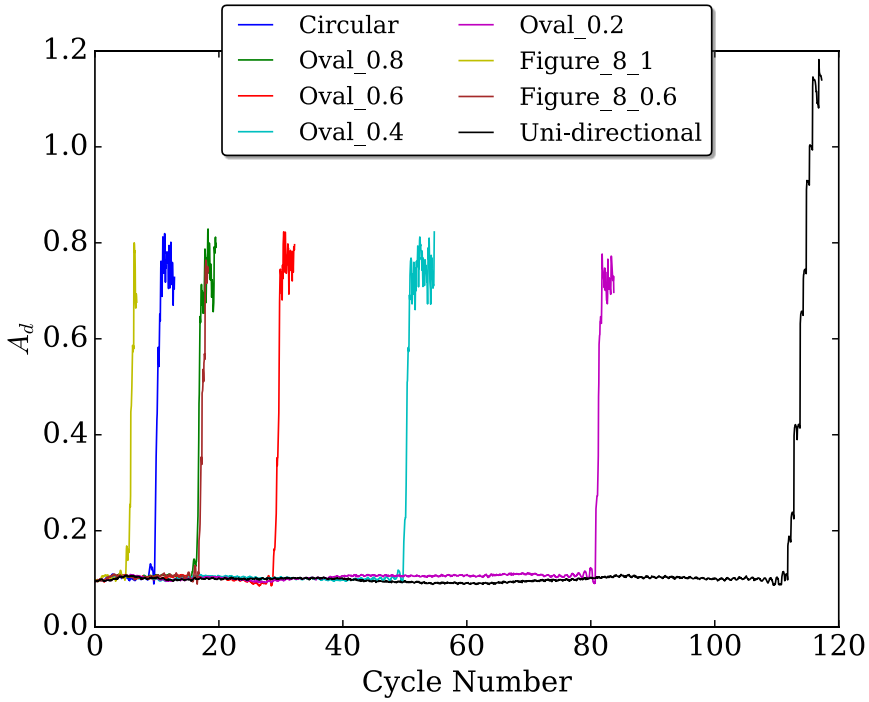
Figure 5.15 Evolution of a_c in different loading paths

5.4.1 Evolution of E_d and A_d

The evolution of E_d and A_d with cycle number in different multi-directional loading paths is shown in Figure 5.16. It is observed that in all the loading cases the evolution in E_d and A_d is constant during the initial cycles of loading where E_d is about 0.311 and A_d is about 0.1. The low value of A_d represents the absence of anisotropy in the particle void cells orientation. Depending on the loading path, after a certain number of cycles, the evolution in E_d shows a sudden decrease in trend whereas the evolution in A_d shows a sudden increase in trend. Eventually, in uni-directional and circular/oval loading paths the evolution in E_d and A_d gets stabilized. In circular/oval loading path, E_d reaches a stable value of around 0.3 and A_d is about 0.75. Relatively a higher decrease in E_d and a higher increase in A_d is observed in the case of uni-directional loading. Note that simulations for figure-8 loading paths aren't continued till the ultimate state in shear strain is reached, thus, stabilized values of E_d and A_d are not known.



(a) Evolution of E_d



(b) Evolution of A_d

Figure 5.16 Evolution of particle-void fabric descriptors in different loading paths

Figure 5.17 shows the evolution of E_d and A_d of different circular/oval loading paths in $E_d - A_d$ space. In case of uni-directional loading the granular packing undergoes full liquefaction. In the post-liquefaction stage, the sample undergoes repeated transition from flow stage to the hardening stage by accumulating shear strains in different cycles. The point at which the behavior changes from a "flow like" state to the "hardening like" state in case of uni-directional loading are referred to as critical points and the coordination number at which it occurs is referred to as Z_{crit} and is unique in all the cycles. The critical points in case of unidirectional loading in different cycles are also shown in Fig. 5.17. It is clear that the critical points delineate a line which separates the flow stage from the hardening stage and is referred to as hardening state line. It is observed that the evolution in E_d and A_d for different circular/oval loading paths always lie above the hardening state line, which indicates that the packing is always in hardening stage.

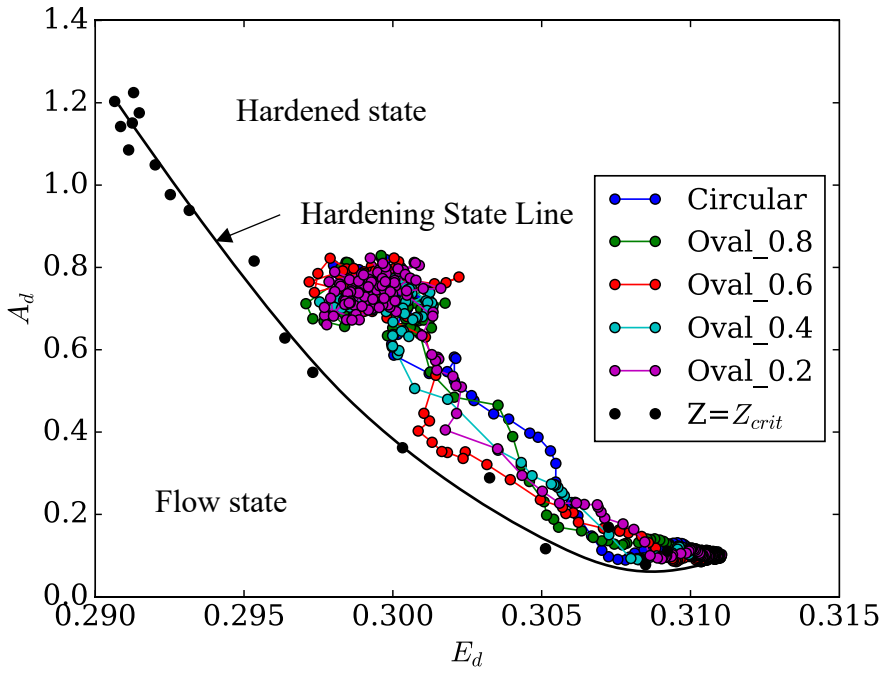
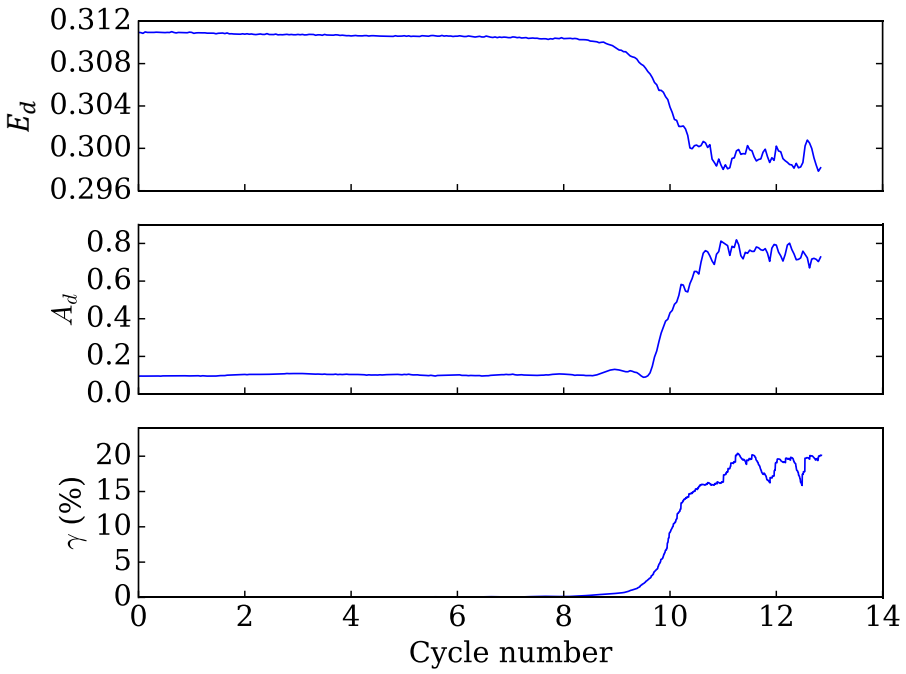
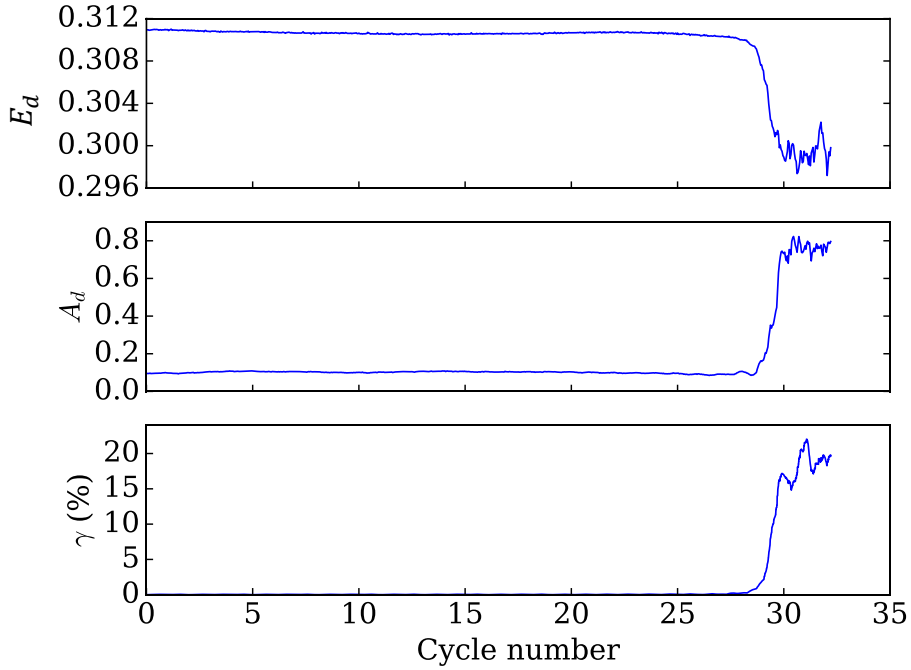


Figure 5.17 $E_d - A_d$ Space

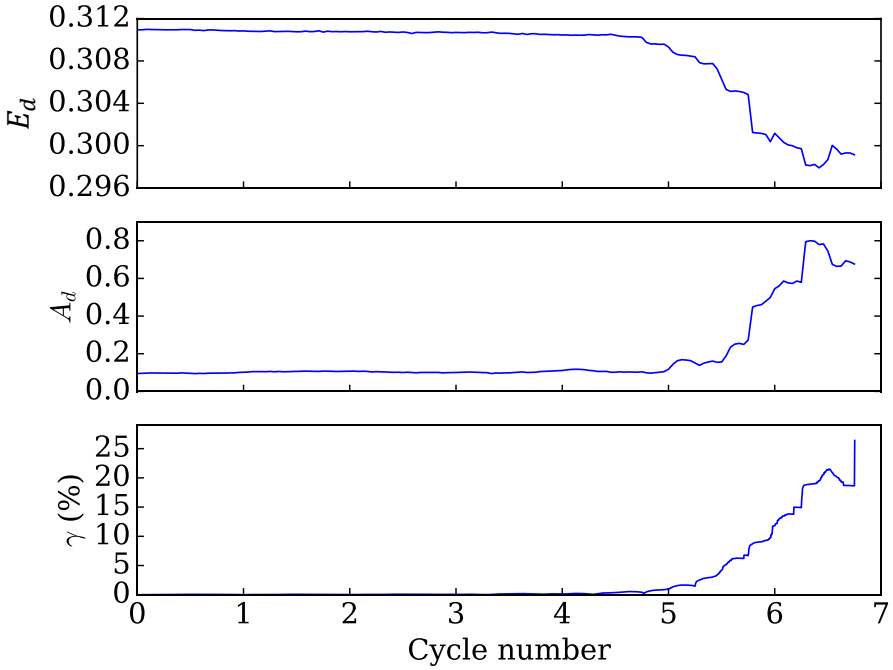
Overall, the evolution of E_d and A_d is analogous to the development of shear strain in the granular assembly. Figure 5.18 shows the evolution of $E_d$and A_d in comparison with the development of total shear strain ($\gamma = \sqrt{\gamma_x^2 + \gamma_y^2}$) against cycle number for different loading paths. In the circular loading path, as shown in Figure 5.18a, the shear strain in the granular assembly start to develop only after about 9 cycles of loading. Interestingly, the evolution in E_d and A_d also starts at the same moment, where E_d starts to decrease and A_d starts to increase. When the total shear strain got stabilized after about 11 cycles of loading, the evolution in E_d and A_d also got stabilized. Similar observations were made in other loading paths as well (refer to Figure 5.18(b,c,d)). This shows a clear correlation between particle-void descriptors and the total shear strain generated in the sample.



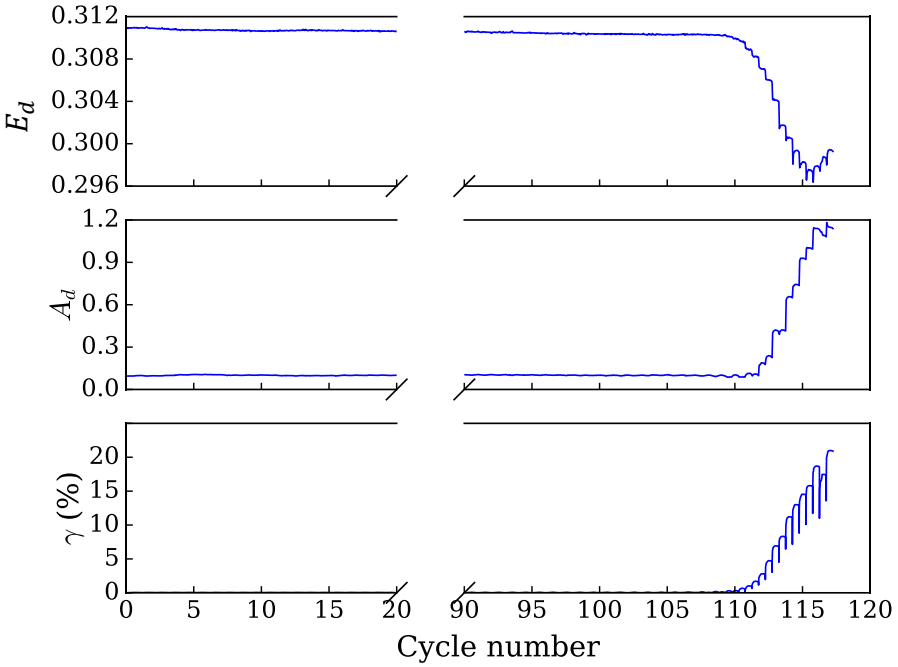
(a) Circular



(b) Oval_0.6



(c)Figure-8_1



(d) Uni-directional

Figure 5.18 Comparison of evolution in particle-void fabric and shear strain generation

5.4.2 Evolution of particle-void fabric tensor

As mentioned in section 3.3.3.3, the particle void fabric tensor ($\boldsymbol{V}$) is defined as:

$$V_{ij} = \frac{1}{N_v}\sum_{k=1}^{N_v} v_i^{(k)} v_j^{(k)} \tag{5.6}$$

where $\boldsymbol{v}^k$ is the orientation of kth particle-void cell and N_v is the total number of particle-void cells. $\boldsymbol{V}$ is a second-order symmetric ($V_{ij} = V_{ji}$) fabric tensor and is the directional indicator of the orientations of particle void cells. The initial sample is isotropically consolidated and the particle-void cells are randomly oriented. The non-diagonal terms are approximately zero, i.e., $V_{xy} = V_{zy} = V_{zx} \approx 0$, and diagonal terms are approximately equal to 1/3, i.e., $V_{xx} = V_{yy} = V_{zz} \approx 1/3$.

Figure 5.19 shows the evolution of each component of fabric tensor (V_{ij}) in a circular loading test. It is observed that the diagonal components of the fabric tensors have no significant evolution and almost remains constant (≈ 0.33) during the whole loading process. Whereas the off-diagonal terms, V_{zx} and V_{zy} have significant variation during the whole loading process. It is due to the fact that shear stress is applied in the XY-plane. The evolution in these components also contributes to the development of anisotropy in void space in the granular assembly. The variation in V_{xy} is insignificant compared to the variation in other off-diagonal components and is ignored.

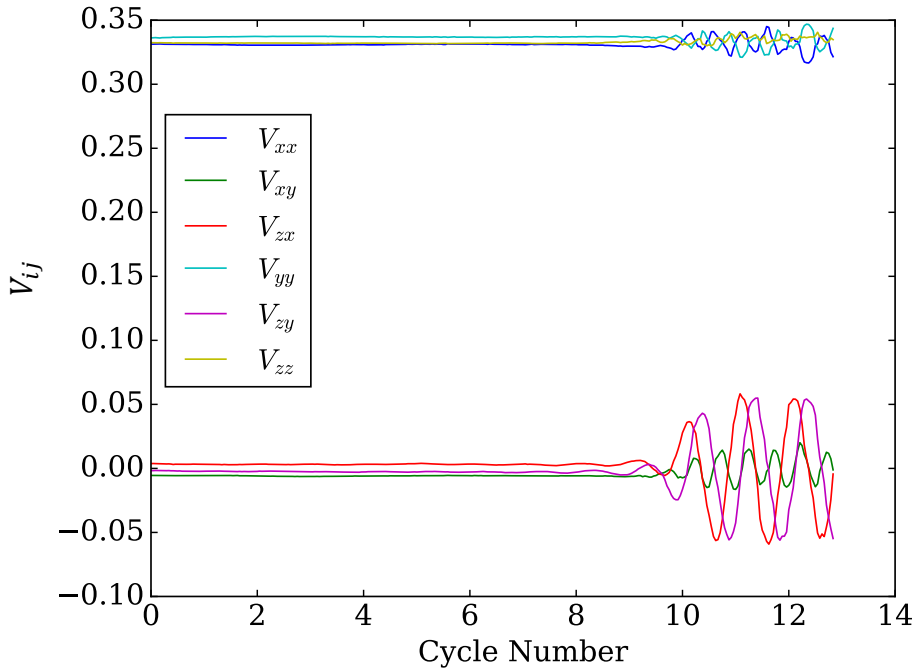
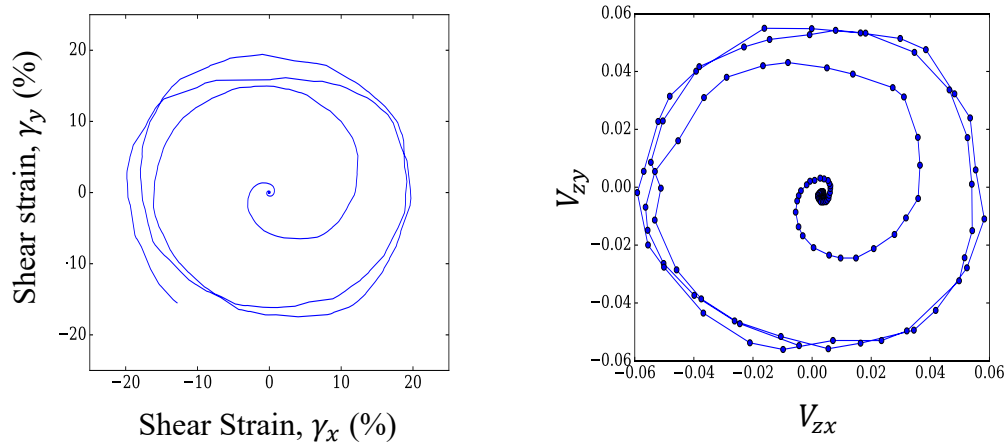


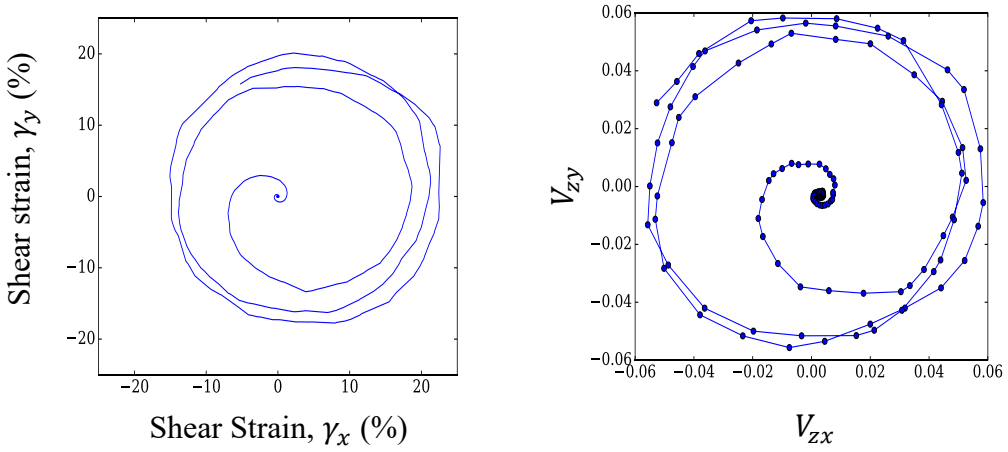
Figure 5.19 Evolution components in particle-void fabric tensor

Figure 5.20 shows the evolution in the fabric tensor components V_{zx} versus V_{zy} in comparison with the development of shear strain γ_x versus γ_y for all the loading paths. In case of uni-directional loading the evolution of V_{zy} is not significant, thus, only the evolution of V_{zx} with cycle number is shown in comparison with the shear strain, γ_x in Figure 5.20h. It is observed that the trajectory of evolution in fabric tensor components is very similar to the development of shear strains in XY-plane. In circular loading path, no significant total shear strain is developed for the initial 9 cycles of loading, but only within 2 cycles of further loading the total shear strain reaches up to 20% and stabilizes (refer to Figure 5.20a). Similarly, the

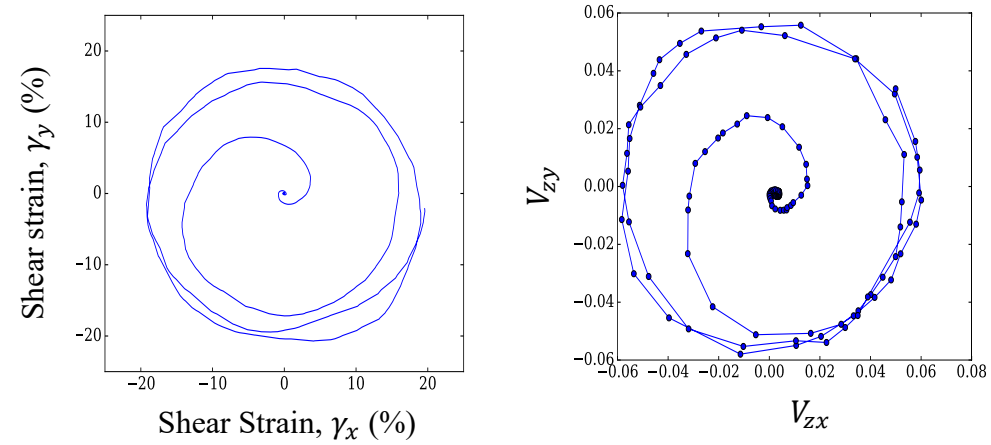
trajectory of evolution in the fabric components is a tiny circle for the initial 9 cycles, and with 2 cycles of further loading, the trajectory of evolution in fabric components reaches to a bigger circle of radius 0.06 and gets stabilized. Similar observations are also made for other loading paths (refer to Figure 5.20(b-h)). This resembles identical evolution behaviors in both the particle void fabric components and the shear strains in XY-plane.



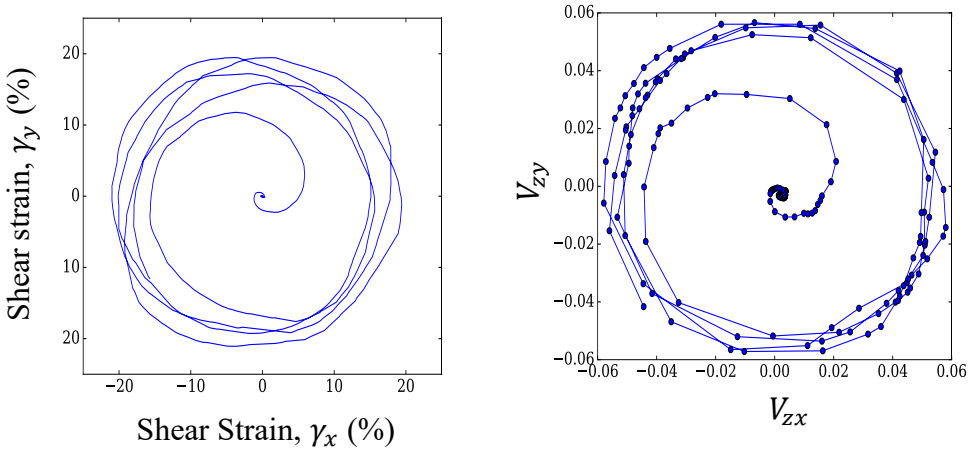
(a) Circular loading path



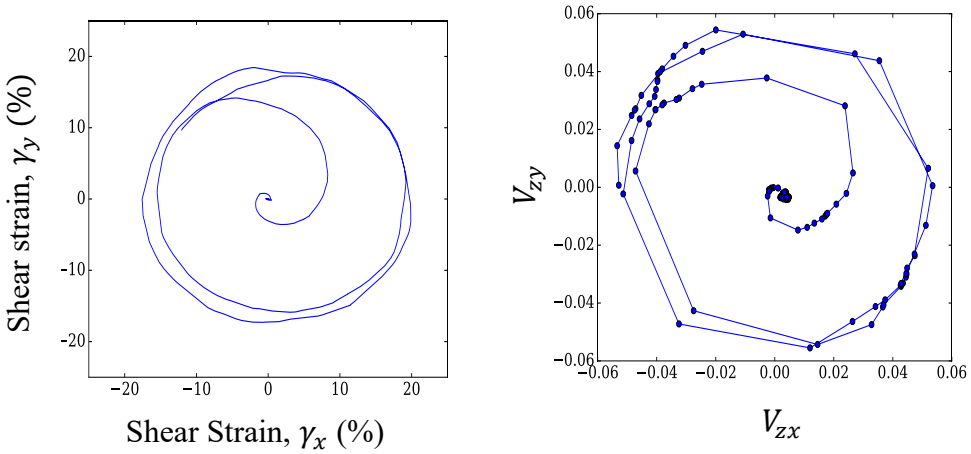
(b) Oval_0.8 loading path



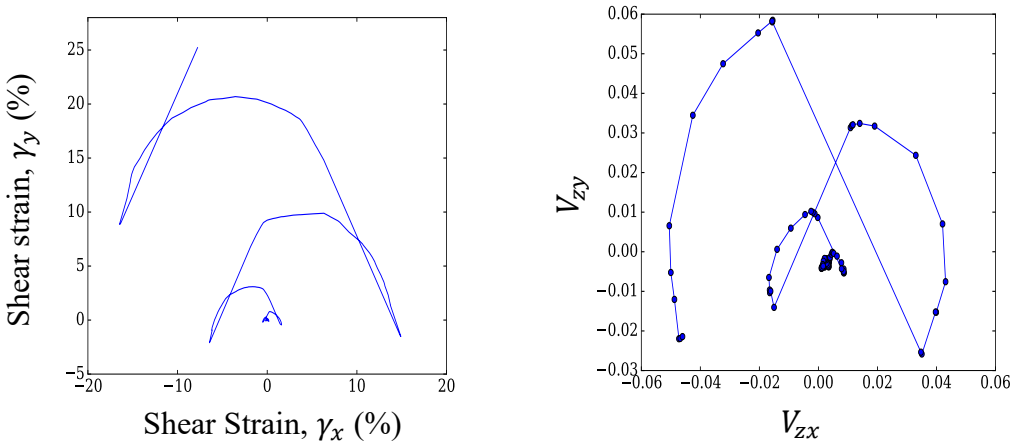
(c) Oval_0.6 loading path



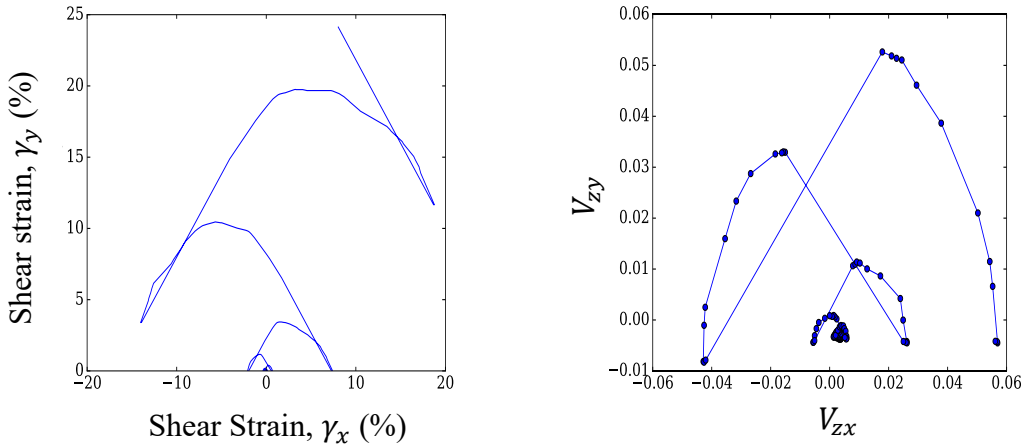
(d) Oval_0.4 loading path



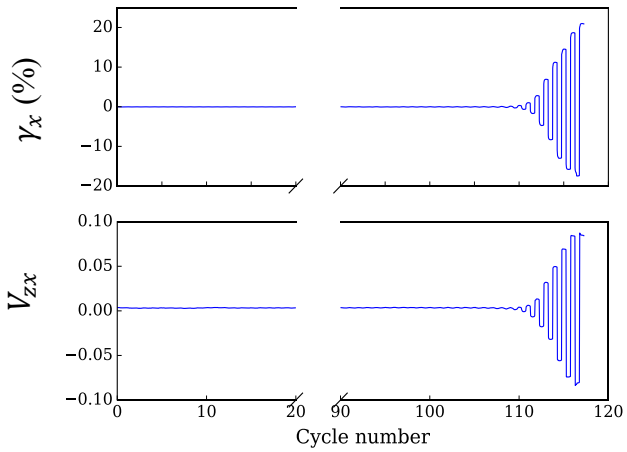
(e) Oval_0.2 loading path



(f) Figure-8_1 loading path



(g) Figure-8_0.6 loading path



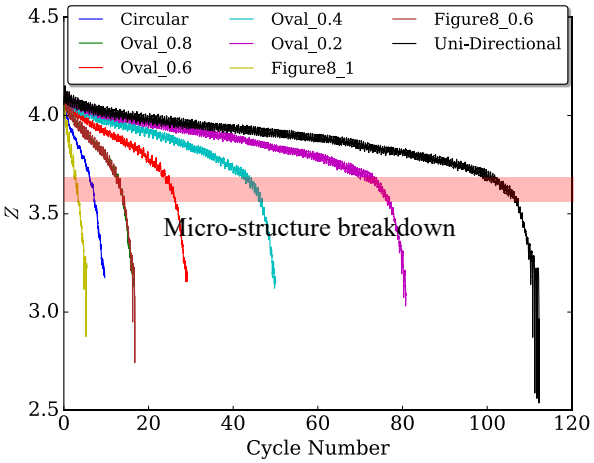
(h) Uni-directional loading path

Figure 5.20 Evolution of particle-void fabric in comparison with generation in shear strains

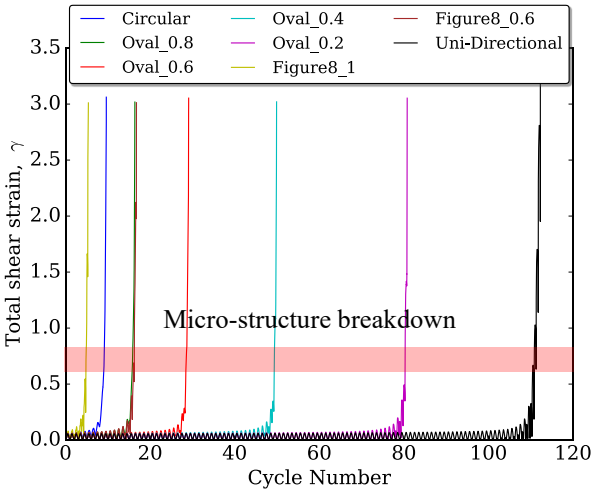
Overall, the development of individual components of shear strain has a clear correlation with the evolution of individual fabric components of the particle-void fabric tensor.

5.5 Understanding liquefaction from microlevel perspective

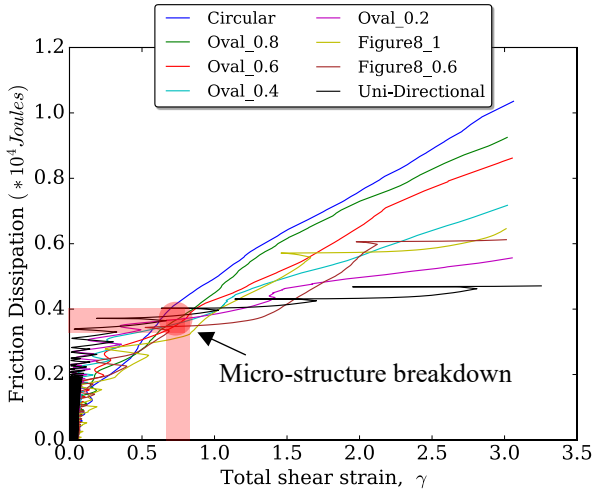
In general, the liquefaction resistance of sands in multi-directional loading paths is quantified based on the failure criterion adopted by Ishihara and Yamazaki (1980), i.e., number of loading cycles required to generate a total shear strain ($\gamma = \sqrt{\gamma_x^2 + \gamma_y^2}$) greater than 3%. This criterion is purely empirical based, no physical basis of explanation is provided for choosing 3% strain as the threshold for liquefaction. In this study, different parameters such as coordination number, total shear strain, degree of anisotropy and coordination number are recorded continuously in the course of different cyclic loading paths. The evolution of above parameters till the 3% total shear strain in different loading paths is shown in Figure 5.21. The evolution of coordination number in different loading paths till 3% shear strain is shown in Figure 5.21a. It is observed that in all the loading paths the rate of decrease in coordination number significantly increases after reaching a coordination number of about (3.6-3.7). Thus, at this range of coordination number "micro-structure breakdown" in the granular packing happens and results in significant reduction in coordination number further. Based on the cycle number where the "micro-structure breakdown" happens a range of shear strains are determined, i.e. around 0.6-0.75% of shear strain, where break down in microstructure initiates in all the loading paths, as shown in Figure 5.21b. Interestingly at the point of "micro-structure breakdown", same amount of friction dissipation, i.e. about $0.38 * 10^4$ joules, and the same amount of degree of anisotropy in contact normal, i.e. about 0.45, are generated in all the loading paths, as shown in Figure 5.21(c,d). Also, the particle-void fabric tends to significantly evolve after reaching the point of "micro-structure breakdown" as shown in Figure 5.21(e,f). Thus, the point of micro-structure breakdown has unique characteristics in terms of micro and macro response of granular soils. Thus, it can be considered as a criterion to decide soil liquefaction.



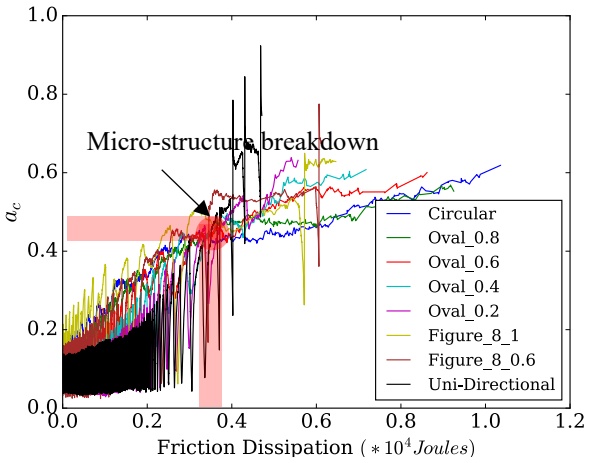
(a) Evolution of coordination number



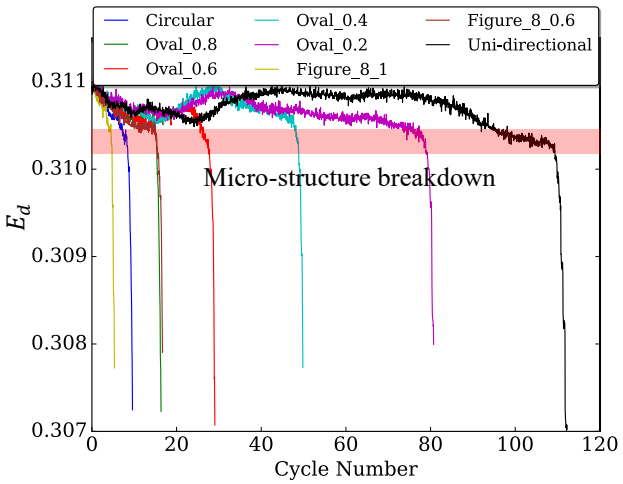
(b) Generation of shear strain



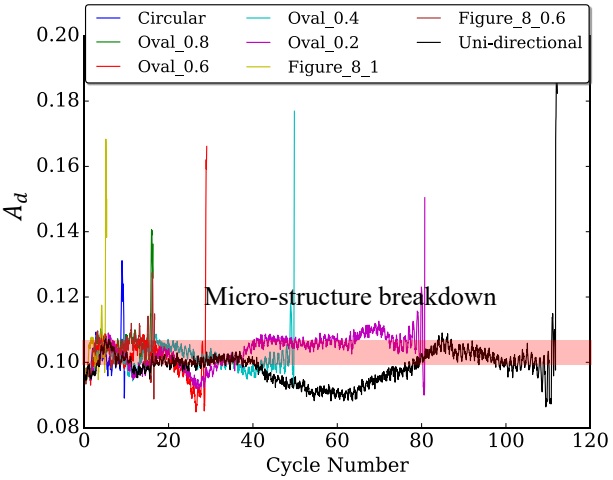
(c) Frictional dissipation Vs shear strain



(d) Frictional dissipation Vs anisotropy degree



(e) Evolution of E_d



(f) Evolution of A_d

Figure 5.21 Variation in different parameters till 3% shear strain

5.6 Conclusions

In summary, stress-controlled multi-directional loading tests on a granular assembly are conducted using DEM. Different multi-directional loading paths, such as circular, oval, figure-8, and uni-directional, are generated.

1) Granular packings subjected to multi-directional loading are more susceptible to liquefaction compared to uni-directional loading. With the increase in aspect ratio (AR), the susceptibility to liquefaction also increases. For the same AR value, figure-8 loading paths are more susceptible to liquefaction compared to oval/circular loading tests.

2) In uni-directional and figure-8 loading paths, the maximum excess pore water ratio (r_u) reaches 1, indicating full liquefaction. Whereas, in circular/oval loading paths, the excess pore water pressure ratio is always smaller than the initial confining pressure. This is due to the existence of load-bearing structure even after liquefaction.

3) The contact-based fabric descriptors (Z, a_c) has a good correlation with the load-bearing capacity of the granular packing. In multi-directional loading, the coordination number from 4.1 gradually decreases with an increase in cycle number and eventually gets stabilized at a value of 3.2 in all the loading paths. In uni-directional and figure-8 loading paths, in the limiting state, as the stress path travels to zero shear stress zone, the load-bearing structure collapses, i.e., coordination number dips to a minimum value. After limited strain, as the contacts reestablish, a new load-bearing structure is formed with the average maximum coordination number of 3.2. The anisotropy degree (a_c) in contact-normals, gradually increases from 0.05 and get stabilized at a value of 0.8 in all the loading paths.

4) The particle-void fabric descriptors (E_d, A_d) has a clear correlation with the deformation behavior of the granular packing. In all the loading paths, the variation in shear strain is negligible for most loading cycles. In the last few cycles, the shear strains tend to develop at an exponential rate. A similar trend is observed in the evolution of E_d and A_d, where E_d decreases and A_d increases. In circular/oval and uni-directional loading paths, the existence of an ultimate state in terms of deformation is observed. Similarly, the evolution of particle-void fabric descriptors also tends to saturate when the ultimate state is reached.

5) The trajectory of evolution in the fabric tensor components has a clear correlation to the macroscopic directional behavior. The evolution in contact-normal fabric tensor in the initial cycles of loading is similar to the applied loading path in XY-plane. In all the circular/oval loading paths it is found that the evolution of fabric tensor eventually gets stabilized to a same trajectory, indicating independence of loading path in limiting state. The evolution in particle-void fabric tensor is similar to the developed shear strains in XY-plane.

6) The point of micro-structure breakdown in the granular soils during cyclic loading can be used as a criterion to decide soil liquefaction. For a given packing, at the point of micro-structure breakdown, a unique range of shear strain and frictional dissipation will be generated and are independent of the loading paths taken.

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CHAPTER 6

Summary and Future Research

6.1 Major contributions and findings

New particle-void fabric descriptors (E_d and A_d) are proposed for a 3D granular packing in order to quantify void-space in both liquefied and non-liquefied states. Special particle-void cells are generated from the Voronoi cells of the granular packing which describes the normalized void space around a particle. Particle-void cells are approximated to an ellipsoidal fitting using Minkowski tensor, $\boldsymbol{W}_0^{20}$. The average measure of local shape elongation factors and the principal direction of local particle void cell gives a scalar global mean elongation factor, E_d and second-order particle-void fabric tensor, $\boldsymbol{V}$ respectively. The anisotropy degree of the tensor $\boldsymbol{V}$ is calculated as A_d. Thus (E_d, A_d) are the proposed as the new particle-void fabric descriptors, and $\boldsymbol{V}$ is the new particle void fabric tensor. The evolution of the new fabric quantities in uni-directional and multi-directional loading paths reveals new insights into cyclic liquefaction phenomenon, and some of them are listed below:

1) The evolution of new particle-void descriptors primarily occurs in the post-liquefaction stage, accompanied by increasing post-liquefaction deformation. After initial liquefaction, the value of E_d decreases and A_d increases gradually cycle after cycle. For a given relative density, the post-liquefaction deformation of the granular packing is strongly correlated to the evolution of the particle-void fabric descriptors. In case of multi-directional loading, the development of deformation in different directions in XY-plane is also well correlated to the evolution of fabric tensor ($\boldsymbol{V}$) components in ZX and ZY- directions.

2) In the post-liquefaction stage of uni-directional loading, the jamming transition could occur in the granular flow, which transforms the granular assemblage from a "fluid-like" state to a "solid-like" state to arrest the flow deformation. The coordination number (Z_{crit}) at which this transition occurs is found to be unique and independent of both loading path and relative density. Given the relative density, unique hardening state line (HSL) is defined in the void-fabric space, that delineates the boundary between a flow stage and a jamming/hardening stage when the soil is fully liquefied.

3) E_d is a good indicator for the compressibility of a granular packing. Granular packing with high E_d value in the liquefied state undergo less volumetric compression compared to the granular packing with low E_d value. E_d is also a good indicator for the relative density of a granular packing. Granular samples consolidated to the same stress state but with different E_d values have different relative densities. Sample with low E_d value has higher relative density and sample with relatively higher E_d value has lower relative density.

4) Depending on the A_d of the packing in a liquefied state, the a_c of the reconsolidated packing is generated. Liquefied samples with high A_d values generate high a_c packing after reconsolidation. Similarly, samples with isotropic particle-void fabric in a liquefied state, i.e., low A_d value generate samples with isotropic contact-normal fabric, i.e., low a_c value after reconsolidation

6.2 Future research

6.2.1 Cyclic liquefaction simulations using real particle shapes

It is evident from the current study that DEM has been a powerful tool to simulate the cyclic liquefaction phenomenon of granular soils. Current DEM simulations ignore the effect of realistic particle shapes, which might significantly influence the density, stiffness, and strength of granular packing. Future research will be focused on running more realistic simulations considering real particle shapes. One way to simulate real particle shapes in DEM is to use "clumped" particles. Figure 6.1 shows an example of a clumped particle with 14 spheres rigidly bonded to approximate the realistic shapes of grains in Toyoura sand (Katagiri 2018). Previous research studies have been focused on obtaining real particle shapes of different sands through

X-ray CT scanning experiments and approximating clumps for the same (e.g., Das 2007; Katagiri et al. 2010; Matsushima 2002). Thus, a more realistic DEM simulation of cyclic liquefaction phenomenon is possible and can provide valuable insights into the fundamental understanding of soil behavior. One challenging task would be the calibration of DEM simulation to experimental results, as the DEM modelling includes different model parameters for particle contact (Young's modulus E, Poisson's ratio v) and inter-particle friction angle (ϕ) which can have an inter-dependent effect on the macroscopic behavior.

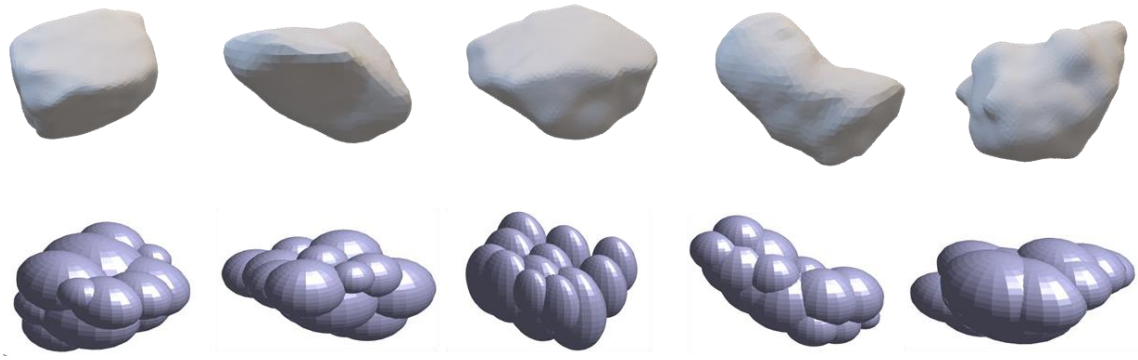
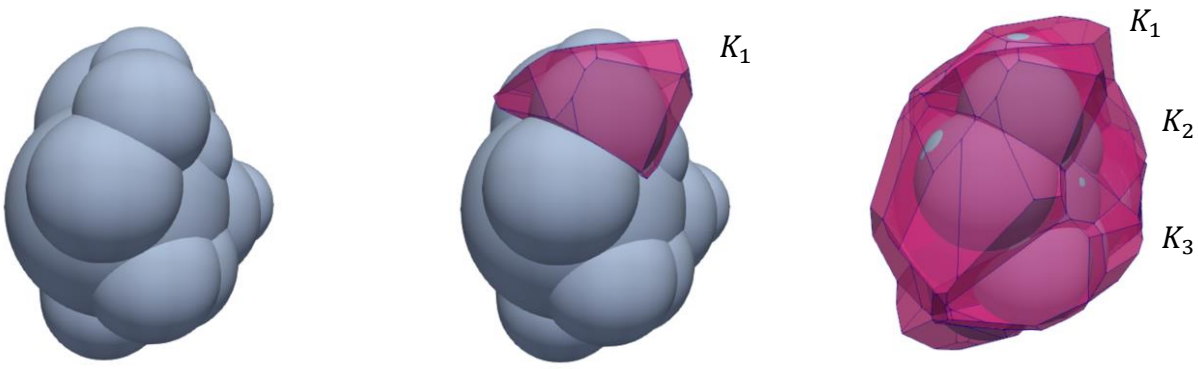


Figure 6.1 Clump particles approximating real particle shapes of Toyoura sand

6.2.2 Void descriptors for granular packing with realistic particles

In this study new particle-void fabric descriptors (E_d and A_d) were introduced by obtaining local shape factors of "particle-void cells" generated from weighted Voronoi tessellation of a polydisperse spherical particle granular assembly. But for a granular assembly with clumped particles, the present method of computing particle void descriptors may not hold good. A new mathematical definition for particle-void descriptors is required and will be a focus of future research. A possible idea for the new definition is presented here.

In the current study, the particle void cells are obtained after the normalization of Voronoi cells radial dimension with the radius of the constituent spherical particle. Figure 6.2(a,b,c) shows a 14-sphered clump particle, Voronoi space of one sphere corresponding to the clump and Voronoi space of all the spheres in a clump respectively. The total Voronoi space around the clump particle is the summation of Voronoi spaces of individual spheres in the clump (refer to Figure 6.2(b,c)). To construct a particle-void cell for a clumped particle, the total Voronoi space must be normalized with the radial measure of the clump. Obtaining the radial measure of the clump can be challenging. Thus, an indirect way by using the moment of inertia of the clump for computing tensorial shape measure of the particle-void cell can be an alternative.



(a) Clumped Particle (b) Voronoi cell of one particle in the clump (c) Total Voronoi cell

Figure 6.2 Generation of Voronoi cell of a clump

It is important to note that individual Voronoi spaces are all convex polyhedrons which are continuous and non-intersecting. However, the total Voronoi space formed would be a concave polyhedron (non-convex body). The Minkowski tensor for a simple non-convex body, as shown in Figure 6.3, can be obtained from the Minkowski tensors of constituent convex bodies using the summation rule as follows (SCHRÖDER-TURK et al. 2010):

$$\boldsymbol{W}_0^{2,0}(L_1 \cup L_2) = \boldsymbol{W}_0^{2,0}(L_1) + \boldsymbol{W}_0^{2,0}(L_2) - \boldsymbol{W}_0^{2,0}(L_1 \cap L_2) \tag{6.1}$$

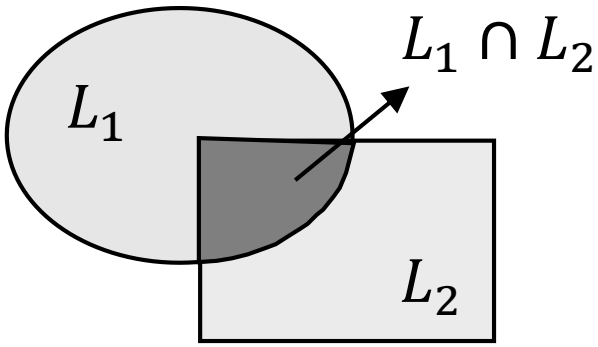


Figure 6.3 Minkowski tensor of a union body

Thus, Minkowski tensor of the total Voronoi space w.r.t. to the centroid of the clump can be obtained by summing up the Minkowski tensors of individual Voronoi spaces at the centroid.

$$\boldsymbol{M}_1 = \sum_{i=1}^{m} [\boldsymbol{W}_0^{20}]_{K_i} \tag{6.2}$$

where $[\boldsymbol{W}_0^{20}]_{K_i}$ is the Minkowski tensor of the K_i Voronoi space (refer to Figure 6.2c). m is the total number of individual Voronoi spaces constituting the total Voronoi space.

Also, the Minkowski tensor of the clumped particle ($\boldsymbol{M_2}$) can be obtained from its moment of inertia tensor ($\boldsymbol{I}$) of the clump as follows:

$$\boldsymbol{M}_2 = \frac{1}{\rho}\left(\frac{tr(\boldsymbol{I})}{2} * \boldsymbol{\delta_{ij}} - \boldsymbol{I}\right) \tag{6.3}$$

where ρ is the density of the clumped particle and $\boldsymbol{\delta}_{ij}$ is the Kronecker delta function.

The normalized Minkowski tensor $\boldsymbol{M_3}$ can be defined as follows:

$$\boldsymbol{M}_3 = \boldsymbol{M}_1 * \boldsymbol{M}_2^{-1} \tag{6.4}$$

Second-order tensor quantity, $\boldsymbol{M}_1$ gives the quantification of volume distribution in the total Voronoi space, whereas, $\boldsymbol{M}_2$ gives the quantification of volume distribution of the clumped particle. Thus, $\boldsymbol{M}_3$ quantifies the volume of the normalized void distribution around the clumped particle. Local shape factors (e_d,$\boldsymbol{v}$) can be obtained from the eigenvalue analysis of $\boldsymbol{M}_3$ tensor and finally global shape descriptors (E_d, A_d) can be obtained after homogenization.

6.2.3 Physics-based constitutive models with various fabric indicators

Fabric indicators, as we know, are the mathematical descriptors for the microscale understanding of granular materials. In this study, the evolution of these indicators in cyclic liquefaction phenomenon is highlighted. The evolution in contact-based fabric indicators was found to have an excellent correlation with the macroscopic behavior of sands in pre-liquefaction stage, such as the reduction in effective stress and transition in soil behavior (contractive and dilative). Similarly, the evolution in new particle-void fabric descriptors was found to have a strong correlation with the macroscopic behavior in the post-liquefaction stage, such as cyclic mobility and transition of the state of sands (flow state and hardening state). These descriptors have a strong physical meaning, and their evolution during the loading process have strong correlations with macroscopic soil behavior. Most of the constitutive models are phenomenological in nature involving various parameters and unverified assumption in model formulation. Incorporating the fabric indicators and defining the rules for their evolution in the available constitutive models may lead to the development of a physics-based constitutive model for soil liquefaction.

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